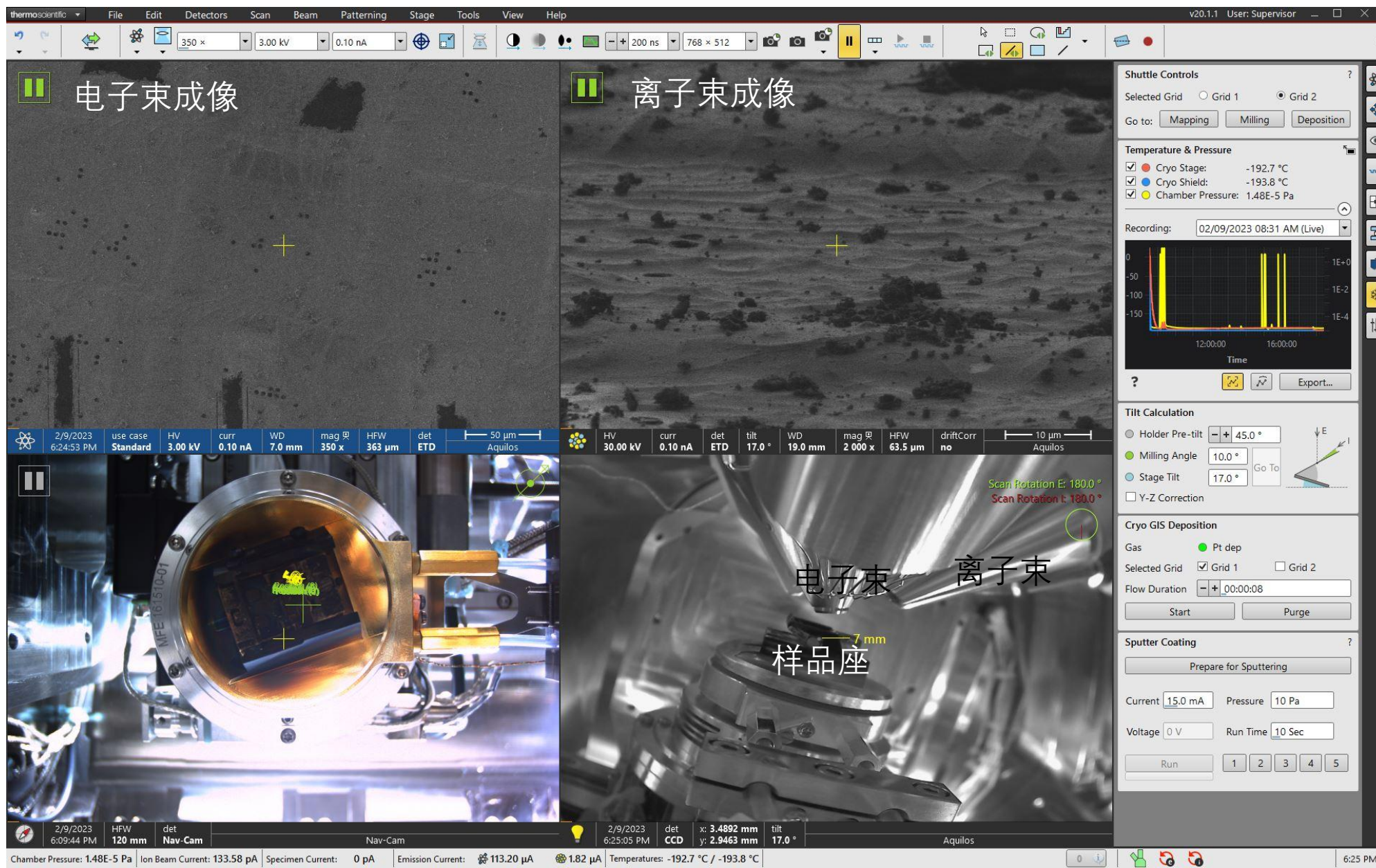


Aquilos2成像特点



用户需求/技术流程

用户手动

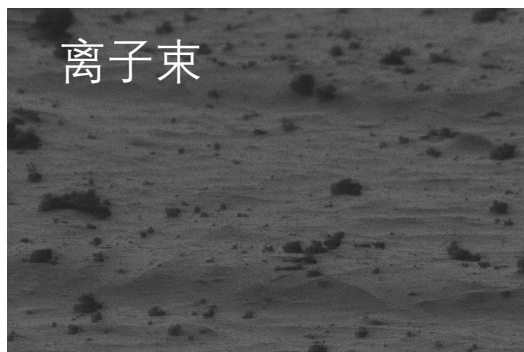
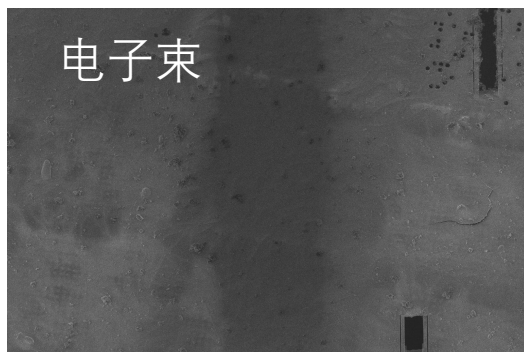
Sputtering
GIS
(sputtering)

Sputter:
30mA, 10Pa, 15sec
15mA, 10Pa, 10sec

GIS:
28°C, 10sec
38°C, 8-10sec
40°C, 4-6sec
45°C, 8-12sec
以喷完后用聚焦离子束小范围聚焦不烧样品为底线

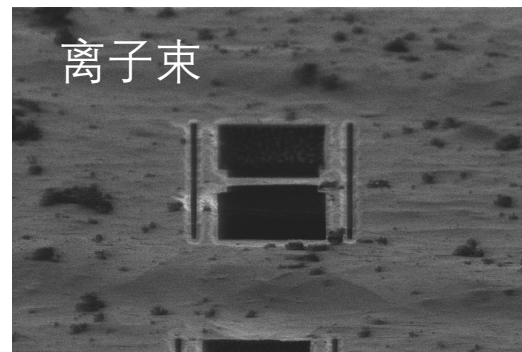
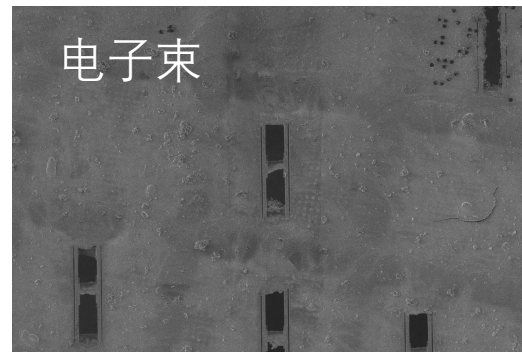
自动

定位
(共焦点调节)



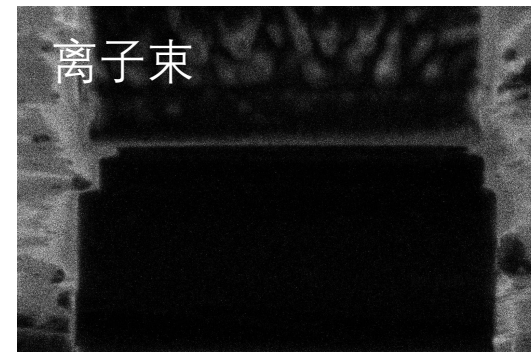
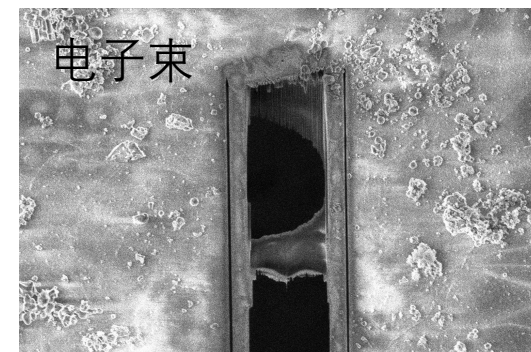
自动

粗减薄
0.3nA束流 -> 2-3 μm
0.1nA束流 -> 0.5-1 μm
(可能伴随小角度倾转)



用户手动

精修至目标厚度
200nm左右
用更低的束流 (可能伴随小角度倾转)



处于共焦点时

不同位点之间移动，离子束与电子束成像可保持同样的工作距离（使用相同的焦距）

电子束与离子束视野中心为同一位点

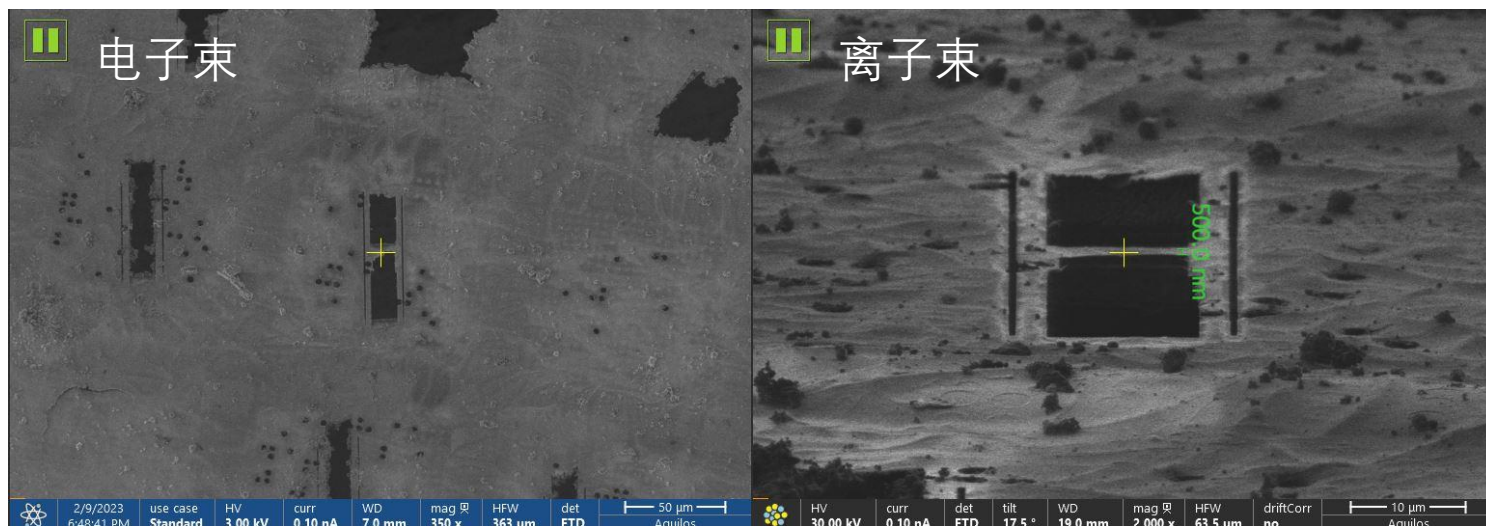
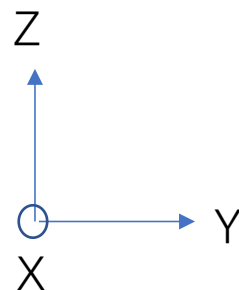
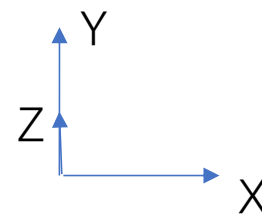
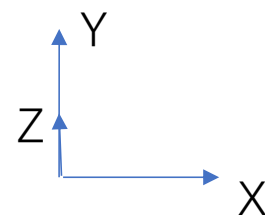
电子束
正焦点

离子束
正焦点

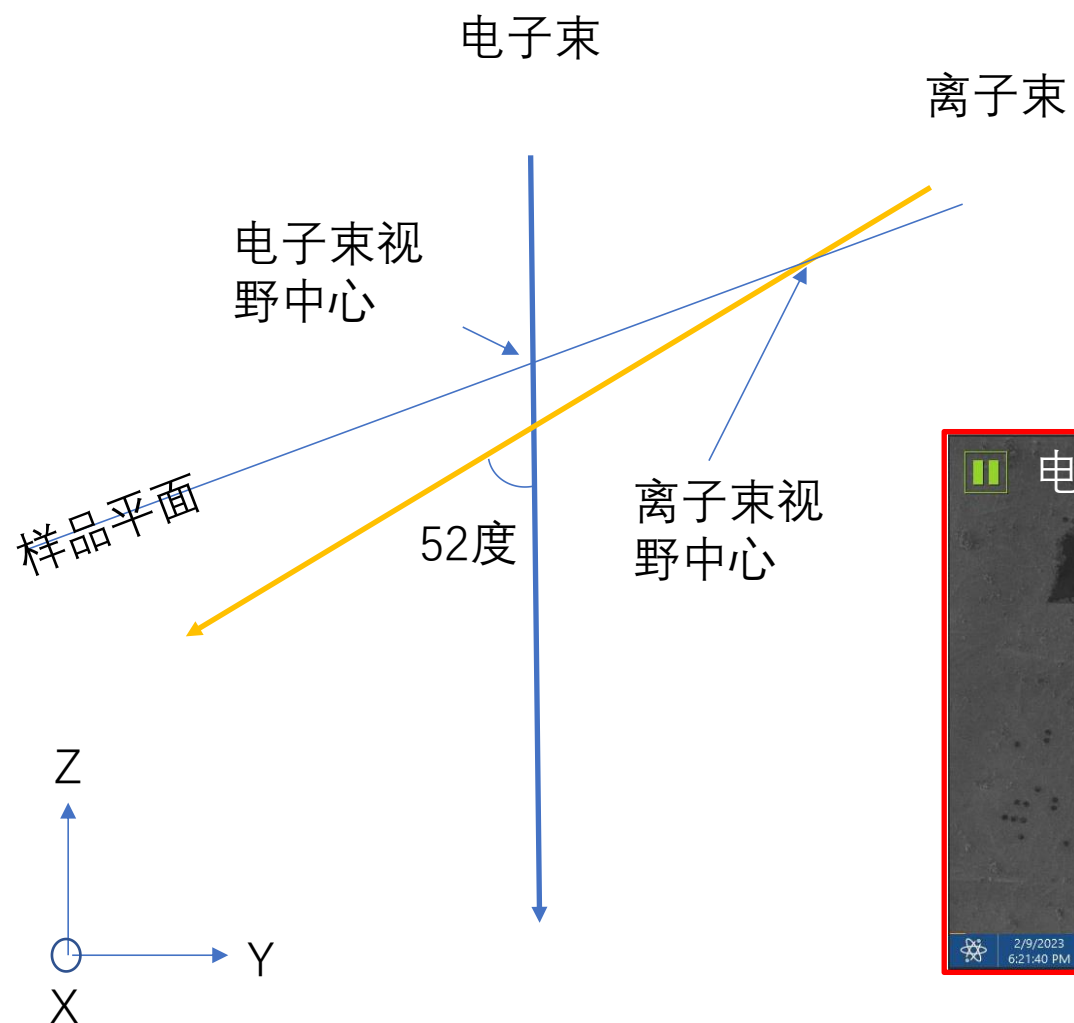
52度

样品平面

样品台倾转时，电子束与离子束目标点基本不偏移

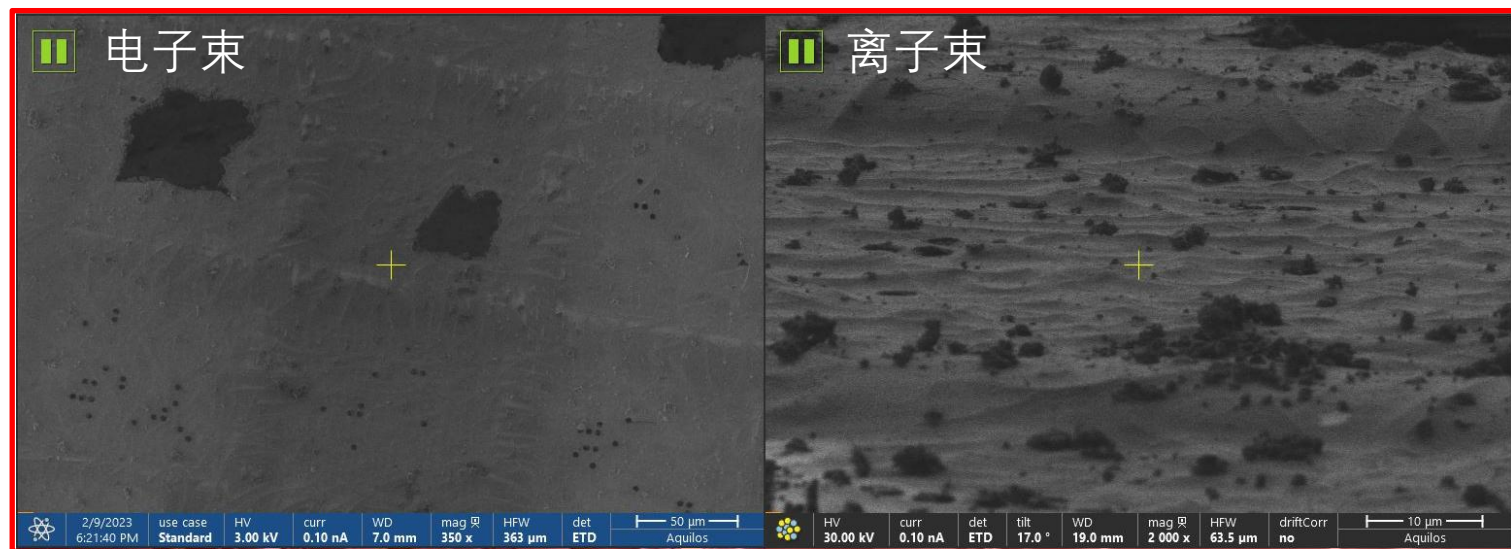


不处于共焦点时-上方

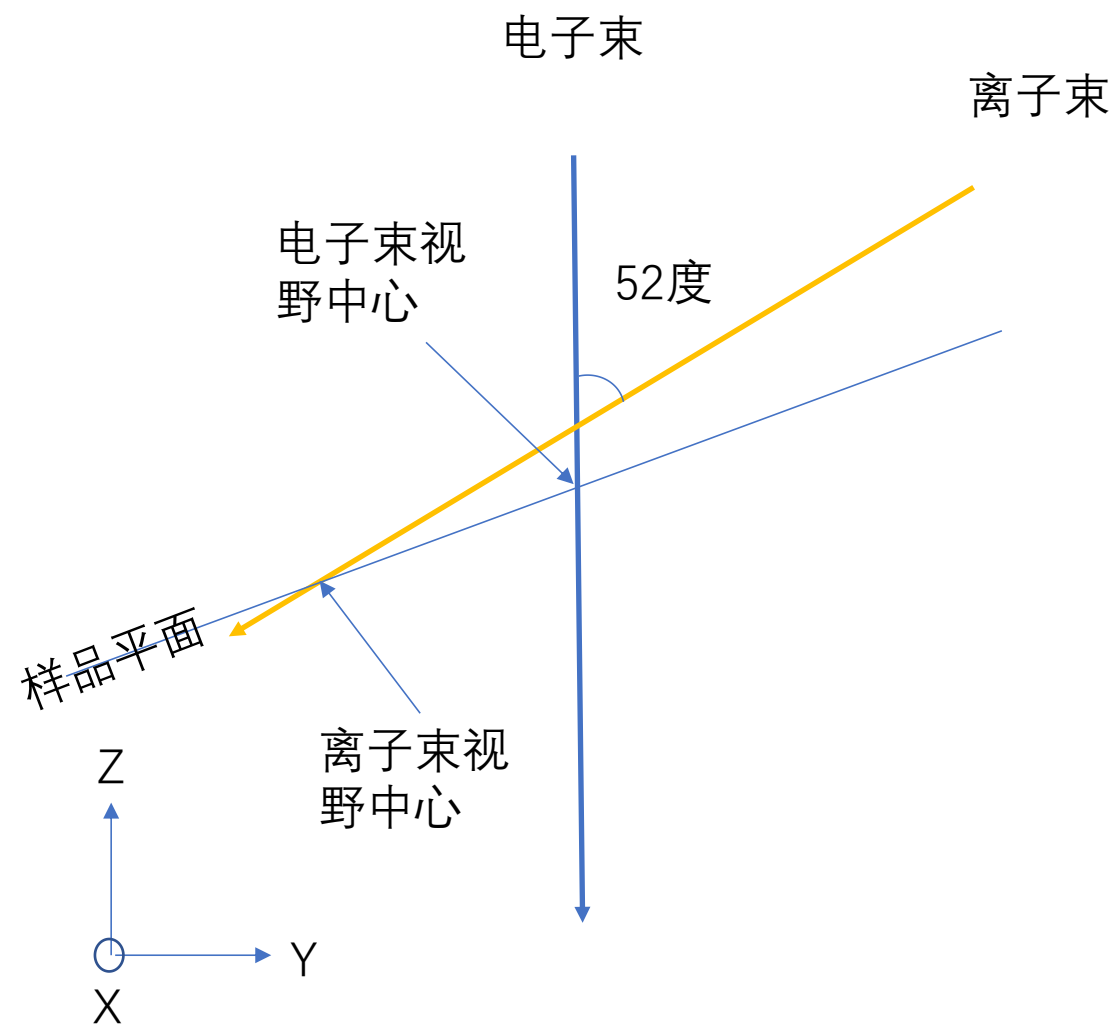


不同位点之间移动，离子束与电子束成像工作距离需要重新调整（焦距有变化）

样品台倾转时，电子束与离子束目标点偏移较大



不处于共焦点时-下方



不同位点之间移动，离子束与电子束成像工作距离需要重新调整（焦距有变化）

样品台倾转时，电子束与离子束目标点偏移较大

当前仪器已有的自动化cryo-FIB减薄工具- AutoTEM Cryo

The screenshot displays the AutoTEM Cryo software interface, which is organized into several panels. On the left is the 'TEMPLATE LIST' panel, showing a search bar and a list of templates: 'Yeast default template Cryo' (highlighted with a red box), 'auto_sleep Cryo', and 'Cell Lamella template Cryo'. The main area is divided into three vertical sections: 'Preparation', 'Milling', and 'Thinning'. The 'Preparation' section includes settings for 'Ion HFW Oversize' (30.0 μm), 'Sample Pretilt' (45.0 $^\circ$), and 'Milling angle' (Not Available). The 'Milling' section includes 'Size' (5.0 μm x 3.0 μm), 'Correction Factor' (0.50), and a list of steps with toggle switches: DELAY, ELECTRON REFERENCE DEFINITION, REFERENCE DEFINITION, STRESS RELIEF CUTS, REFERENCE REDEFINITION 1, ROUGH MILLING, ROUGH MILLING - ELECTRON IMAGE, REFERENCE REDEFINITION 2, MEDIUM MILLING, MEDIUM MILLING - ELECTRON IMAGE, FINE MILLING, FINE MILLING - ELECTRON IMAGE, FINER MILLING, and FINER MILLING - ELECTRON IMAGE. The 'Thinning' section includes 'Final Thickness' (250.0 nm), 'Enable Windows' (disabled), and a list of steps with toggle switches: DELAY, POLISHING 1, POLISHING 1 - ELECTRON IMAGE, POLISHING 2, and POLISHING 2 - ELECTRON IMAGE. A red box highlights the 'EUCENTRIC TILT' toggle switch in the 'Preparation' section, which is currently turned on. A red text overlay in the bottom left corner of the 'Preparation' section reads: '采用倾转样品台寻找共焦点，耗时长、样品台损耗大而且精度不理想'. A yellow box highlights the 'Thinning' section, with a yellow text overlay in the center reading: '非必要'. The bottom status bar shows the version '2.2.0 (core: 10.0.3)'.

采用倾转样品台寻找共焦点，耗时长、样品台损耗大而且精度不理想

非必要

用户需求/技术流程

用户手动

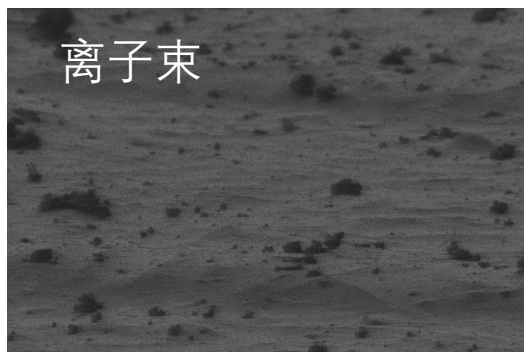
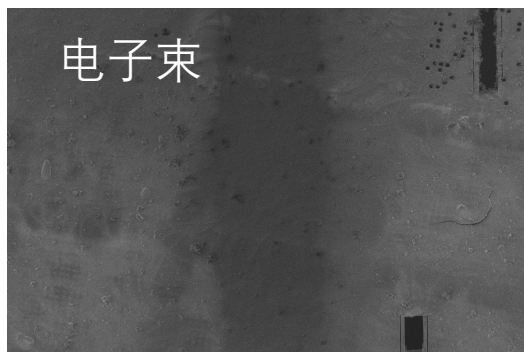
Sputtering
GIS
(sputtering)

Sputter:
30mA, 10Pa, 15sec
15mA, 10Pa, 10sec

GIS:
28°C, 10sec
38°C, 8-10sec
40°C, 4-6sec
45°C, 8-12sec
以喷完后用聚焦离子
束小范围聚焦不烧样
品为底线

自动

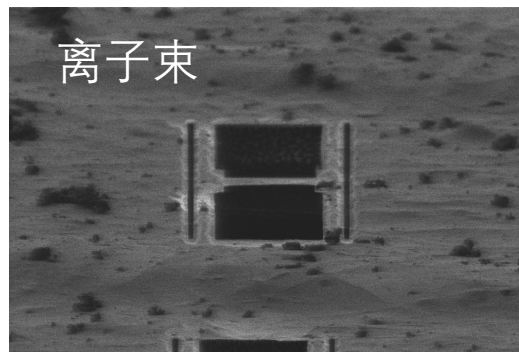
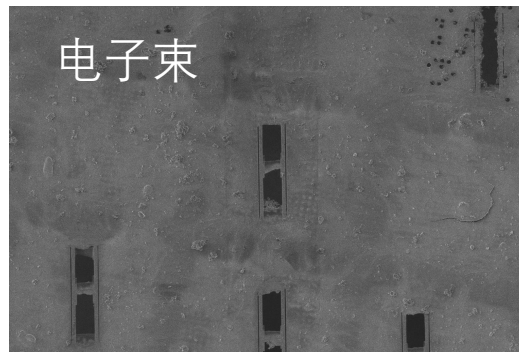
定位
(共焦点调节)



自动

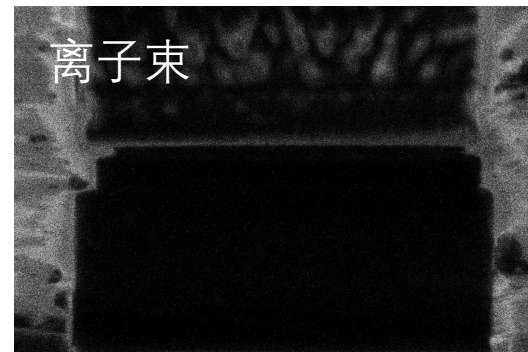
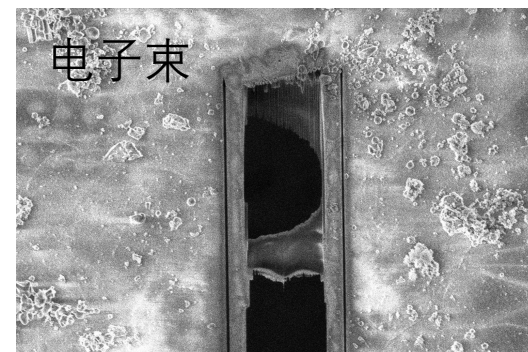
粗减薄

0.3nA束流 -> 2-3 μm
0.1nA束流 -> 0.5-1 μm
(可能伴随小角度倾转)

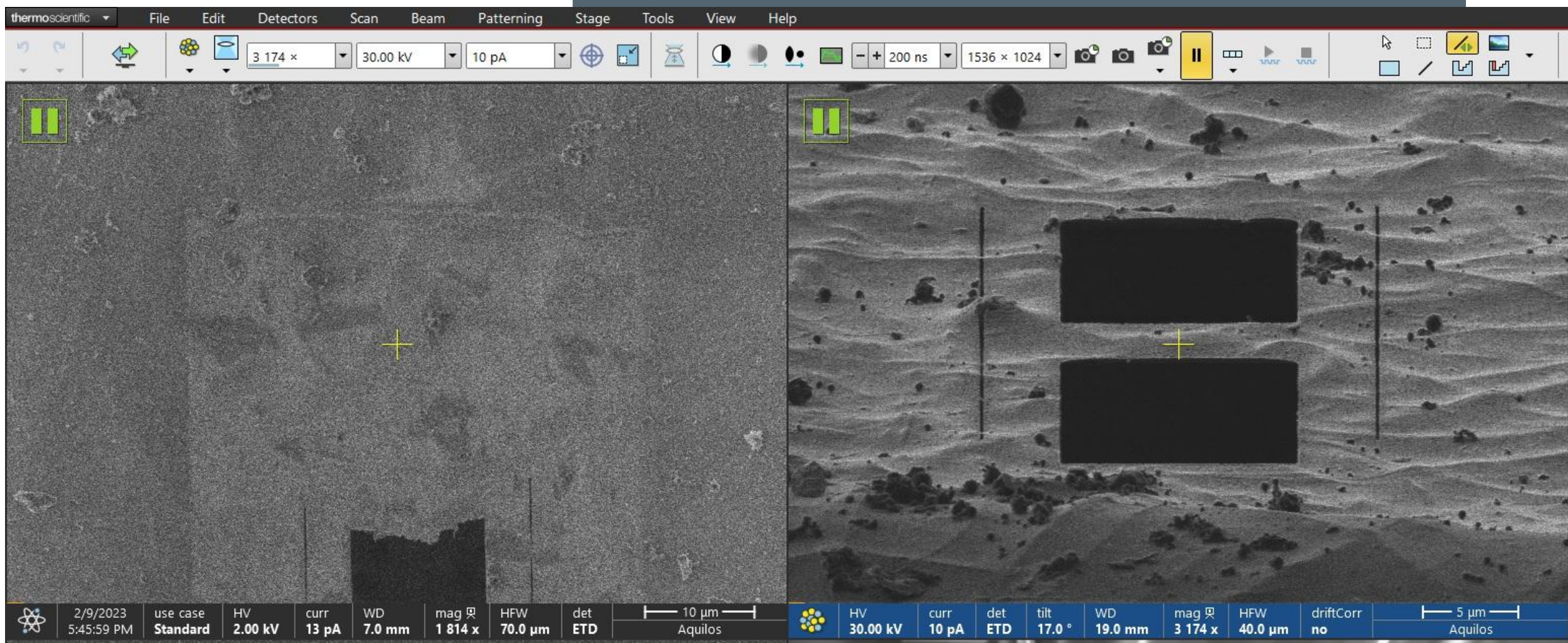
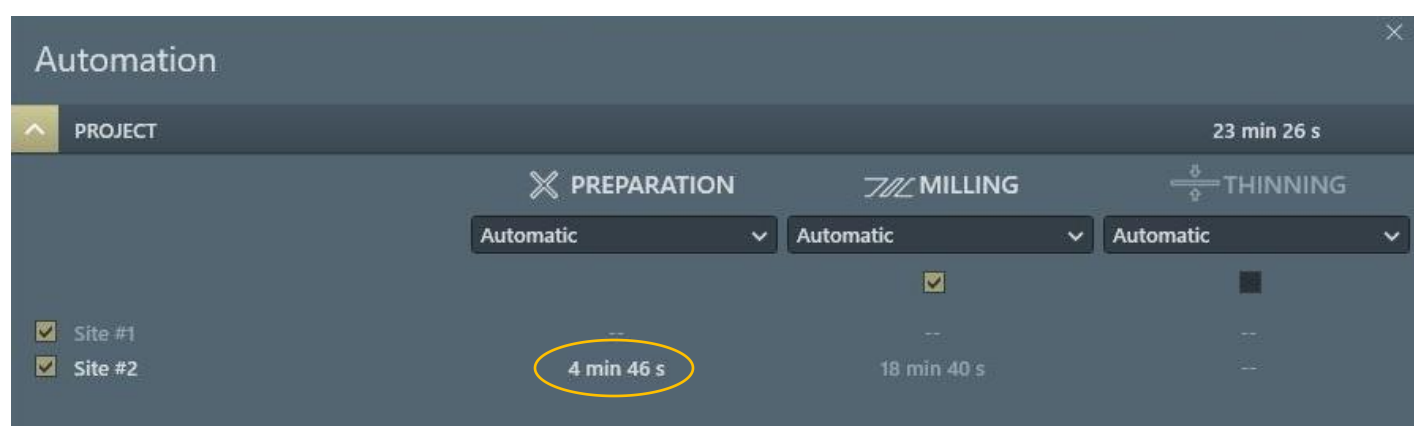


用户手动

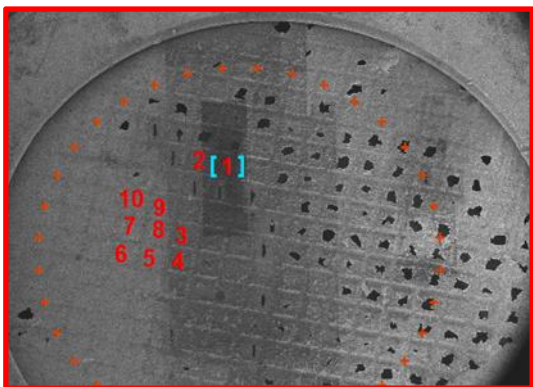
精修至目标厚度
200nm左右
用更低的束流 (可能伴随
小角度倾转)



采用倾转样品台寻找共
焦点，耗时长、样品台
损耗大而且精度不理想

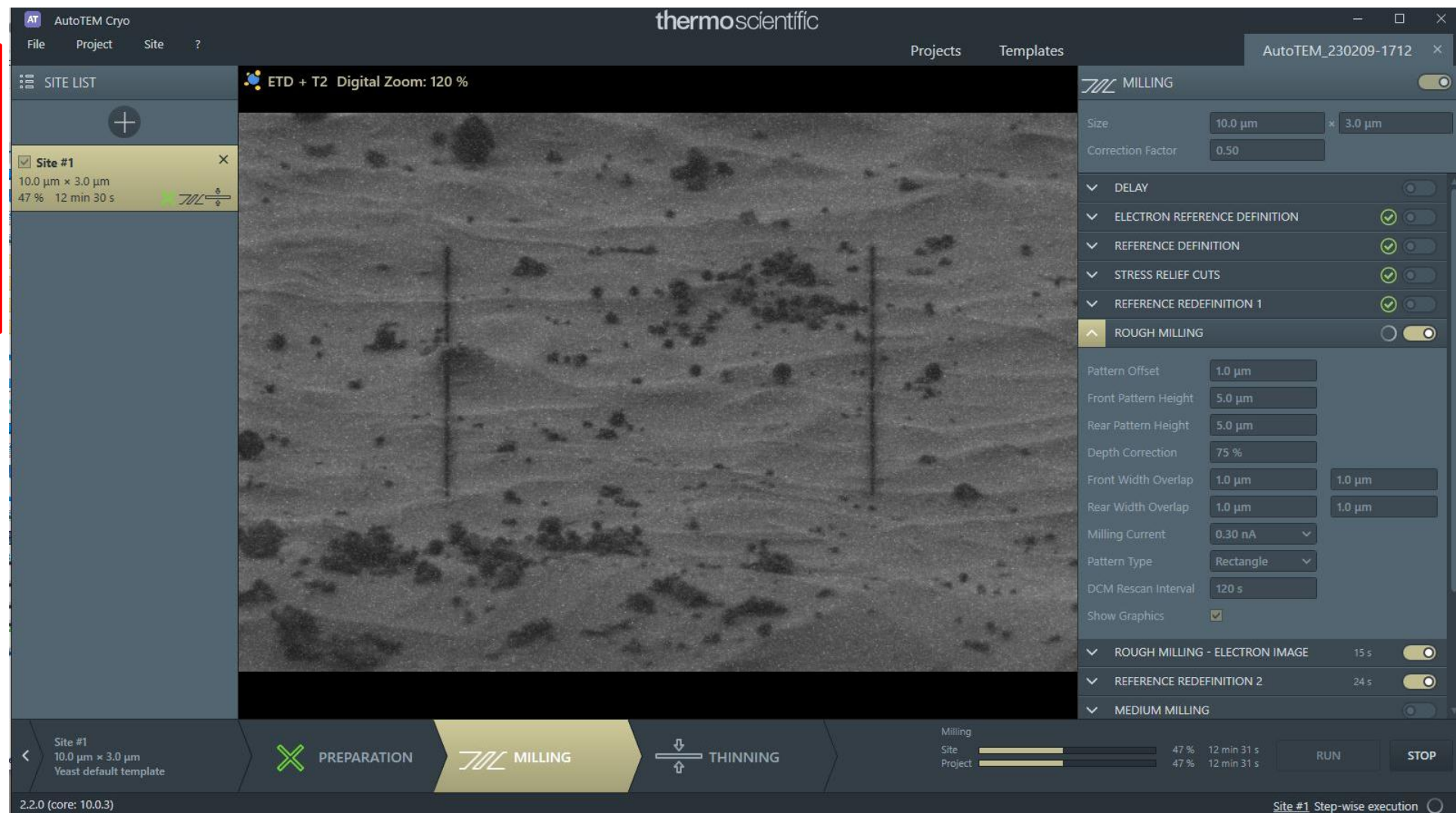


当前仪器已有的自动化cryo-FIB减薄工具- AutoTEM Cryo



位点添加效率低:

- (1) 导航由电子束
(角度) 地图发起, 需
经过4分钟的共焦点搜
索后才能确定离子束切
割方向的位点质量。
- (2) 缺少从Aquilos2
地图与Titan3地图的衔
接对照

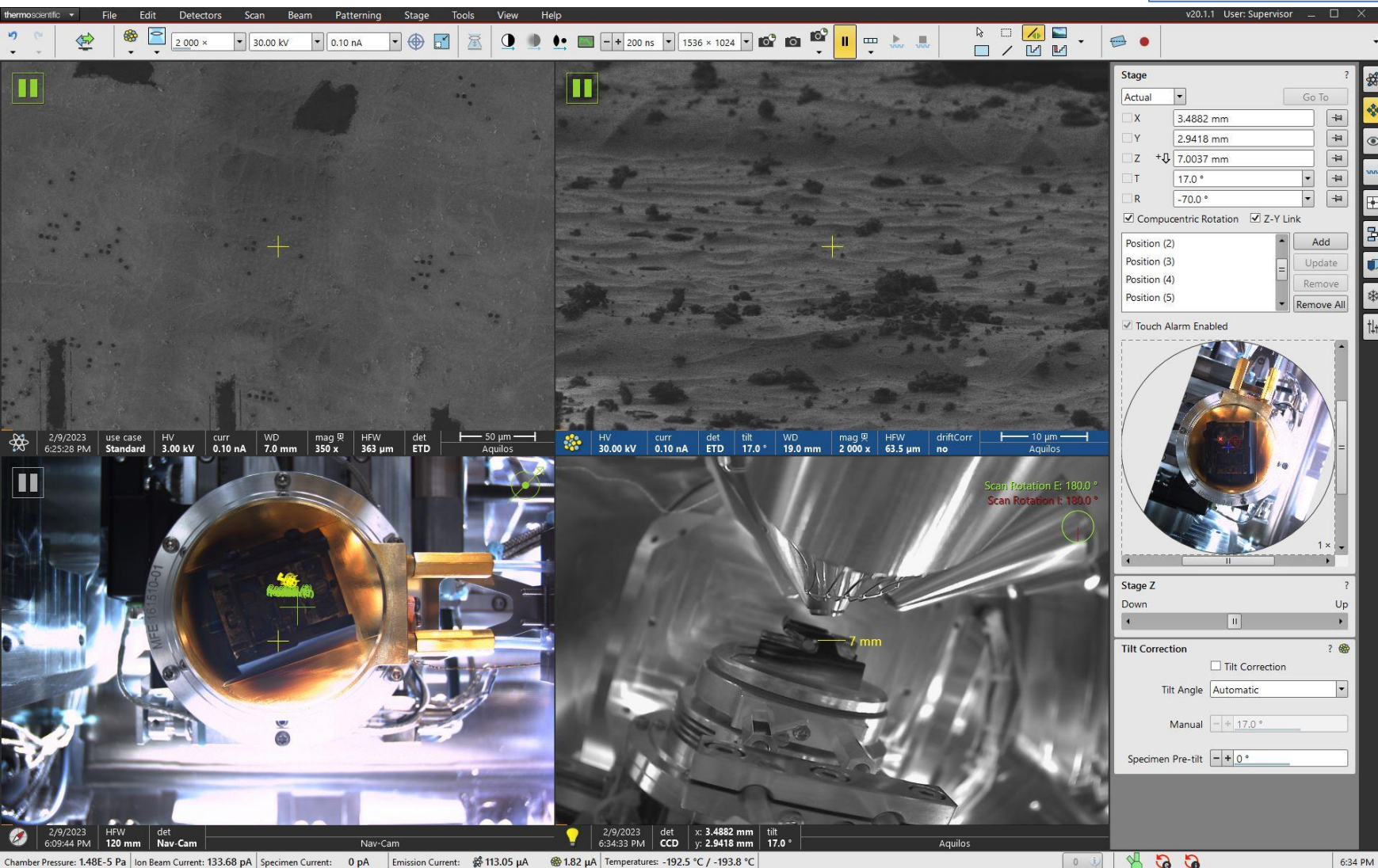


当前所有用户都使用的手动减薄

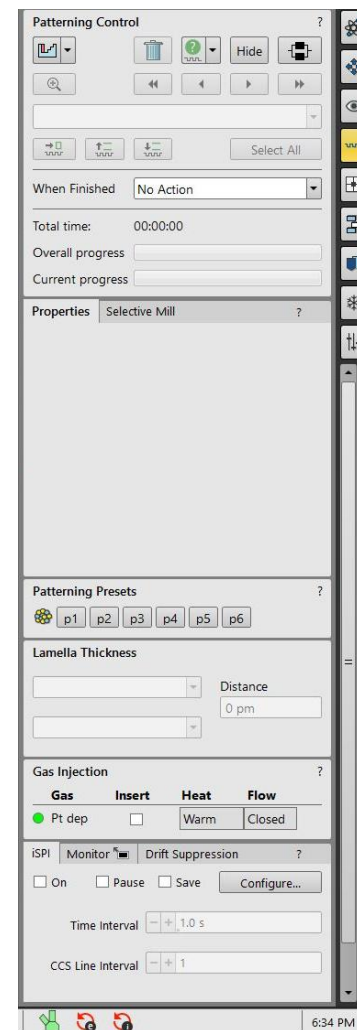
- 优点：
 - 定位速度快
 - 成功率高
- 缺点：
 - 离子束角度发起定位，缺少电子束角度（地图）辅助，缺少对重复位点的信息获取
 - 不在共焦点上进行减薄
 - 需要人在仪器旁持续操作
 - 依赖UI界面，需要多个窗口来回切换，重复性操作多，效率低
 - 缺乏Aquilos2地图与Titan3地图的衔接

当前UI界面

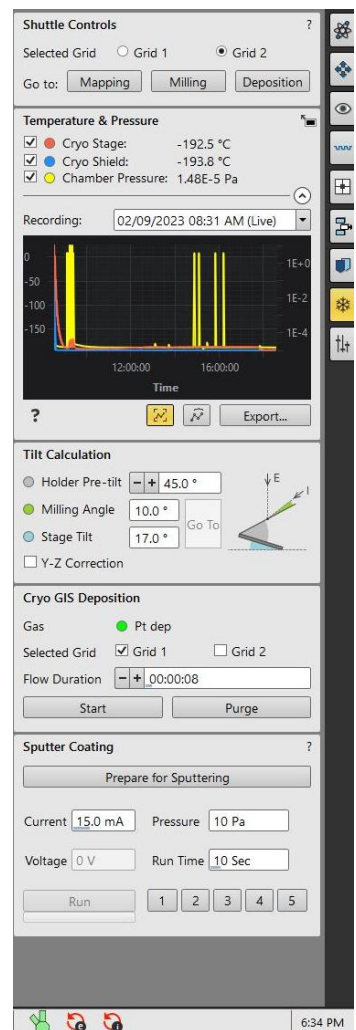
样品台控制



减薄图案控制



样品台倾转控制

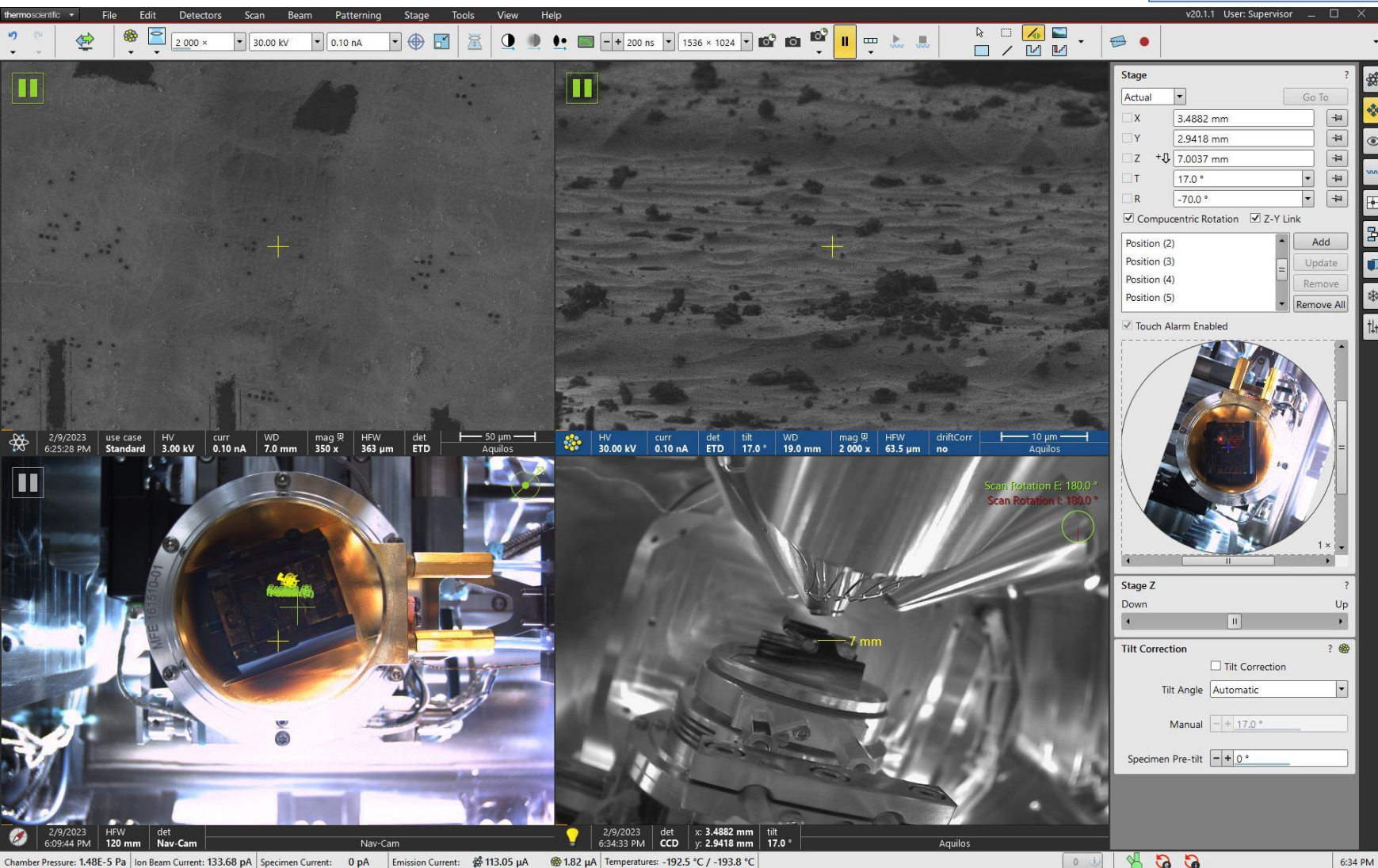


当前大量用户都使用的手动减薄

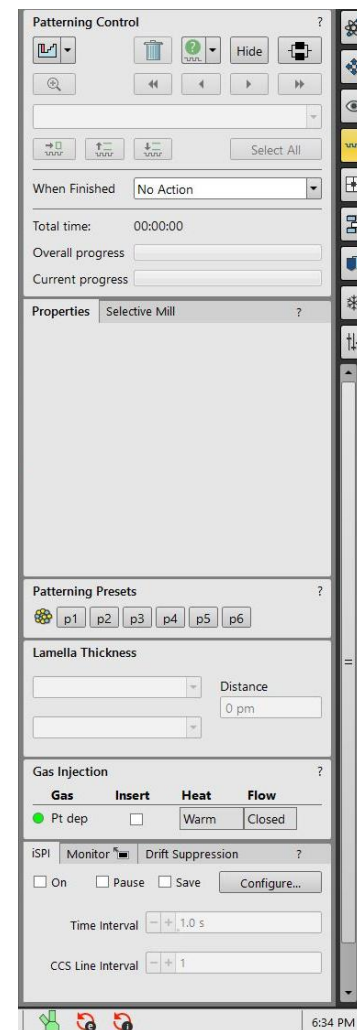
- 优点：
 - 定位速度快
 - 成功率高
- 缺点：
 - 离子束角度发起定位，缺少电子束角度（地图）辅助，缺少对重复位点的信息获取
 - 不在共焦点上进行减薄
 - 需要人在仪器旁持续操作
 - 依赖UI界面，需要多个窗口来回切换，重复性操作多，效率低
 - 缺乏Aquilos2地图与Titan3地图的衔接

当前UI界面

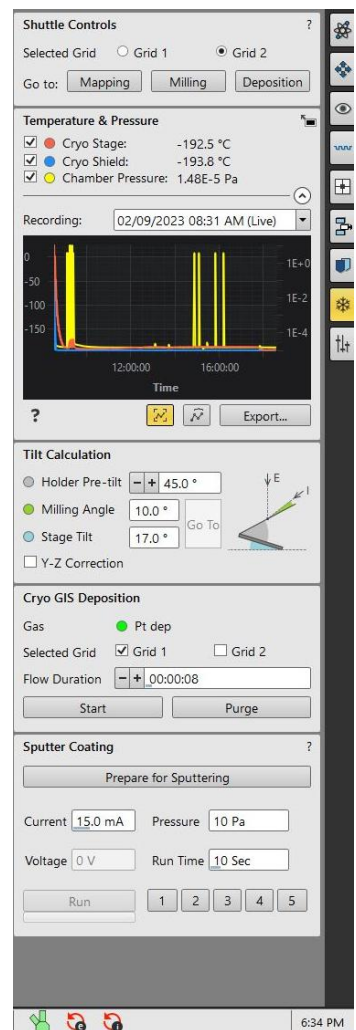
样品台控制



减薄图案控制



样品台倾转控制



需要改进的要点

- 导航从离子束角度发起，电子束角度（地图）辅助，以缩短共焦点调节及定位的耗时
- 增加Aquilos2地图与Titan3地图的衔接
- 重新设计辅助工具，减少多个窗口间的切换

基于赛墨飞提供的AutoScript进行辅助工具编写



AutoScript

AutoScript

AutoScript is a software suite that allows to *perform microscopy analysis and related data processing tasks* in an Python 3 language environment.

The suite contains Python 3 distribution with an interpreter and a rich set of widely used packages like `opencv-python`, `numpy` or `scipy`. Most importantly, it also contains a set of special packages related to microscopy analysis, particularly `autoscript_sdb_microscope_client` with **SDB microscope control features** and `autoscript_toolkit` with **domain-specific data processing capabilities** like fine-tuned computer vision algorithms.

This is the **reference manual** for both forementioned *essential packages*. Please explore the following sections to find detailed information:

[SDB microscope client](#)

[Toolkit](#)

A **custom task library** with a minimum number of dependencies.

Numpy & SciPy

- Linear algebra & Scientific computing

OpenCV

- Computer Vision Library for real-time image processing

QT4

- Design of simple dialogs to move away from the command line with minimal effort.

Matplotlib

- Plotting images and charts

Libraries often used but not in custom library.

Pandas

- Database / Data analysis

SciKit Image

- Image processing / Machine learning

Tensorflow

- Deep Learning



处于共焦点时

本项目利用这一要点进行共焦点调节

电子束
正焦点

离子束
正焦点

电子束与离子束视野
中心为同一位点

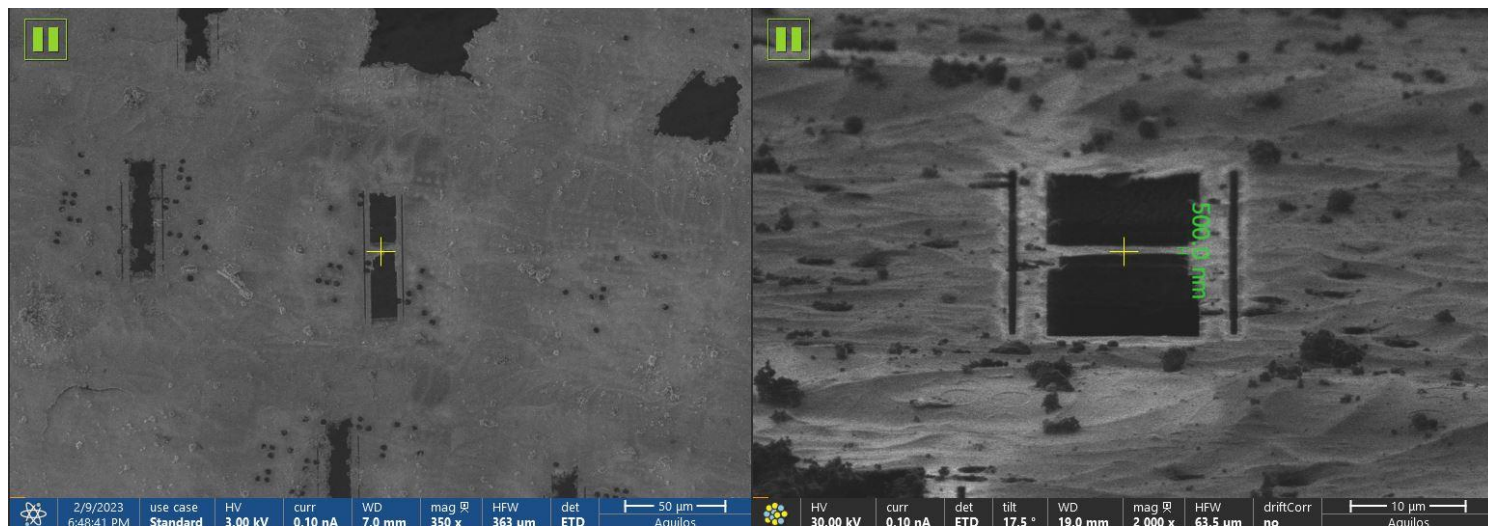
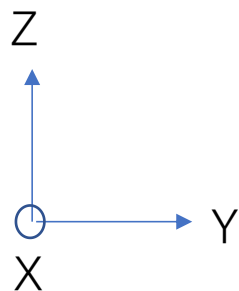
不同位点之间移动，离子束与
电子束成像可保持同样的工作
距离（使用相同的焦距）

52度

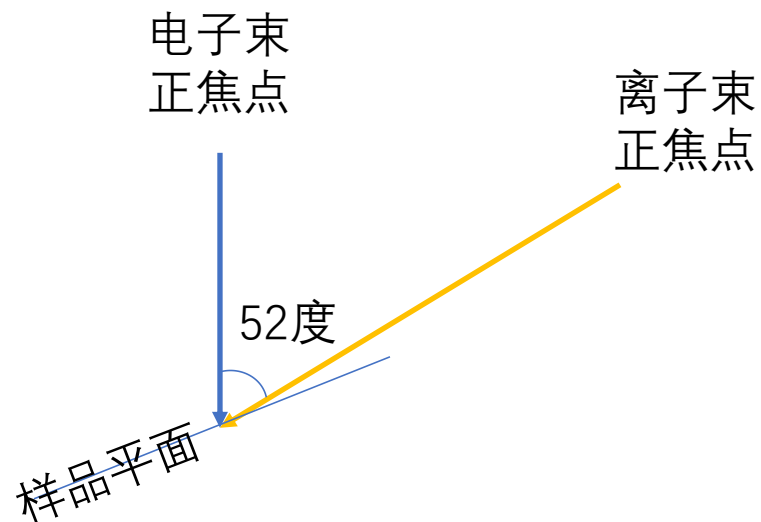
样品平面

样品台倾转
时电子束与
离子束目标
点基本不偏
移

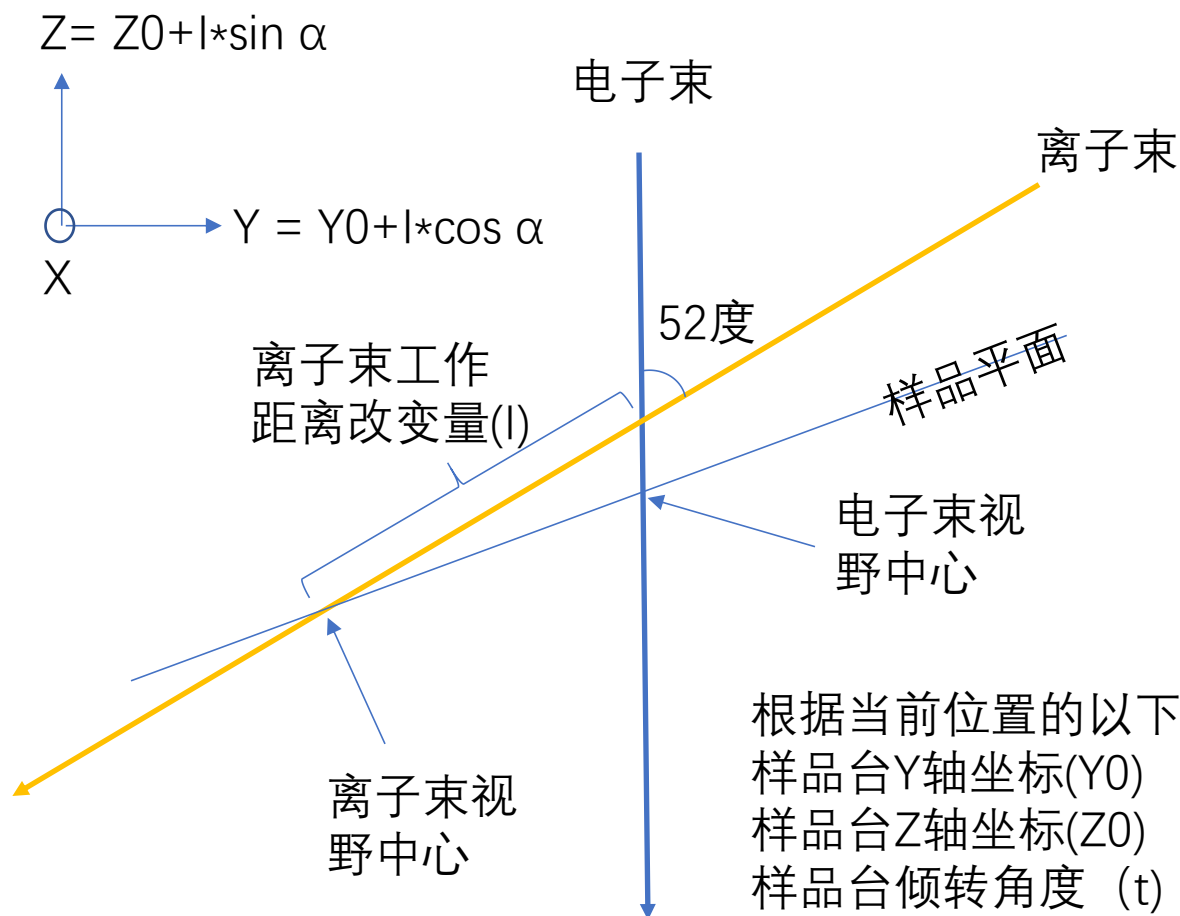
AutoTEM Cryo利用这
一要点进行共焦点调节



由离子束成像正焦时的工作距离改变量推算到达共焦点时的样品台Y和Z调整量



记录处于共焦点时的以下条件：
电子束工作距离（焦距）
电子束工作距离（焦距）
样品台倾转角度（t）



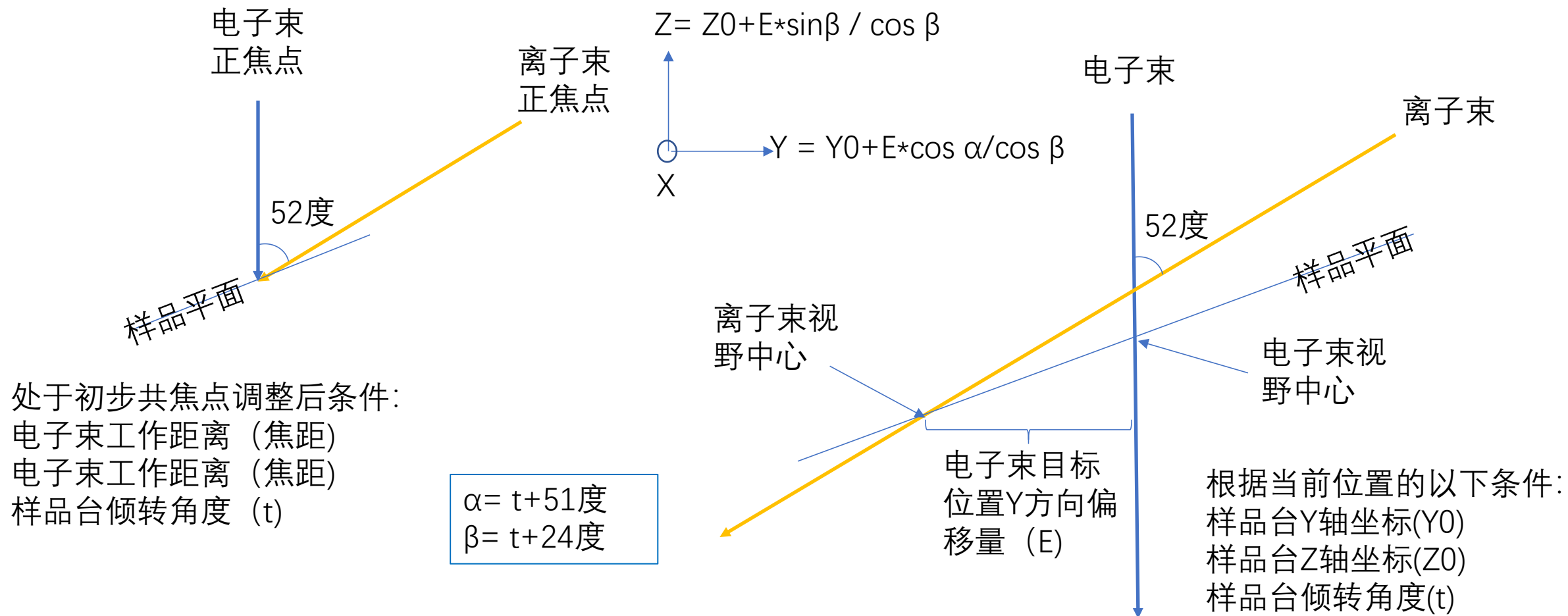
根据当前位置的以下条件：
样品台Y轴坐标(Y0)
样品台Z轴坐标(Z0)
样品台倾转角度（t）

离子束与样品台y方向夹角(α) = t + 39度

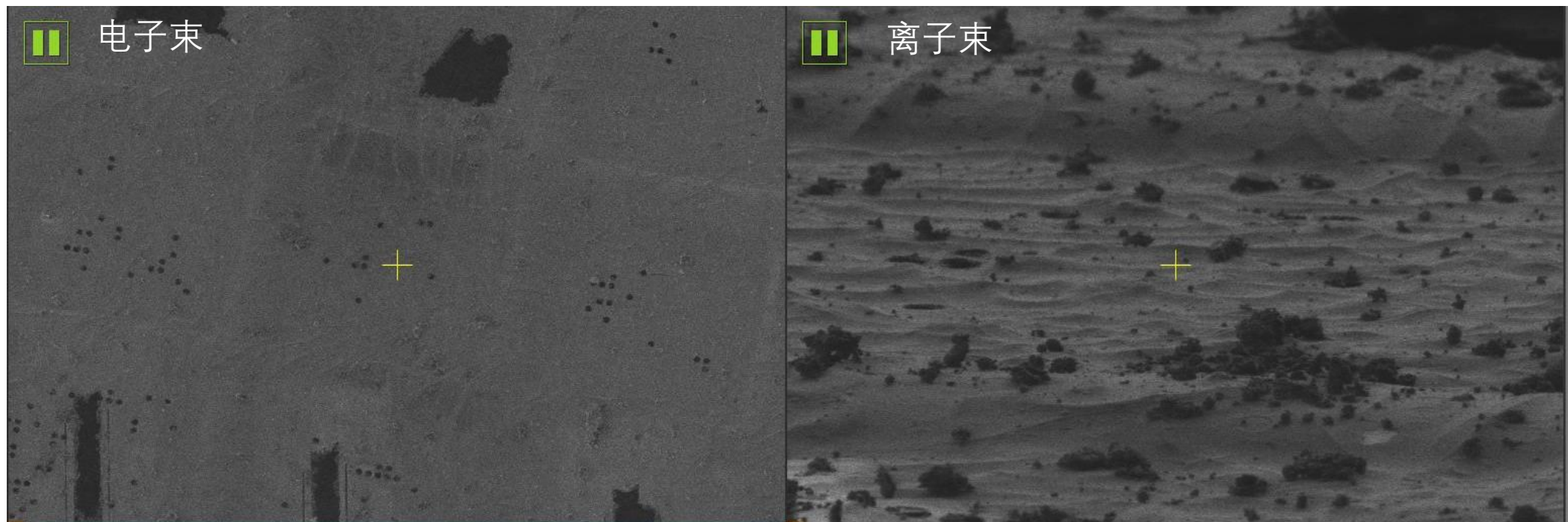
实际应用效果



由电子束目标位置Y方向偏移量推算到达共焦点时的样品台Y和Z调整量



实际应用效果



最多经过两轮调整，达到共焦点

半自动共焦点调整

记录处于共焦点时的以下条件：
电子束工作距离（焦距）
电子束工作距离（焦距）
样品台倾转角度

根据离子束工作距离改变量，调整至共焦点，并记录当前样品台坐标和减薄信息

根据电子束目标位置Y方向偏移量，调整至共焦点，并记录当前样品台坐标和减薄信息

user_folder D:/SharedData/XiaoJunHuang/auto_script/auto_slice_example

set reference take map image load map

name, stage_x, stage_y, stage_z, milling_angle, thinned

point_001	3.49	2.89	6.92	10.0	N
-----------	------	------	------	------	---

add point update point E correct delete point goto point update pattern

settings for taking image

	Magnification	high tension	beam current	dwelt time	resolution
electron	350x	3.0 kV	0.10 nA	1 us	1536x1024
ion	2000x	30.0 kV	0.10 nA	100 ns	1536x1024

auto milling current position auto milling all points stop

=====settings for milling with ion beam=====

name, position, type, high tension, beam current, milling angle, x, y, z

add pattern change pattern delete pattern ☒ include side show pattern load pattern

=====manual thin toolset=====

tilt to milling angle 1	15.3	show select pattern	ccs side pattern
tilt to milling angle 2	14.7	clear pattern	+ pattern
take E image	save E image	save I image	take I image
pause	stop		
start milling			

半自动共焦点调整

记录处于共焦点时的以下条件：
电子束工作距离（焦距）
电子束工作距离（焦距）
样品台倾转角度

Name	Date modified	Type
7degree	09/02/2023 15:23	File folder
8degree	09/02/2023 15:21	File folder
9degree	09/02/2023 15:19	File folder
10degree	03/02/2023 10:32	File folder
11degree	09/02/2023 15:22	File folder
12degree	09/02/2023 15:24	File folder
13degree	09/02/2023 15:25	File folder
14degree	09/02/2023 15:26	File folder
15degree	09/02/2023 15:27	File folder
16degree	09/02/2023 15:28	File folder
17degree	09/02/2023 15:29	File folder
18degree	09/02/2023 15:29	File folder
19degree	09/02/2023 15:30	File folder
20degree	09/02/2023 15:30	File folder

Name	Date modified	Type	Size
default_json	09/02/2023 15:23	File	1 KB

user_folder

D:/SharedData/XiaoJunHuang/auto_script/auto_slice_example

set reference

take map image

load map

name, stage_x, stage_y, stage_z, milling_angle, thinned

point_001 3.49 2.89 6.92 10.0 N

add point

update point

E correct

delete point

goto point

update pattern

settings for taking image

electron

Magnification

350x

high tension

3.0 kV

beam current

0.10 nA

dwelt time

1 us

resolution

1536x1024

ion

2000x

30.0 kV

0.10 nA

100 ns

1536x1024

auto milling current position

auto milling all points

stop

settings for milling with ion beam

name, position, type, high tension, beam current, milling angle, x, y, z

add pattern

change pattern

delete pattern

include side

show pattern

load pattern

manual thin toolset

tilt to milling angle 1

15.3

show select pattern

ccs side pattern

tilt to milling angle 2

14.7

clear pattern

+ pattern

take E image

save E image

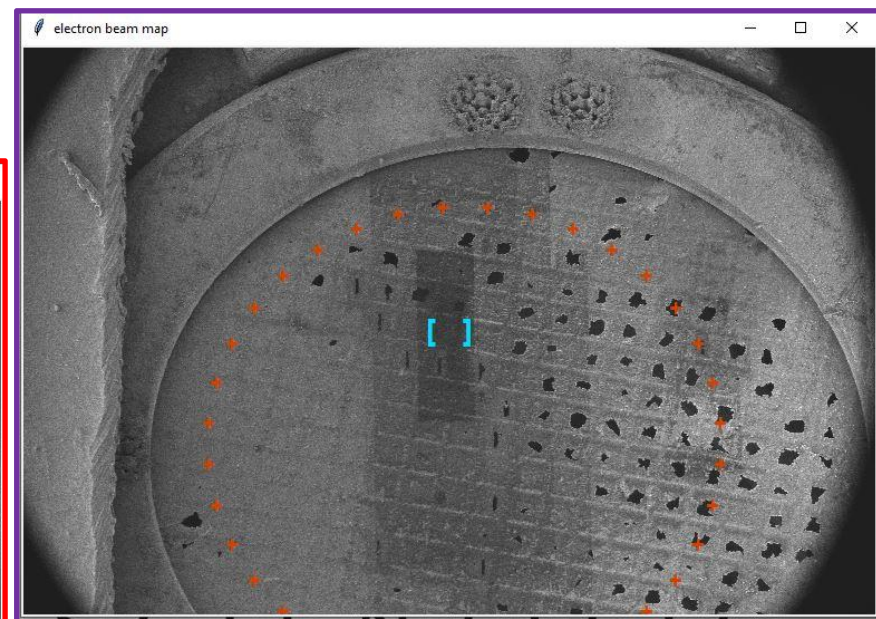
save I image

take I image

pause

stop

start milling



user_folder D:/SharedData/XiaoJunHuang/auto_script/auto_slice_example

set reference take map image load map

name, stage_x, stage_y, stage_z, milling_angle, thinned

point_001 3.49 2.89 6.92 10.0 N

add point update point E correct delete point goto point update pattern

settings for taking image

	Magnification	high tension	beam current	dwel time	resolution
electron	350x	3.0 kV	0.10 nA	1 us	1536x1024
ion	2000x	30.0 kV	0.10 nA	100 ns	1536x1024

=====

auto milling current position auto milling all points stop

=====settings for milling with ion beam=====

name, position, type, high tension, beam current, milling angle, x, y, z

add pattern change pattern delete pattern ☒ include side show pattern load pattern

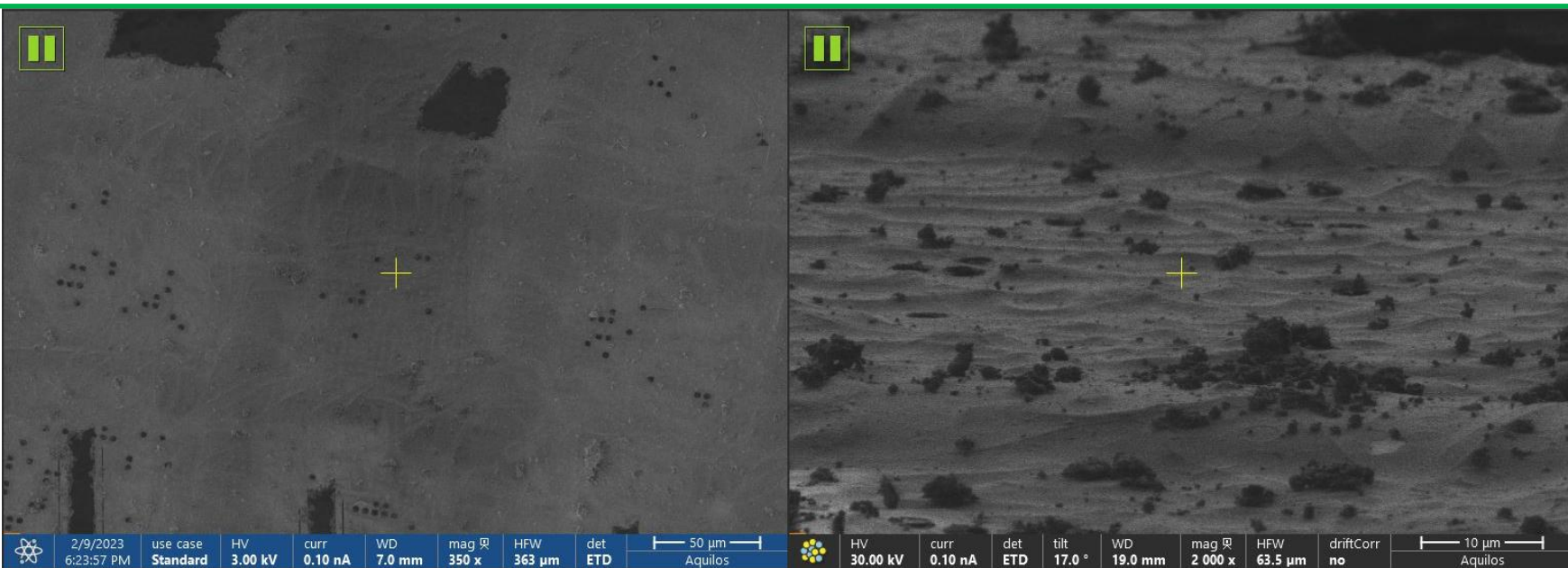
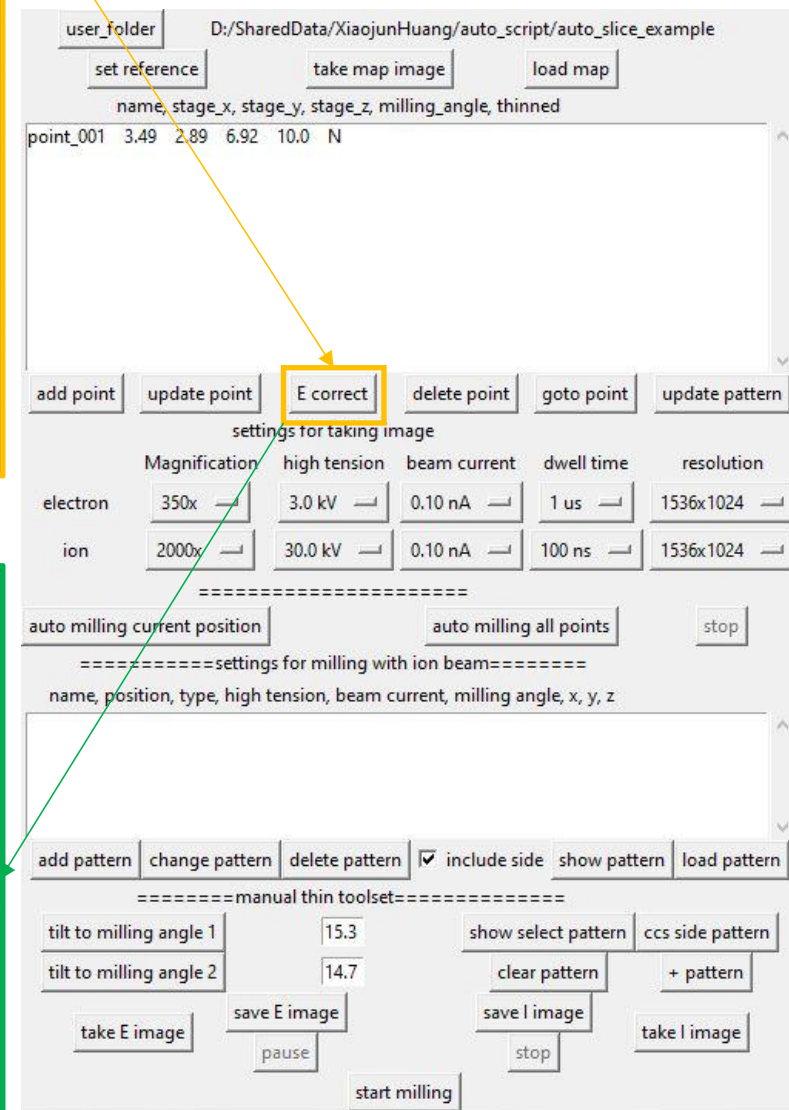
=====manual thin toolset=====

tilt to milling angle 1	15.3	show select pattern	ccs side pattern
tilt to milling angle 2	14.7	clear pattern	+ pattern
take E image	save E image	save I image	take I image
	pause	stop	
start milling			

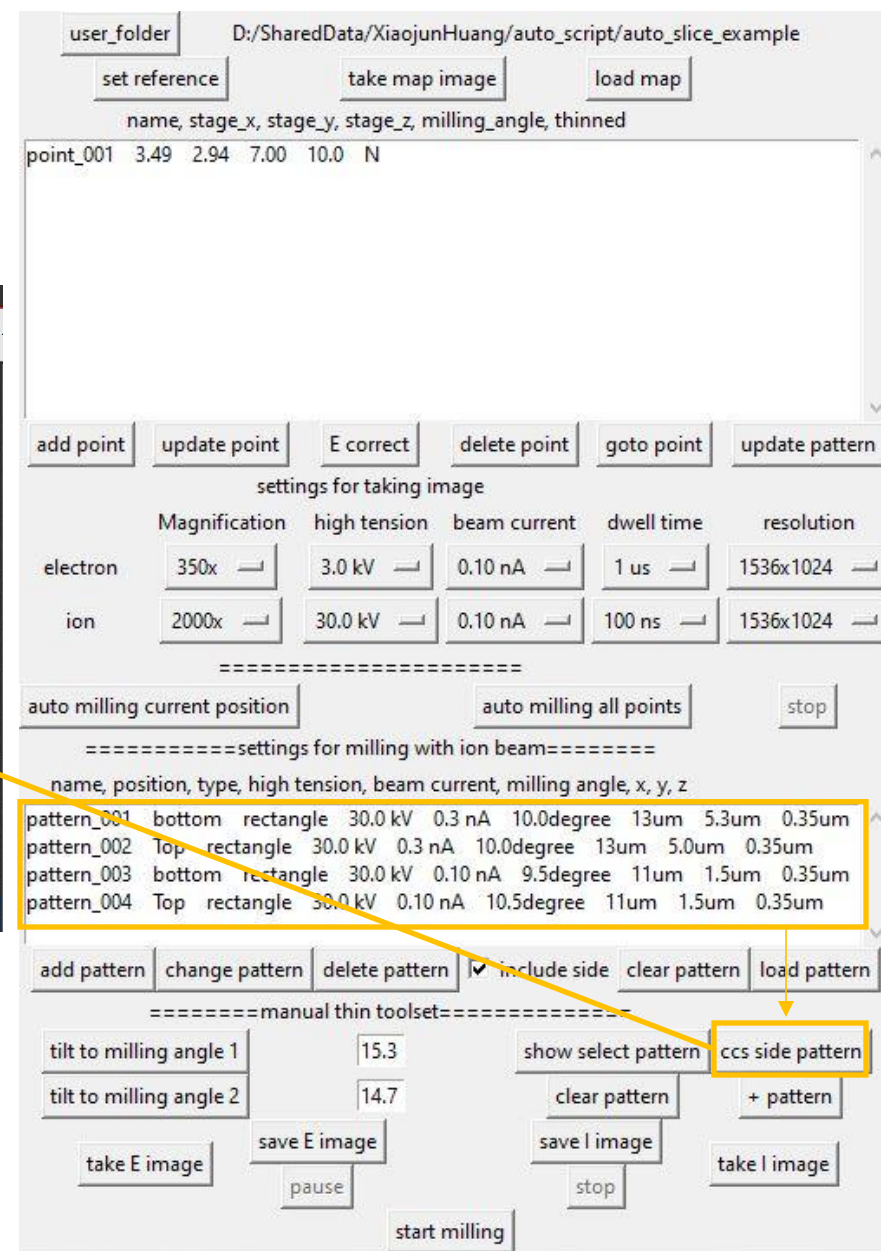
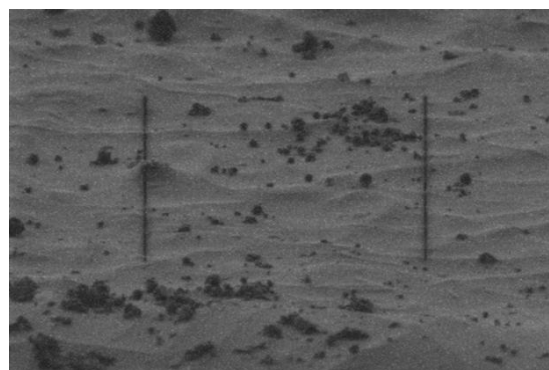
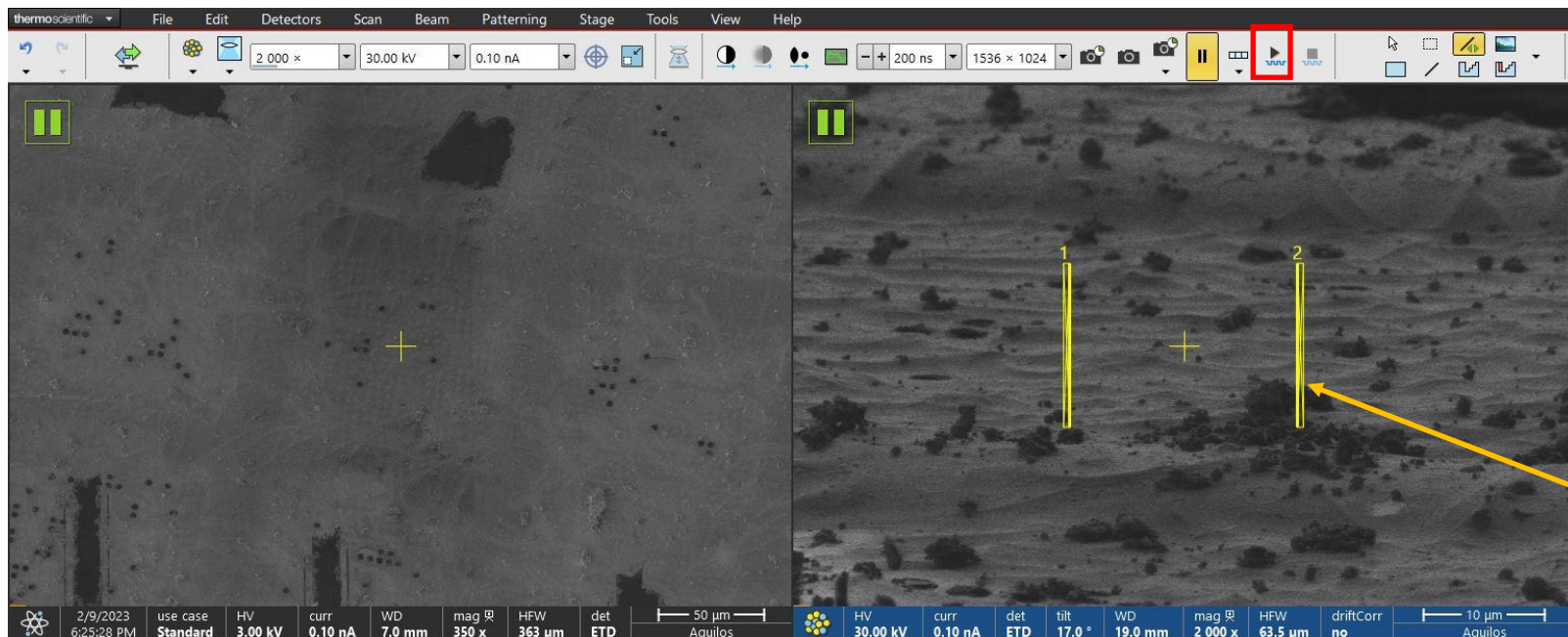




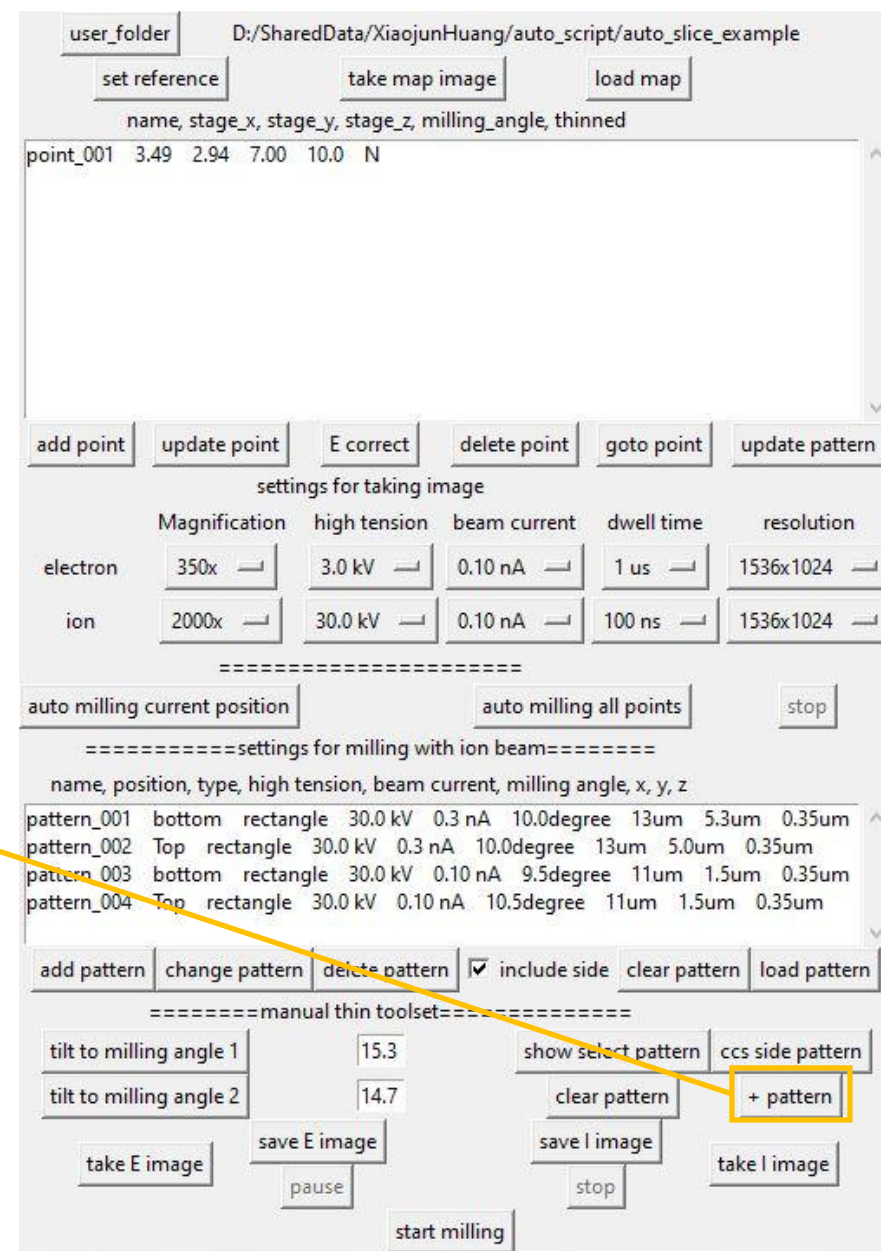
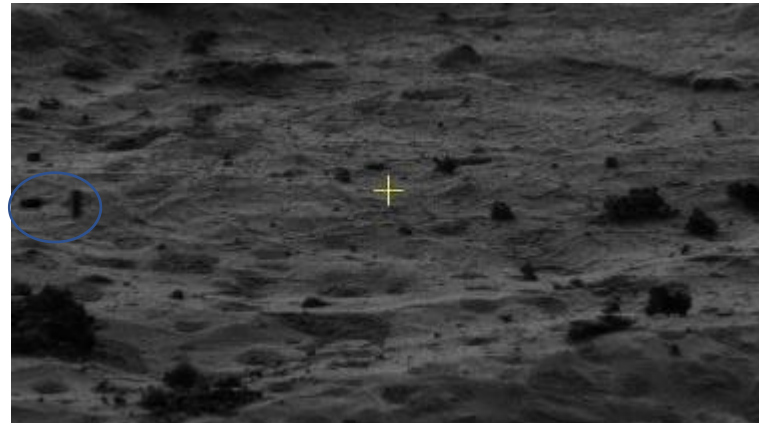
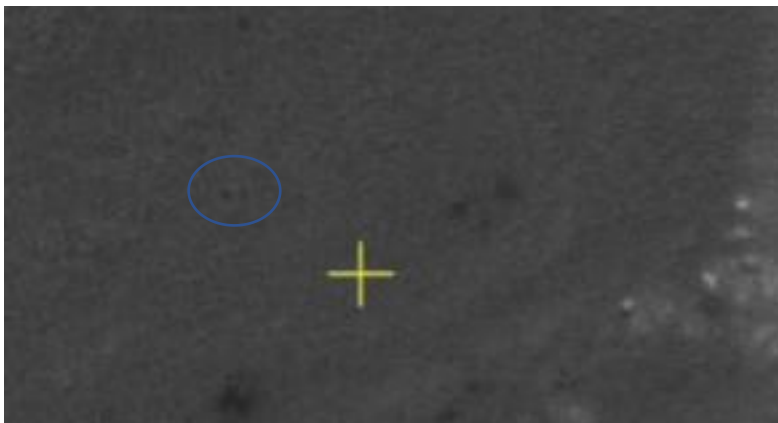
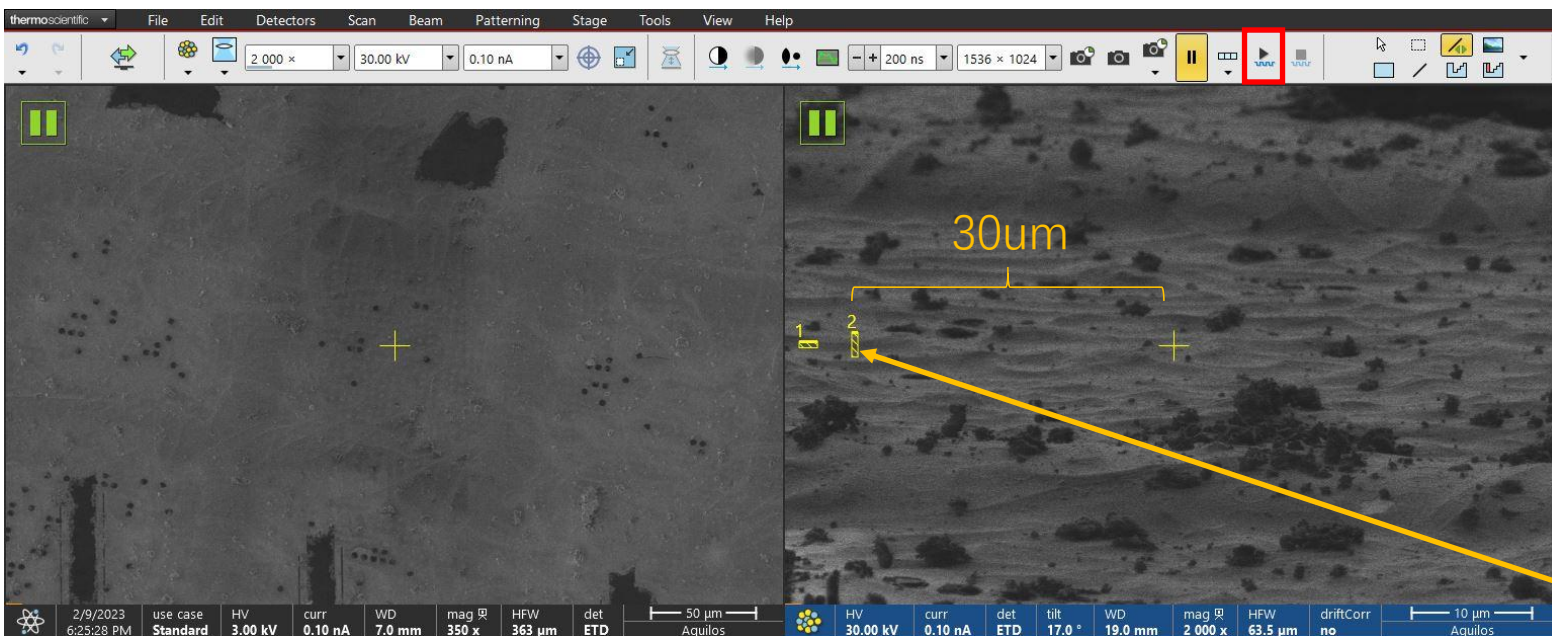
可循环做E correct直至电子束视野中心对准目标位置



当在电子束窗口无法分辨目标位置时1



当在电子束窗口无法分辨目标位置时2



user_folder
D:/SharedData/XiaoJunHuang/auto_script/auto_slice_example

set reference
take map image
load map

name, stage_x, stage_y, stage_z, milling_angle, thinned

point_001	3.49	2.94	7.00	10.0	N
point_002	3.61	2.92	6.99	10.0	Y
point_003	3.49	2.94	7.00	10.0	N

add point
update point
E correct
delete point
goto point
update pattern

settings for taking image

	Magnification	high tension	beam current	dwel time	resolution
electron	350x	3.0 kV	0.10 nA	1 us	1536x1024
ion	2000x	30.0 kV	0.10 nA	100 ns	1536x1024

=====

auto milling current position
auto milling all points
stop

=====settings for milling with ion beam=====

name, position, type, high tension, beam current, milling angle, x, y, z

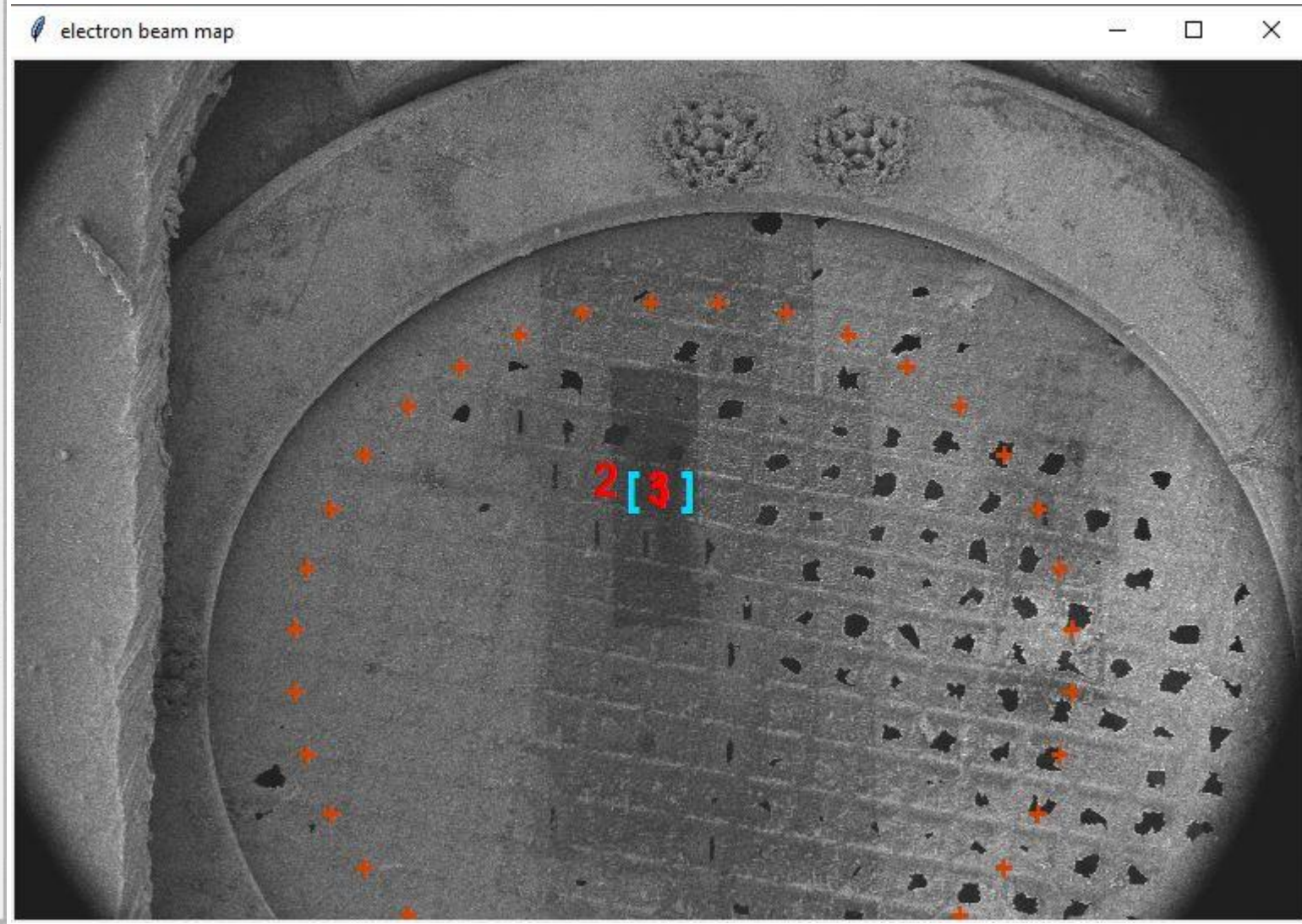
pattern_001	bottom	rectangle	30.0 kV	0.3 nA	10.0degree	13um	5.3um	0.35um
pattern_002	Top	rectangle	30.0 kV	0.3 nA	10.0degree	13um	5.0um	0.35um
pattern_003	bottom	rectangle	30.0 kV	0.10 nA	9.5degree	11um	1.5um	0.35um
pattern_004	Top	rectangle	30.0 kV	0.10 nA	10.5degree	11um	1.5um	0.35um

add pattern
change pattern
delete pattern
☒ include side
show pattern
load pattern

=====manual thin toolset=====

tilt to milling angle 1	15.3	show select pattern	ccs side pattern
tilt to milling angle 2	14.7	clear pattern	+ pattern
take E image	save E image	save I image	take I image
pause	stop		
start milling			

提供电子束角度（地图）辅助，便于探测重复位点



需要改进的要点

- 导航从离子束角度发起，电子束角度（地图）辅助，以缩短共焦点调节及定位的耗时
- 增加Aquilos2地图与Titan3地图的衔接
- 重新设计辅助工具，减少多个窗口间的切换

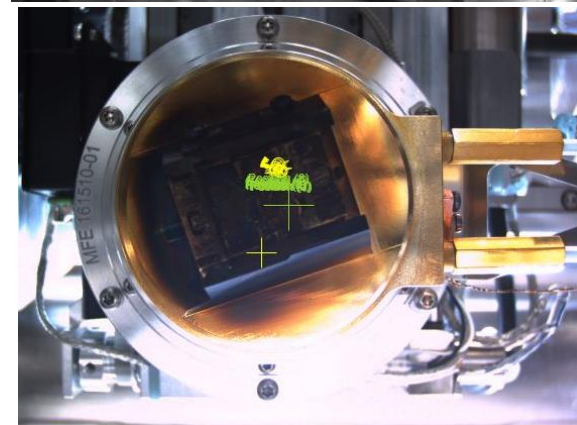
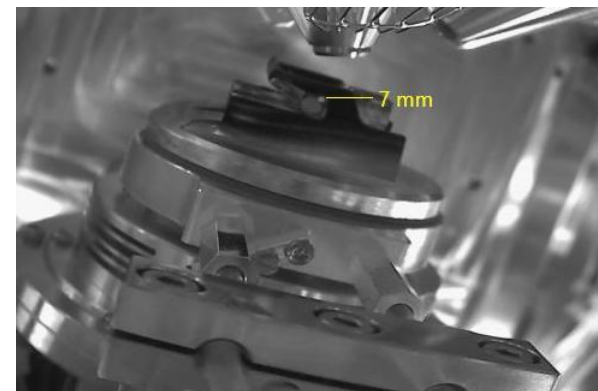
Titan3和Aquilos的坐标转换误差总结



X方向变化在 ± 0.01 mm
Y方向变化在 ± 0.01 mm
角度变化优于 $\pm 5^\circ$

这个误差量已经不适合直接提供点对点的坐标转换结果了

Titan3和Aquilos的坐标转换误差

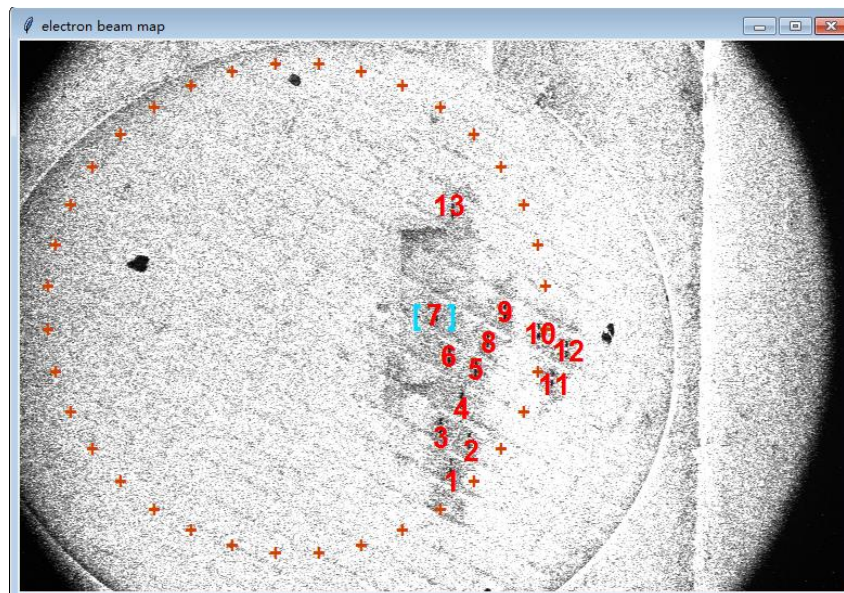


同一个载网三次电镜样品台装载的位置重复性 -1号座grid1位置

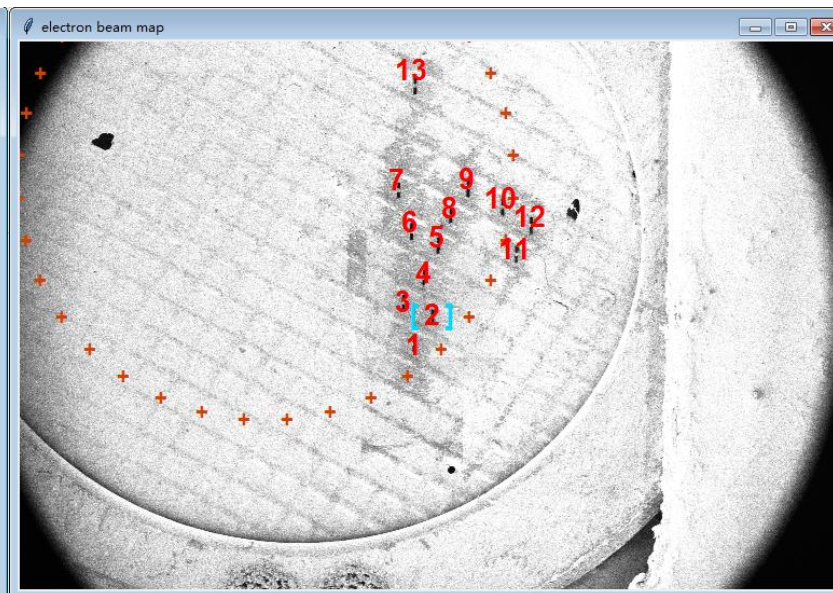
Aquilos
样品座



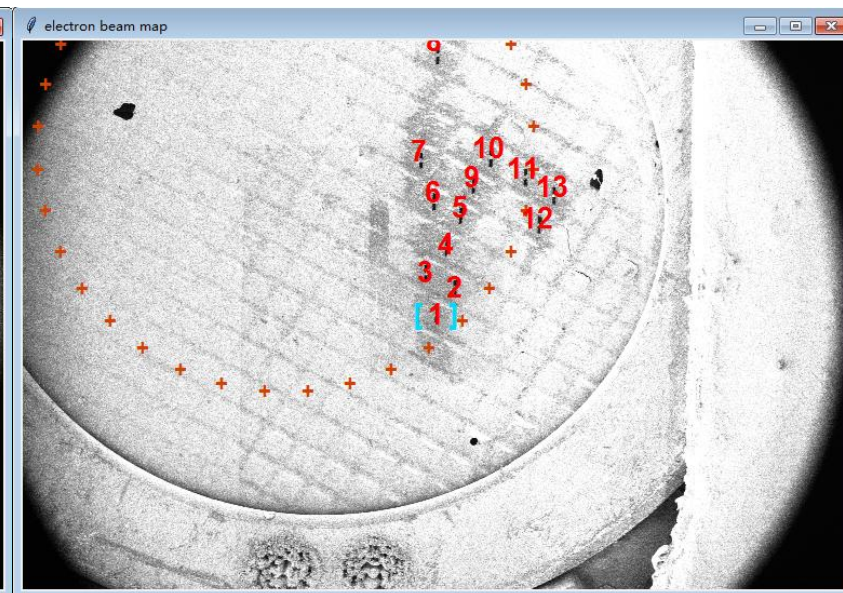
Aquilos
样品台



第一次



第二次



第三次

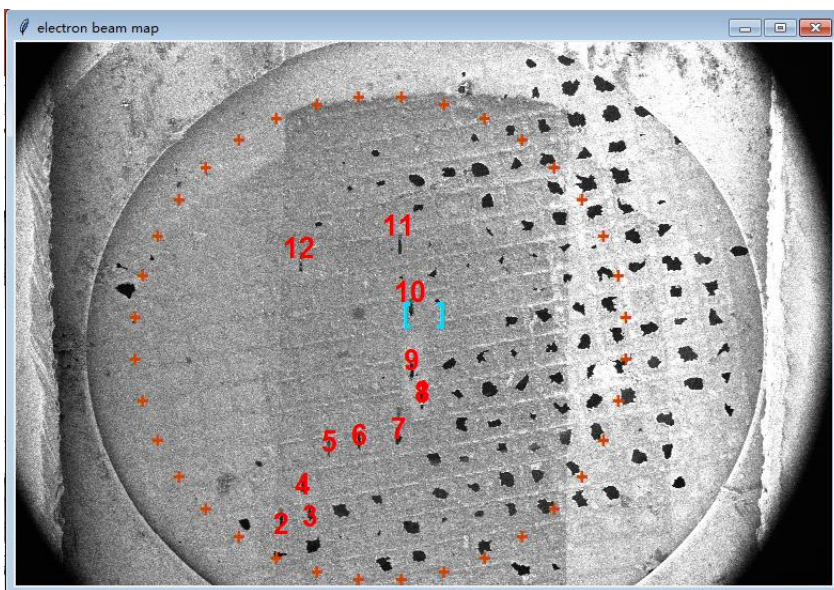
X方向变化在 ± 0.01 mm
Y方向变化在 ± 0.01 mm
角度未有可观测到的变化

同一个载网三次电镜样品台装载的位置重复性 -1号座grid2位置

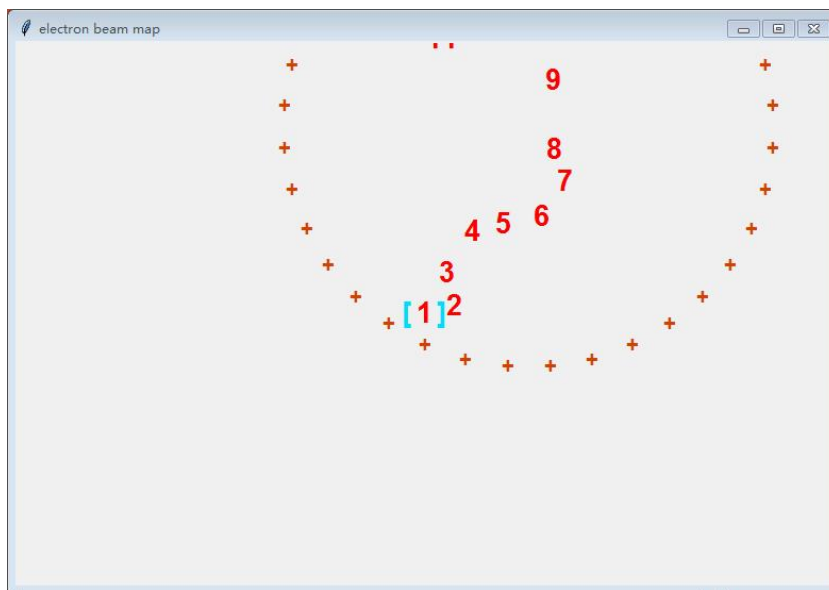
Aquilos
样品座



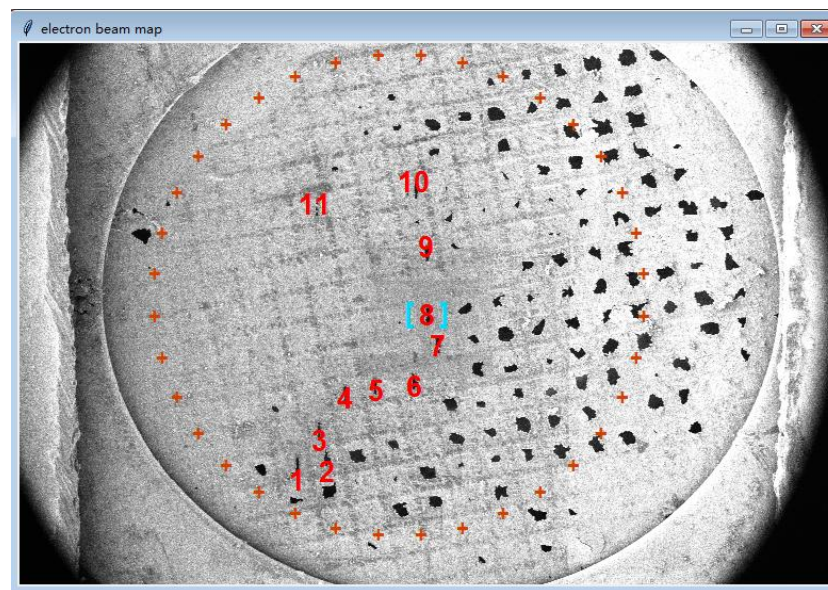
Aquilos
样品台



第一次



第二次



第三次

X方向变化在 ± 0.01 mm
Y方向变化在 ± 0.005 mm
角度未有可观测到的变化

同一个载网在不同样品座装载的位置重复性 -grid1位置

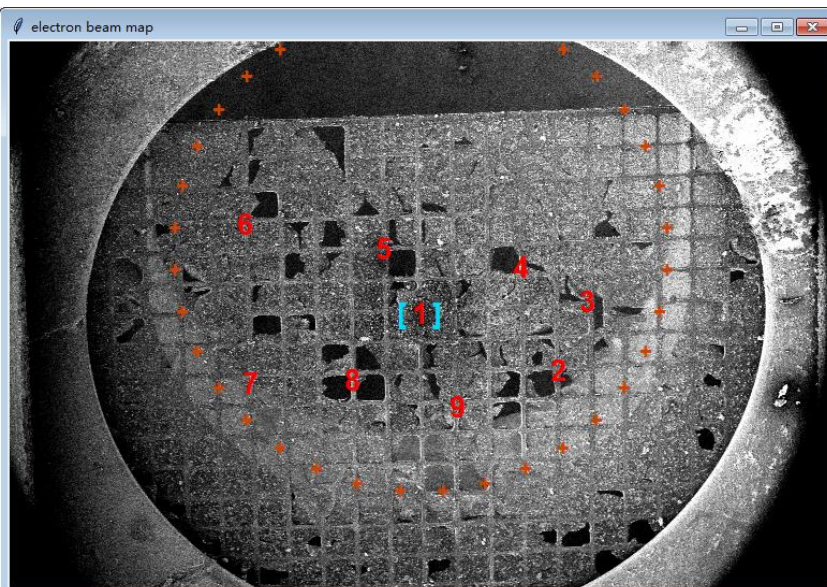
载网



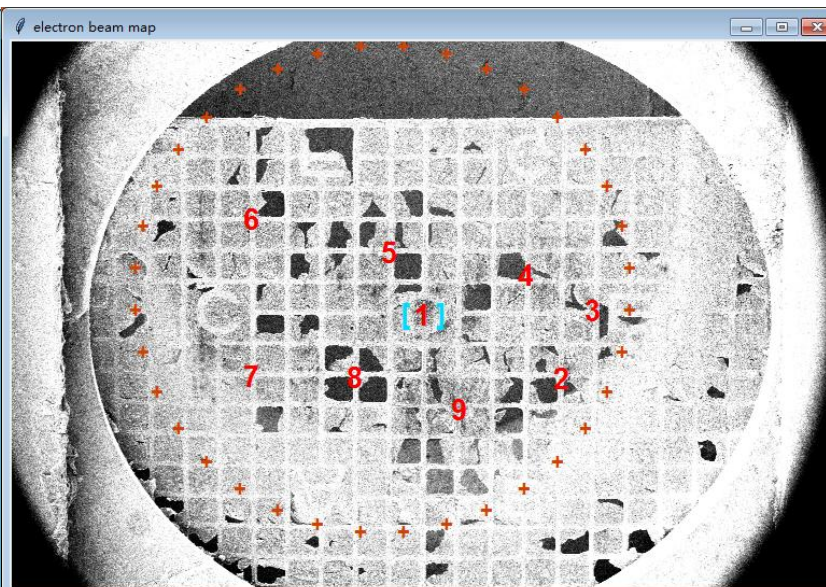
Aquilos
样品座



Aquilos
样品台

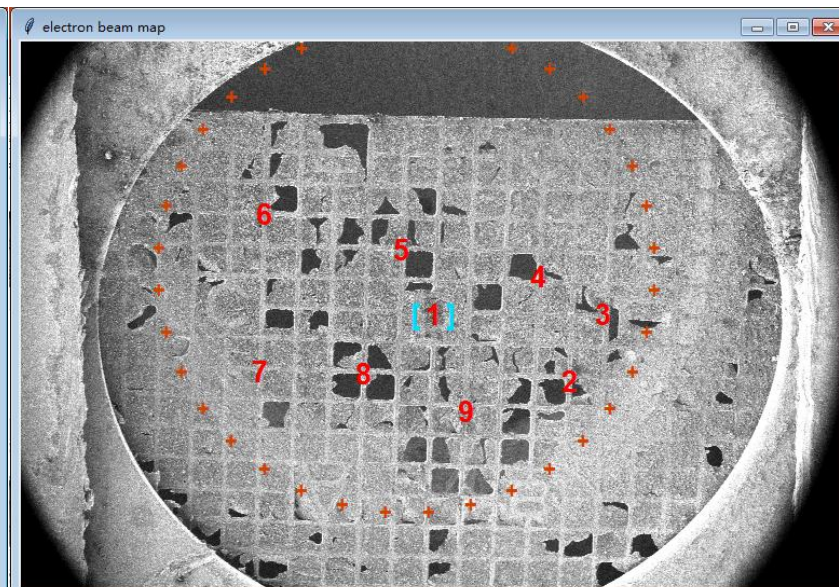


0号座



1号座

X方向变化在 ± 0.1 mm
Y方向变化在 ± 0.1 mm
角度变化在 $\pm 3^\circ$



2号座

同一个载网在不同样品座装载的位置重复性 -grid2位置

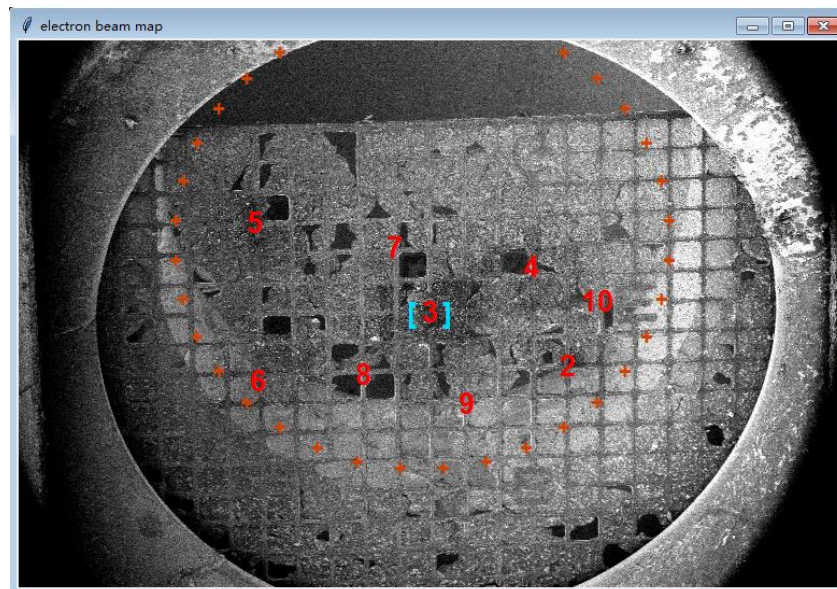
载网



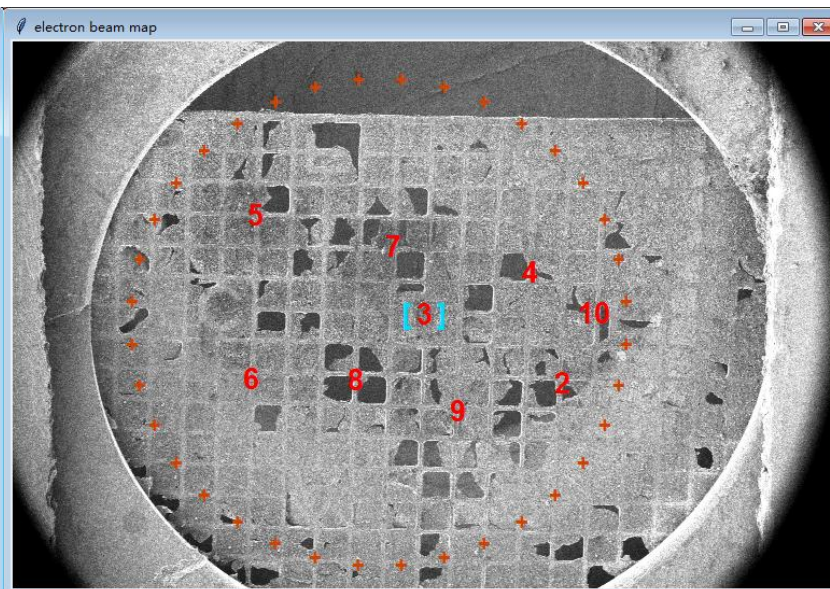
Aquilos
样品座



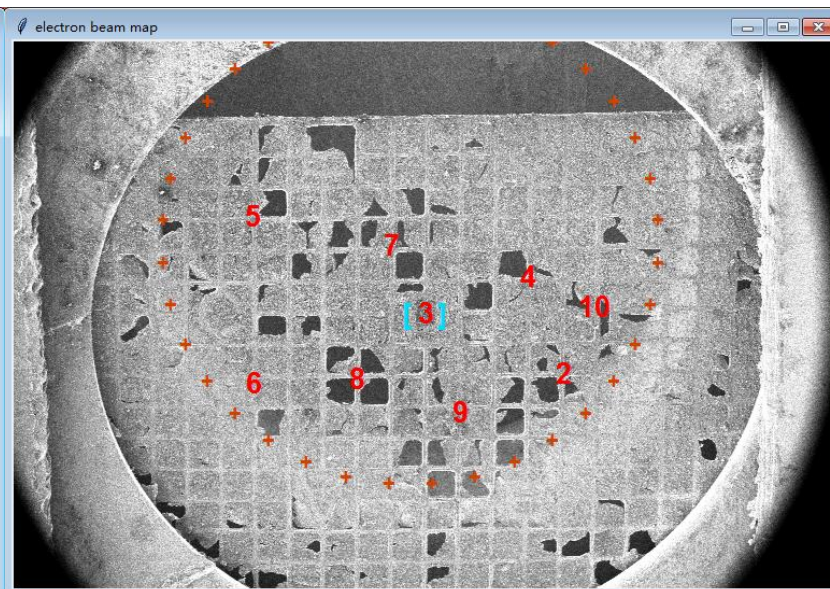
Aquilos
样品台



0号座



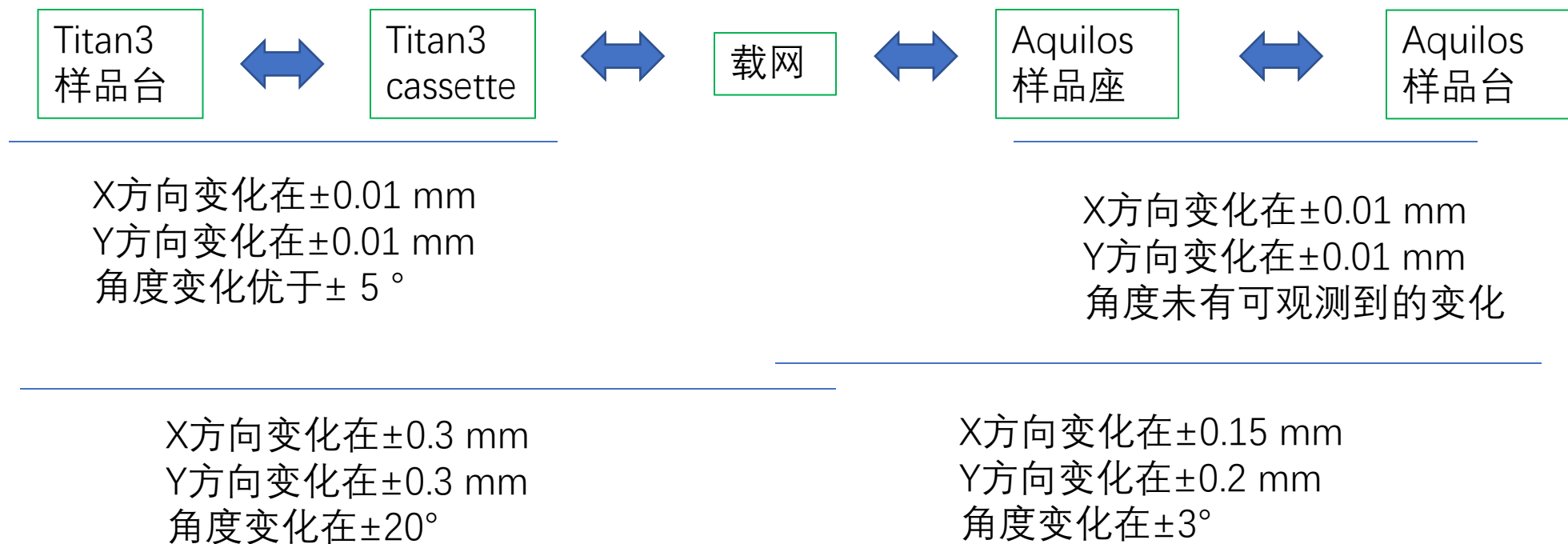
1号座



2号座

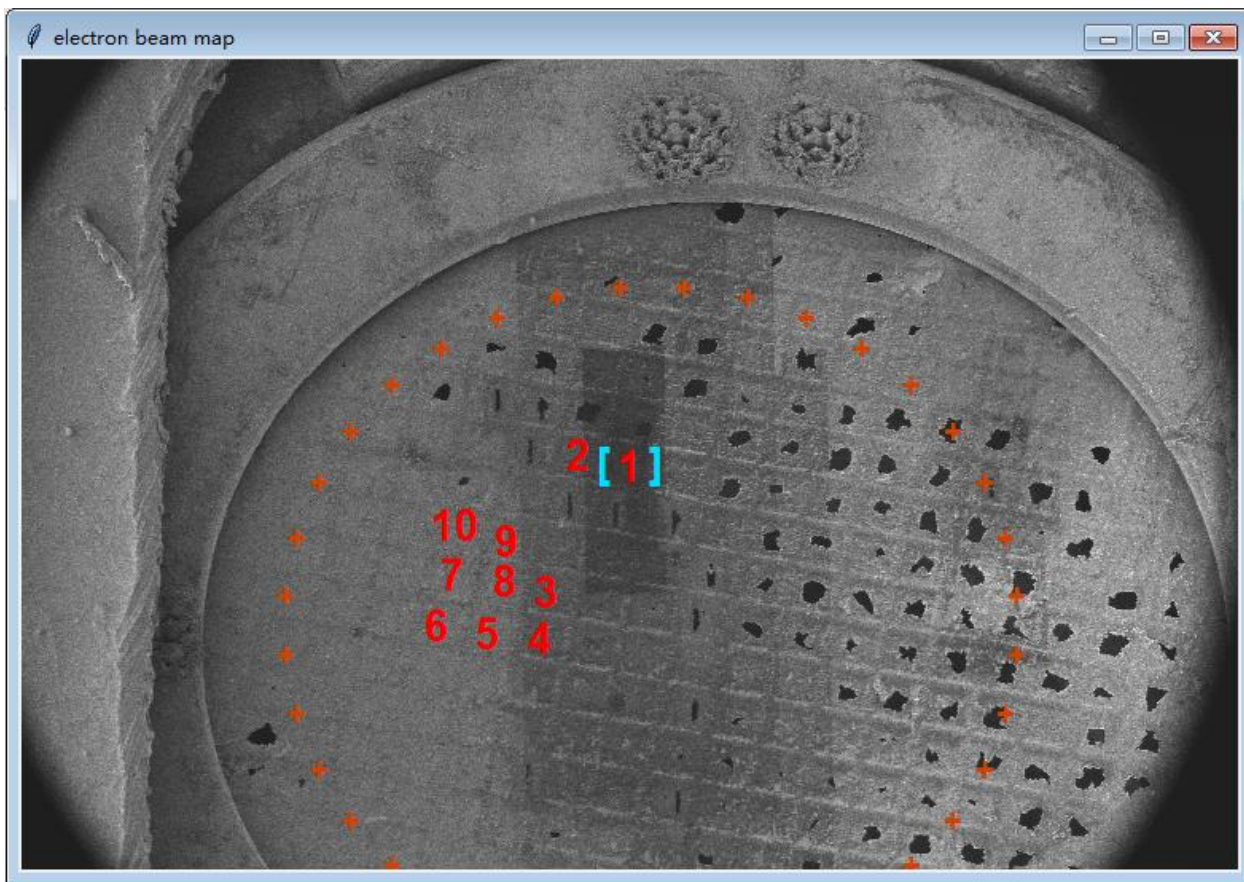
X方向变化在 ± 0.15 mm
Y方向变化在 ± 0.2 mm
角度变化在 $\pm 3^\circ$

Titan3和Aquilos的坐标转换误差总结

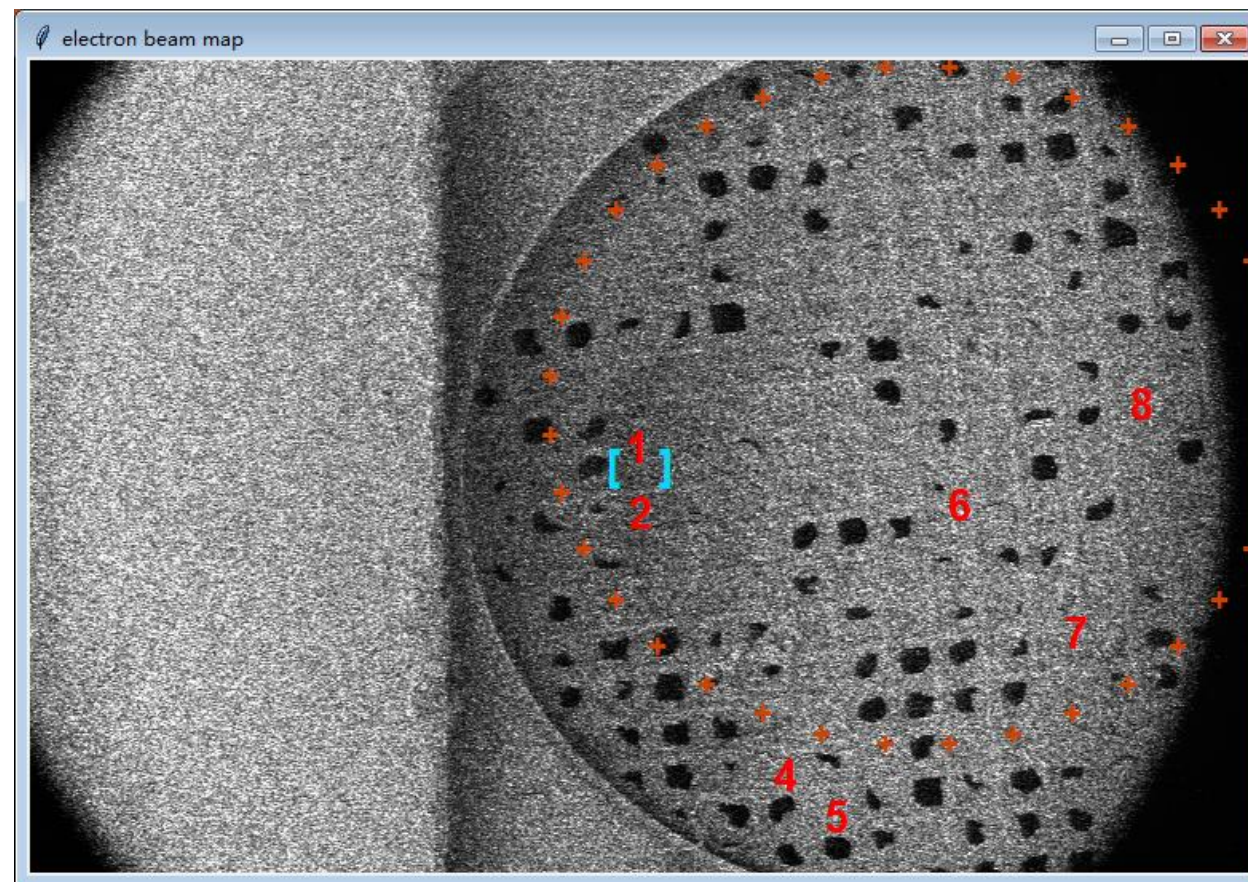


这个误差量已经不适合直接提供点对点的坐标转换结果了

在Aquilos上



所有点都能在Titan3上用于数据收集



4和5无法在Titan3上用于数据收集

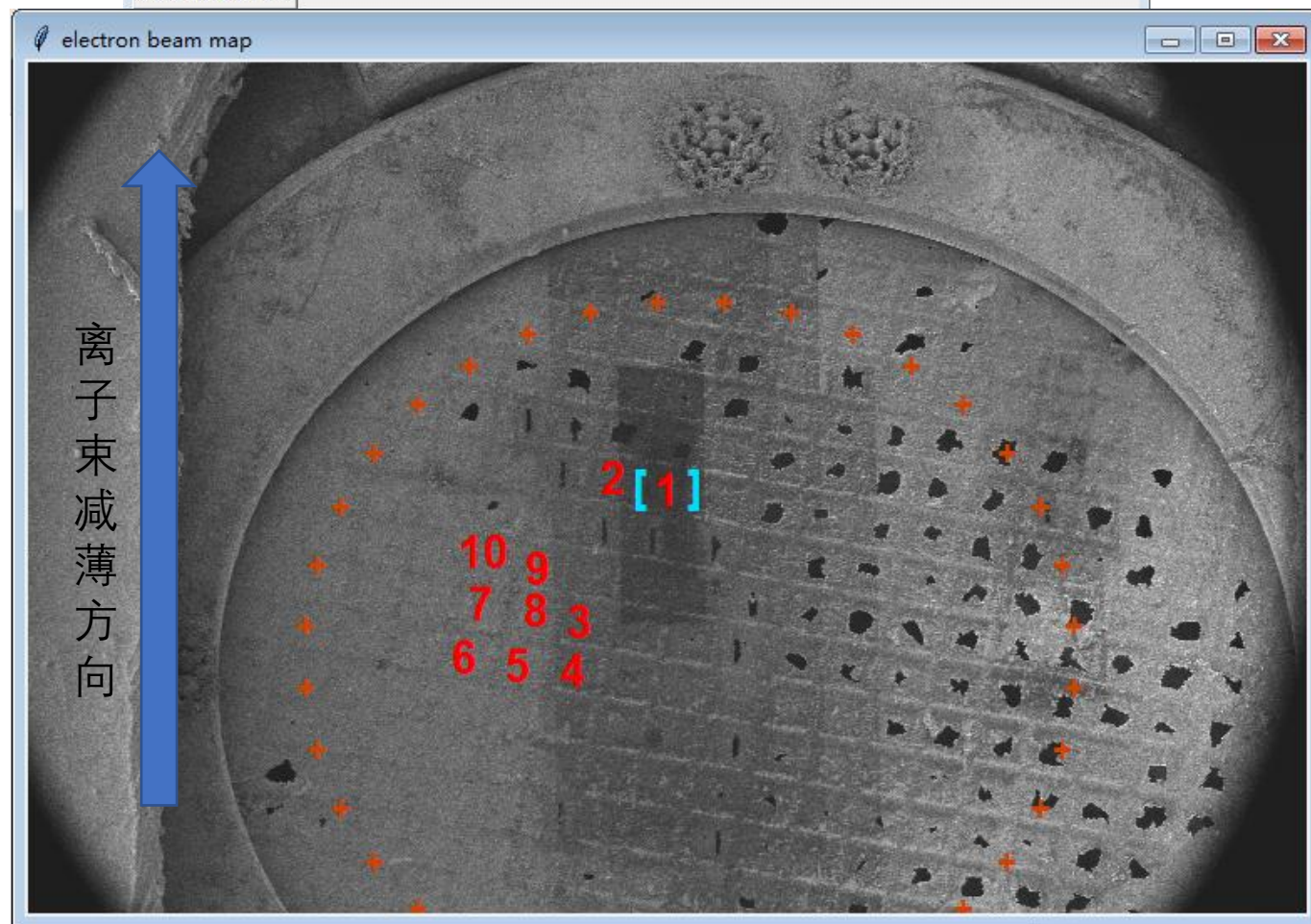
在Titan3上可直接读取到
存储在supportPC中的切割位点
信息

进入Titan3数据处理服务器上直接挂载/aquilos

```
(anaconda3-2020.02) [capture@titan3-processor ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        252G   0  252G   0% /dev
tmpfs           252G  144K  252G   1% /dev/shm
tmpfs           252G   29M  252G   1% /run
tmpfs           252G   0  252G   0% /sys/fs/cgroup
/dev/mapper/centos-root 50G   38G   13G  75% /
/dev/sdc1       894G   49G  845G   6% /ssd_data1
/dev/sdd1       894G  251M  894G   1% /ssd_data2
/dev/sde1       27T   21T   5.5T  80% /data
/dev/sdb2      1014M  157M  858M  16% /boot
/dev/sdb1       200M   12M  189M   6% /boot/efi
/dev/mapper/centos-home 392G  150G  243G  39% /home
//11.11.31.1/X  1.6T  273G  1.3T  18% /k2_x
//192.168.1.231/E 1.7T  1.6G  1.7T   1% /aquilos
//10.0.30.6/k2_x 7.3T  1.9T  5.5T  26% /K2_Arctica
tmpfs           51G   12K   51G   1% /run/user/42
tmpfs           51G    0   51G   0% /run/user/1001
/dev/sdf1       3.7T  3.2T  490G  87% /media
(anaconda3-2020.02) [capture@titan3-processor ~]$
```

运行命令aquilos_map

```
(anaconda3-2020.02) [capture@titan3-processor ~]$ cd /aquilos/
[capture@titan3-processor aquilos]$ aquilos_map
```



在Titan3上可直接读取到
存储在supportPC中的切割位点
信息

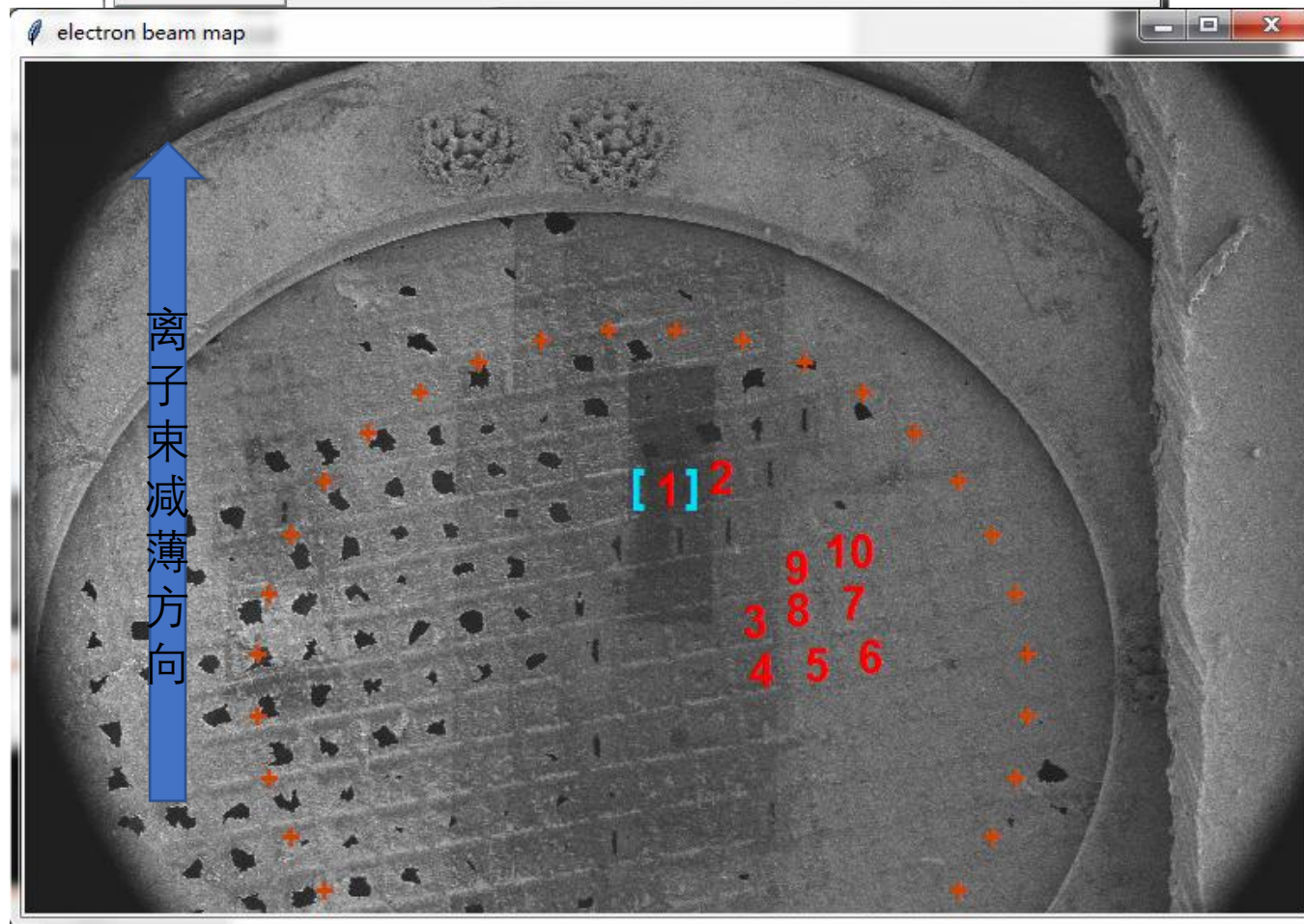
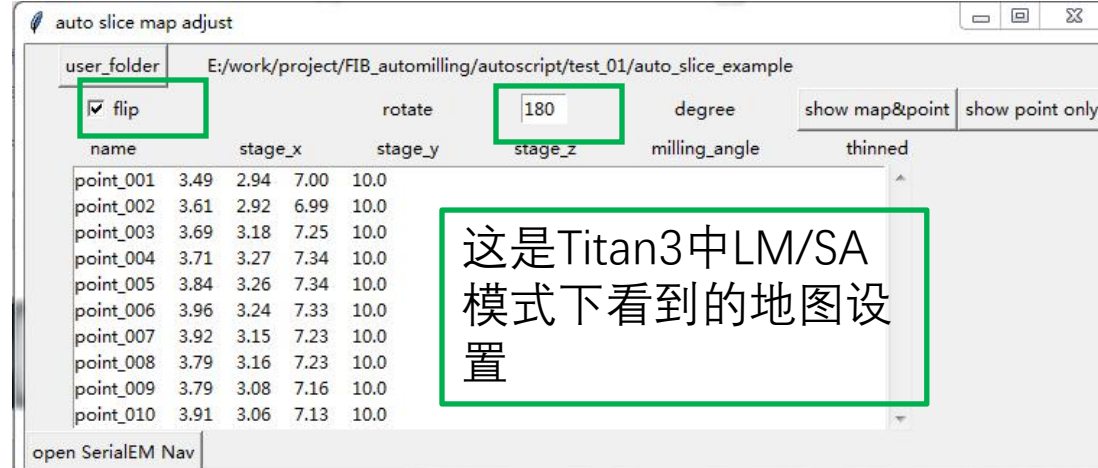


进入Titan3数据处理服务器上直接挂载/aquilos

```
(anaconda3-2020.02) [capture@titan3-processor ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        252G   0  252G   0% /dev
tmpfs           252G  144K  252G   1% /dev/shm
tmpfs           252G   29M  252G   1% /run
tmpfs           252G   0  252G   0% /sys/fs/cgroup
/dev/mapper/centos-root 50G   38G   13G  75% /
/dev/sdc1       894G   49G  845G   6% /ssd_data1
/dev/sdd1       894G  251M  894G   1% /ssd_data2
/dev/sde1       27T   21T   5.5T  80% /data
/dev/sdb2      1014M  157M  858M  16% /boot
/dev/sdb1       200M   12M  189M   6% /boot/efi
/dev/mapper/centos-home 392G  150G  243G  39% /home
//11.11.31.1/X  1.6T  273G  1.3T  18% /k2_x
//192.168.1.231/E 1.7T  1.6G  1.7T   1% /aquilos
//10.0.30.6/k2_x 7.3T  1.9T  5.5T  26% /K2_Artica
tmpfs           51G   12K   51G   1% /run/user/42
tmpfs           51G    0   51G   0% /run/user/1001
/dev/sdf1       3.7T  3.2T  490G  87% /media
(anaconda3-2020.02) [capture@titan3-processor ~]$
```

运行命令aquilos_map

```
(anaconda3-2020.02) [capture@titan3-processor ~]$ cd /aquilos/
[capture@titan3-processor aquilos]$ aquilos_map
```



在Titan3上可直接读取到
存储在supportPC中的切割位点
信息

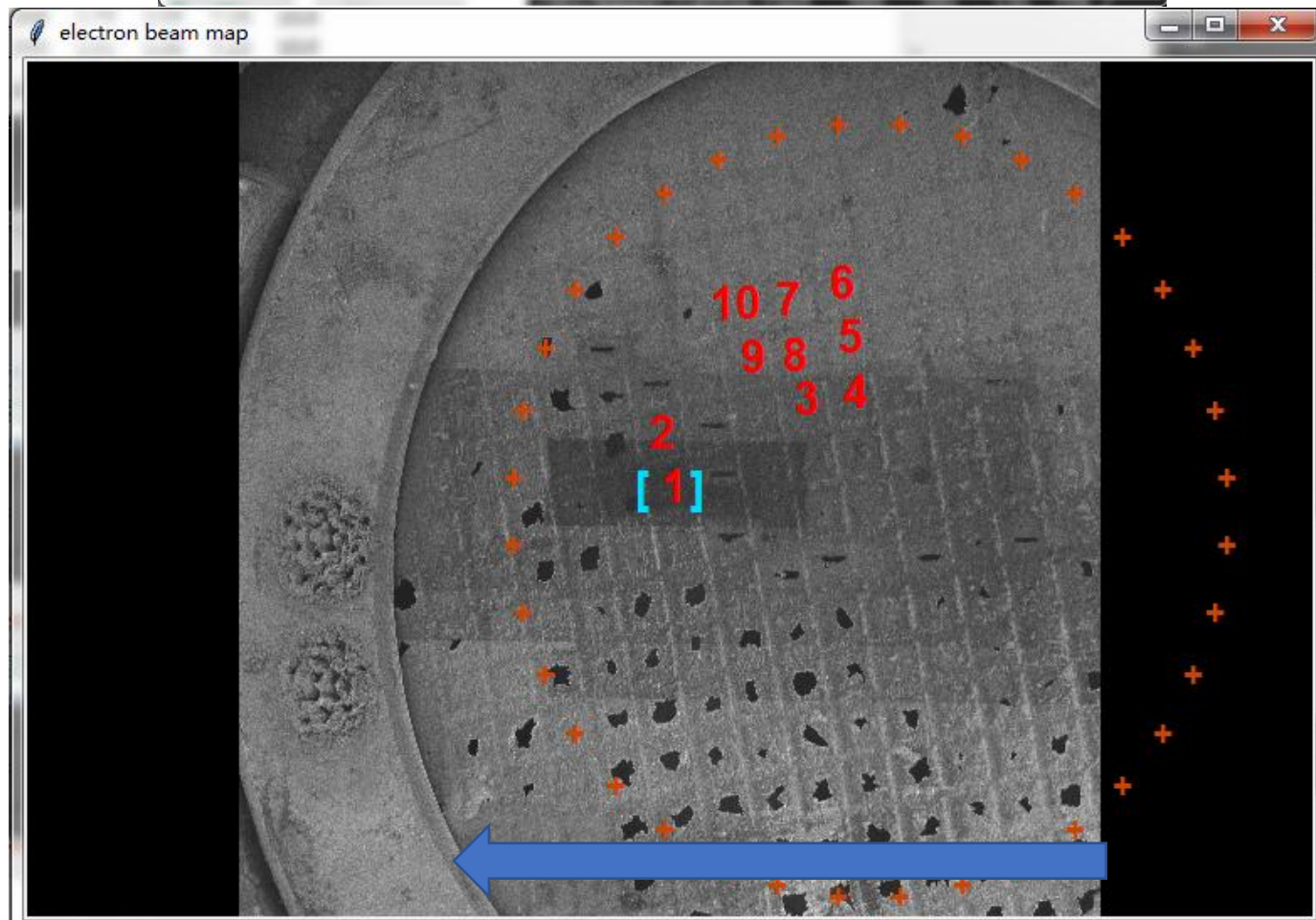
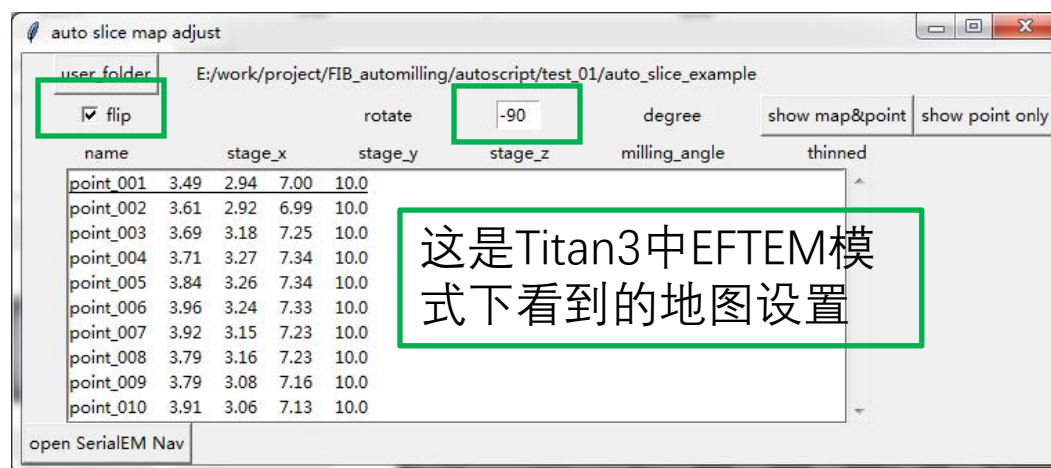


进入Titan3数据处理服务器上直接挂载/aquilos

```
(anaconda3-2020.02) [capture@titan3-processor ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        252G   0  252G   0% /dev
tmpfs           252G  144K  252G   1% /dev/shm
tmpfs           252G   29M  252G   1% /run
tmpfs           252G   0  252G   0% /sys/fs/cgroup
/dev/mapper/centos-root 50G   38G   13G  75% /
/dev/sdc1       894G   49G  845G   6% /ssd_data1
/dev/sdd1       894G  251M  894G   1% /ssd_data2
/dev/sde1       27T   21T  5.5T  80% /data
/dev/sdb2      1014M  157M  858M  16% /boot
/dev/sdb1       200M   12M  189M   6% /boot/efi
/dev/mapper/centos-home 392G  150G  243G  39% /home
//11.11.31.1/X   1.6T  273G  1.3T  18% /k2_x
//192.168.1.231/E 1.7T  1.6G  1.7T   1% /aquilos
//10.0.30.6/k2_x 7.3T  1.9T  5.5T  26% /K2_Artica
tmpfs           51G   12K   51G   1% /run/user/42
tmpfs           51G    0   51G   0% /run/user/1001
/dev/sdf1       3.7T  3.2T  490G  87% /media
(anaconda3-2020.02) [capture@titan3-processor ~]$
```

运行命令aquilos_map

```
(anaconda3-2020.02) [capture@titan3-processor ~]$ cd /aquilos/
[capture@titan3-processor aquilos]$ aquilos_map
```



需要改进的要点

- 导航从离子束角度发起，电子束角度（地图）辅助，以缩短共焦点调节及定位的耗时
- 增加Aquilos2地图与Titan3地图的衔接
- 重新设计辅助工具，减少多个窗口间的切换

user_folder

D:/SharedData/XiaojunHuang/auto_script/auto_slice_example

set reference

take map image

load map

name, stage_x, stage_y, stage_z, milling_angle, thinned

point_001	3.49	2.94	7.00	10.0	N
point_002	3.61	2.92	6.99	10.0	Y
point_003	3.49	2.94	7.00	10.0	N

add point

update point

E correct

delete point

goto point

update pattern

settings for taking image

	Magnification	high tension	beam current	dwelt time	resolution
electron	350x	3.0 kV	0.10 nA	1 us	1536x1024
ion	2000x	30.0 kV	0.10 nA	100 ns	1536x1024

=====

auto milling current position

auto milling all points

stop

=====settings for milling with ion beam=====

name	position	type	high tension	beam current	milling angle	x	y	z
pattern_001	bottom	rectangle	30.0 kV	0.3 nA	10.0degree	13um	5.3um	0.35um
pattern_002	Top	rectangle	30.0 kV	0.3 nA	10.0degree	13um	5.0um	0.35um
pattern_003	bottom	rectangle	30.0 kV	0.10 nA	9.5degree	11um	1.5um	0.35um
pattern_004	Top	rectangle	30.0 kV	0.10 nA	10.5degree	11um	1.5um	0.35um

add pattern

change pattern

delete pattern

☒ include side

show pattern

load pattern

=====manual thin toolset=====

tilt to milling angle 1	15.3	show select pattern	ccs side pattern
tilt to milling angle 2	14.7	clear pattern	+ pattern

take E image

save E image

pause

start milling

save I image

stop

take I image

样品台控制

Stage

Actual

Go To

X

3.4882 mm

+

-

Y

2.9418 mm

+

-

Z

7.0037 mm

+

-

T

17.0 °

+

-

R

-70.0 °

+

-

☒ Compucentric Rotation

☒ Z-Y Link

Position (2)

Position (3)

Position (4)

Position (5)

Add

Update

Remove

Remove All

☒ Touch Alarm Enabled

Stage Z

Down

Up

Tilt Correction

☐ Tilt Correction

Tilt Angle

Automatic

Manual

-

+

17.0 °

Specimen Pre-tilt

-

+

0 °

减薄图案控制

Patterning Control

Selected Grid

Go to: Mapping

Milling

Deposition

When Finished

No Action

Total time: 00:00:00

Overall progress

Current progress

Properties

Selective Mill

Patterning Presets

p1

p2

p3

p4

p5

p6

Lamella Thickness

Distance

0 pm

Gas Injection

Gas

Insert

Heat

Flow

Pt dep

Warm

Closed

iSPI

Monitor

Drift Suppression

On

Pause

Save

Configure...

Time Interval

1.0 s

CCS Line Interval

1

Shuttle Controls

Selected Grid

Grid 1

Grid 2

Go to: Mapping

Milling

Deposition

Temperature & Pressure

☒ Cryo Stage: -192.5 °C

☒ Cryo Shield: -193.8 °C

☒ Chamber Pressure: 1.48E-5 Pa

Recording: 02/09/2023 08:31 AM (Live)

Tilt Calculation

Holder Pre-tilt

-

+

45.0 °

Milling Angle

10.0 °

Go To

Stage Tilt

17.0 °

Go To

☐ Y-Z Correction

Cryo GIS Deposition

Gas

Pt dep

Selected Grid

☒ Grid 1

☐ Grid 2

Flow Duration

00:00:08

Start

Purge

Sputter Coating

Prepare for Sputtering

Current

15.0 mA

Pressure

10 Pa

Voltage

0 V

Run Time

10 Sec

Run

1

2

3

4

5

样品台倾转控制

user_folder D:/SharedData/XiaoJunHuang/auto_script/auto_slice_example

set reference take map image load map

name, stage_x, stage_y, stage_z, milling_angle, thinned

point_001 3.49 2.89 6.92 10.0 N

add point update point E correct delete point goto point update pattern

settings for taking image

	Magnification	high tension	beam current	dwel time	resolution
electron	350x	3.0 kV	0.10 nA	1 us	1536x1024
ion	2000x	30.0 kV	0.10 nA	100 ns	1536x1024

=====

auto milling current position auto milling all points stop

=====settings for milling with ion beam=====

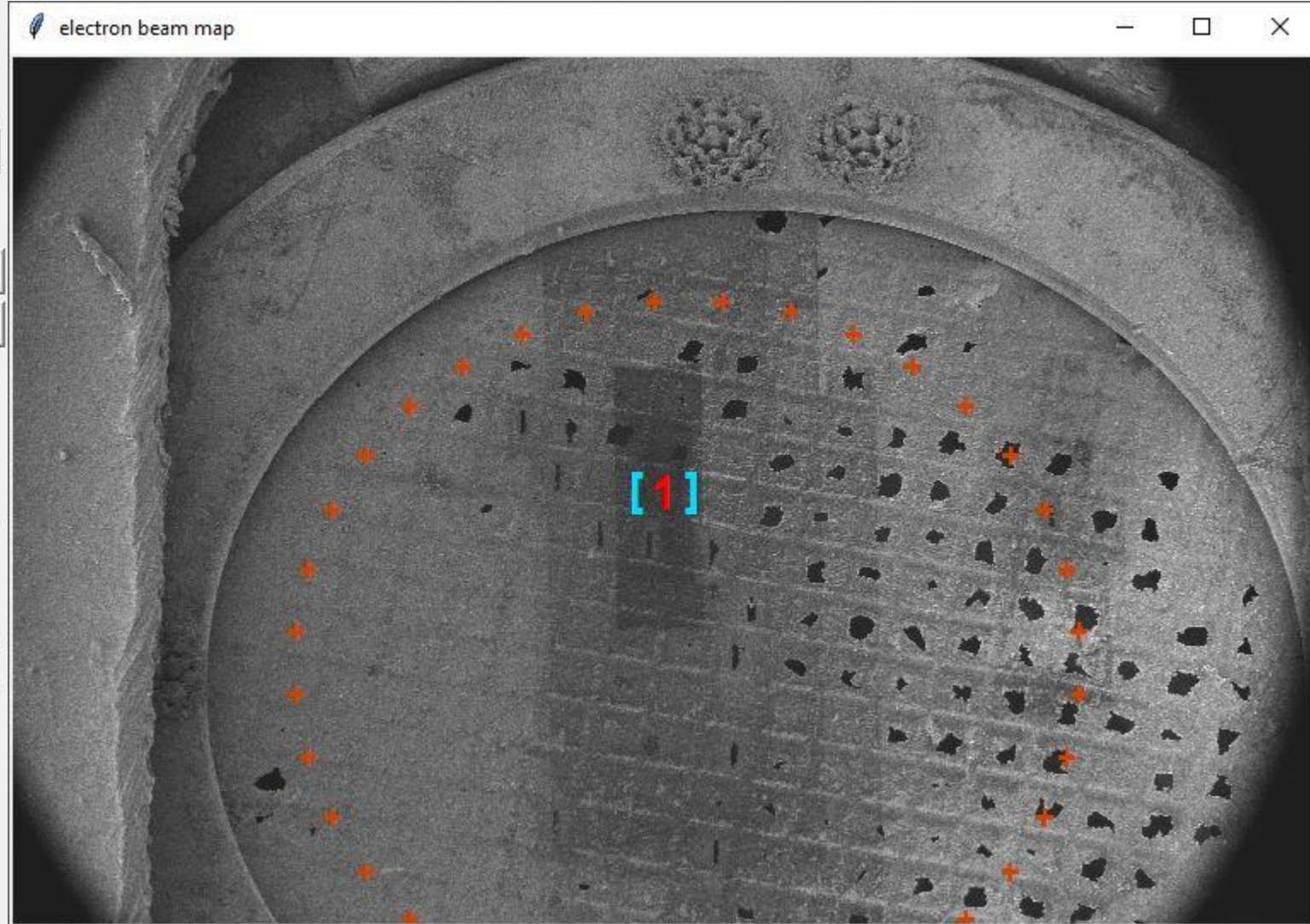
name, position, type, high tension, beam current, milling angle, x, y, z

add pattern change pattern delete pattern ☒ include side show pattern load pattern

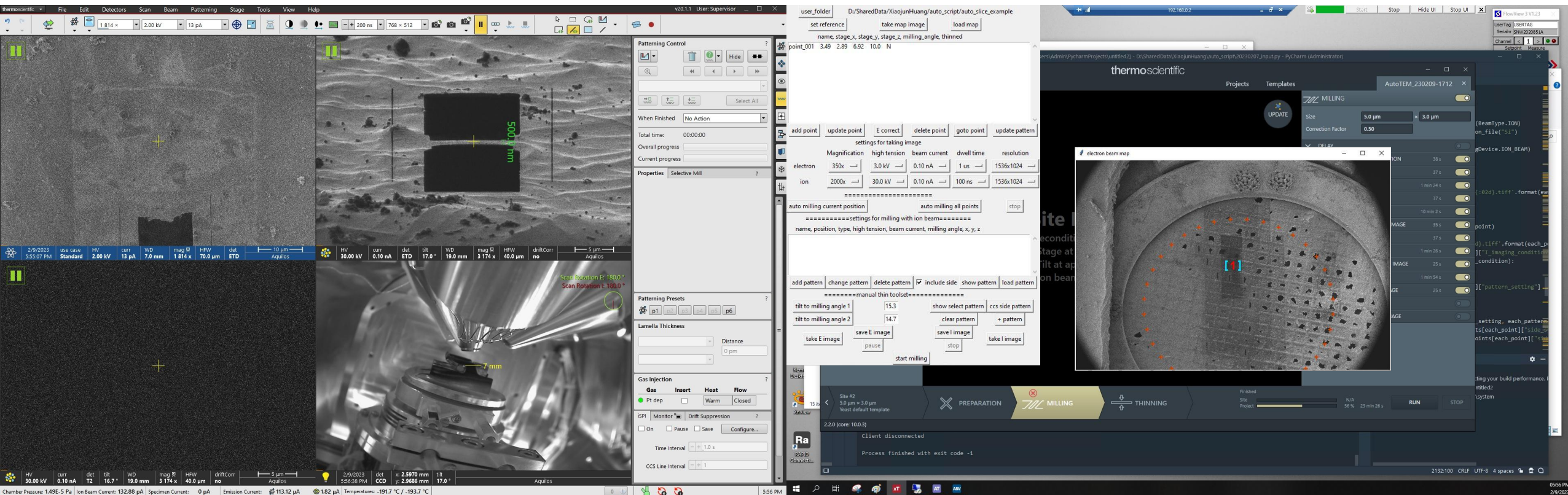
=====manual thin toolset=====

tilt to milling angle 1	15.3	show select pattern	ccs side pattern
tilt to milling angle 2	14.7	clear pattern	+ pattern
take E image	save E image	save I image	take I image
pause	stop		
start milling			

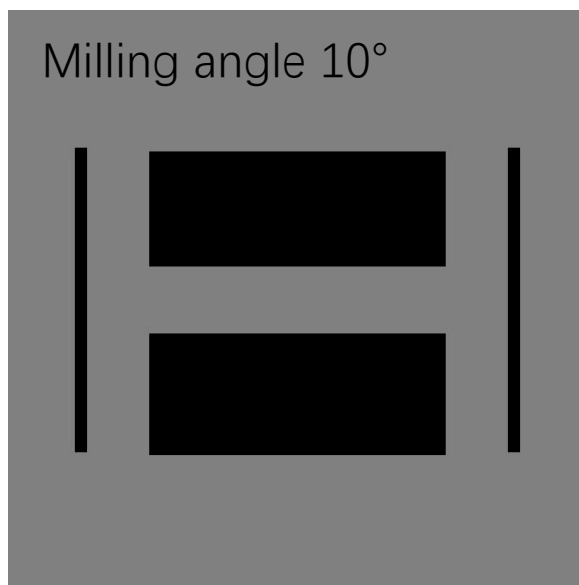
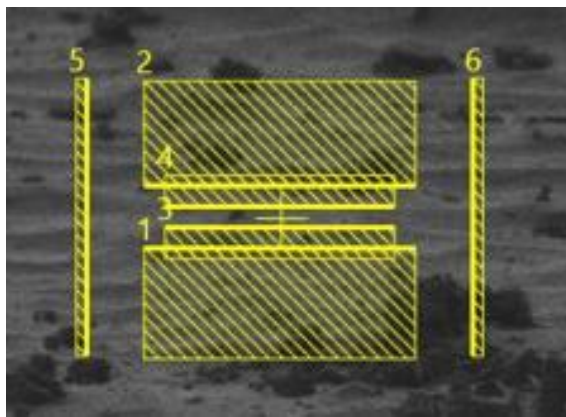
增加全载网地图窗口，以便从电子束角度提供定位辅助
即使不使用自动减薄，手动减薄也能借助该辅助定位功能



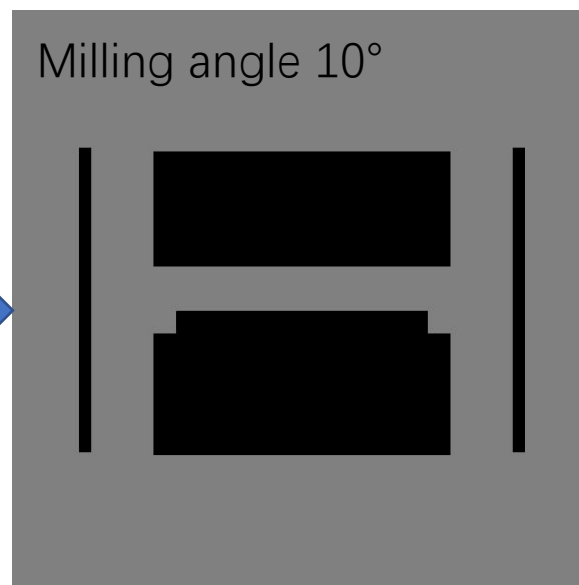
与UI界面的兼容，仅运行在supportPC上
将supportPC以远程桌面形式登陆，可以拥有双屏工作优势
与AutoTEM Cryo可以同时工作（例如，利用AutoTEM Cryo来实现自动进入sleep状态）



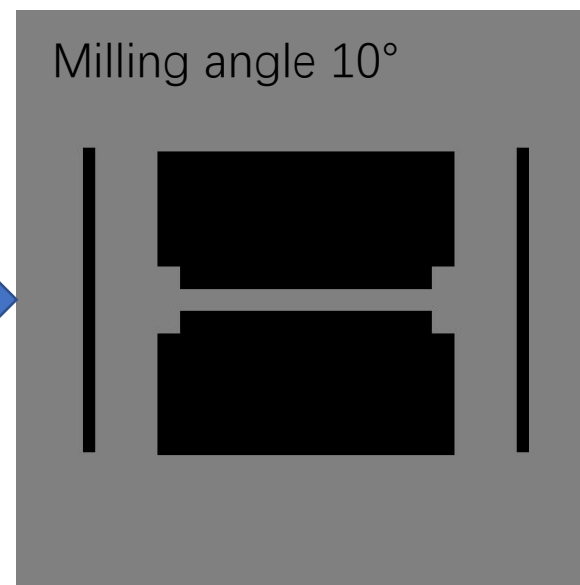
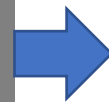
自动减薄策略-固定倾转角



30kV 0.3nA -> 3 um

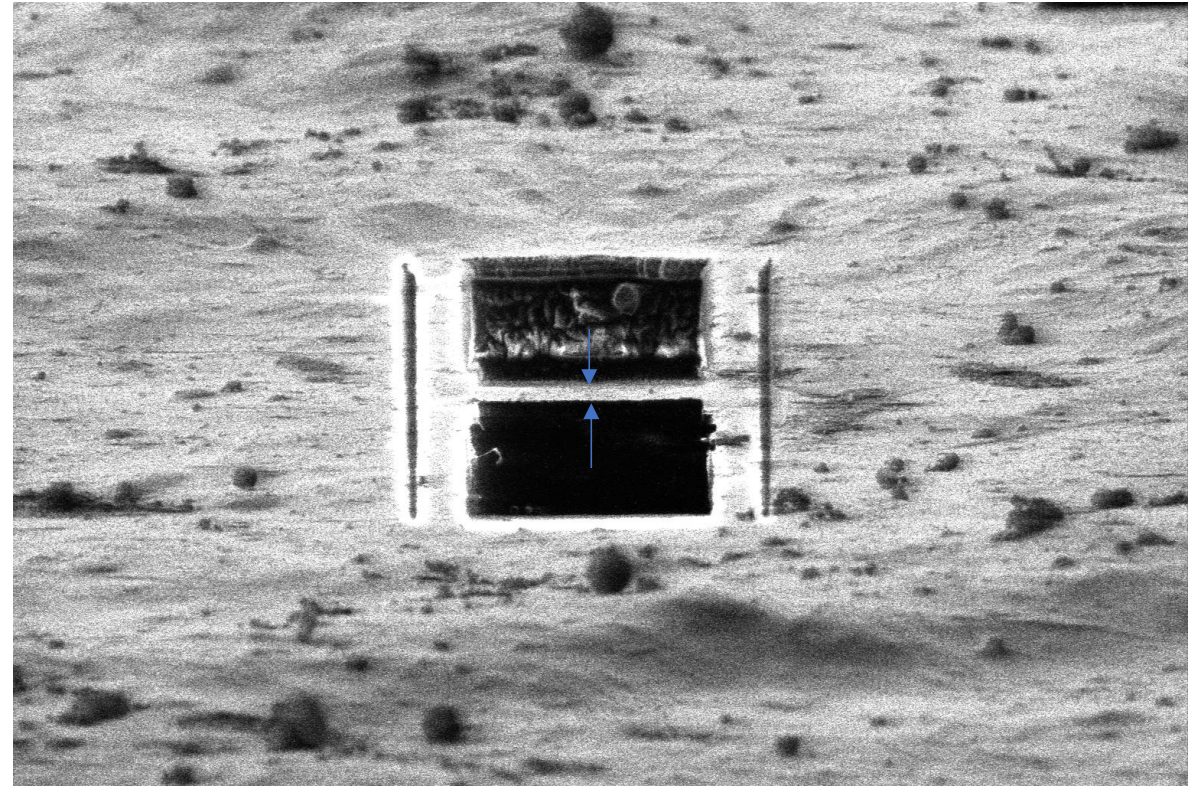


30kV 0.1nA -> 1 um



自动减薄的效果

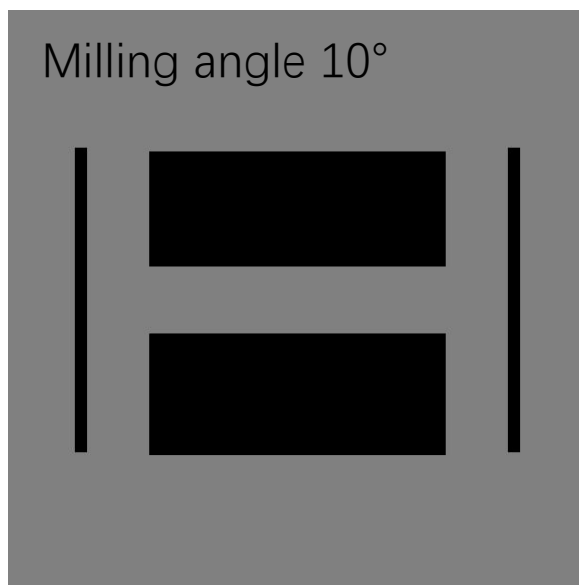
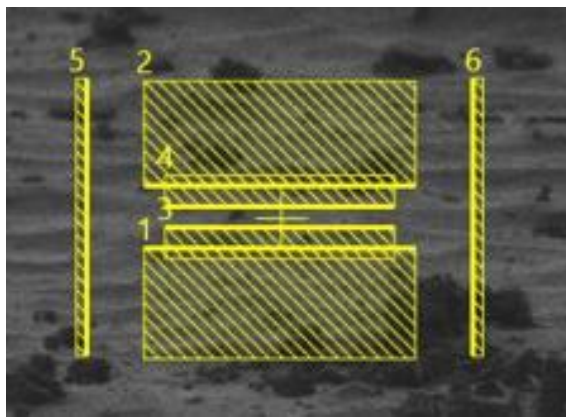
- Milling angle 10°
- Ion beam 30kV 0.3nA $\rightarrow$ 3 μm (thickness)
- Ion beam 30kV 0.1nA $\rightarrow$ 1 μm (thickness)
- measured thickness: $753 \pm 45 \text{ nm}$



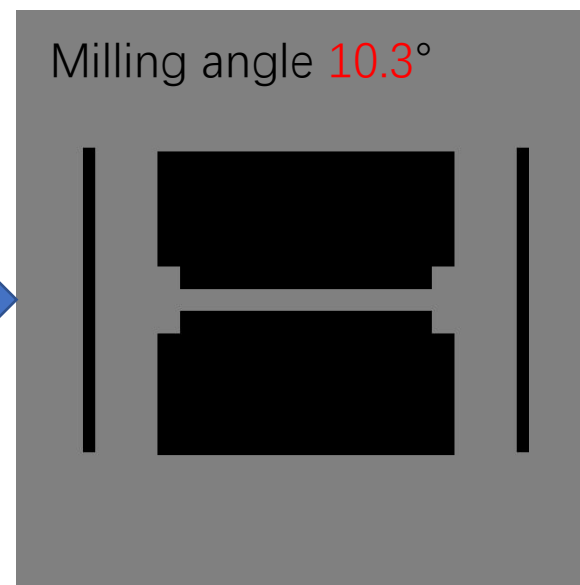
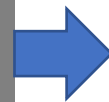
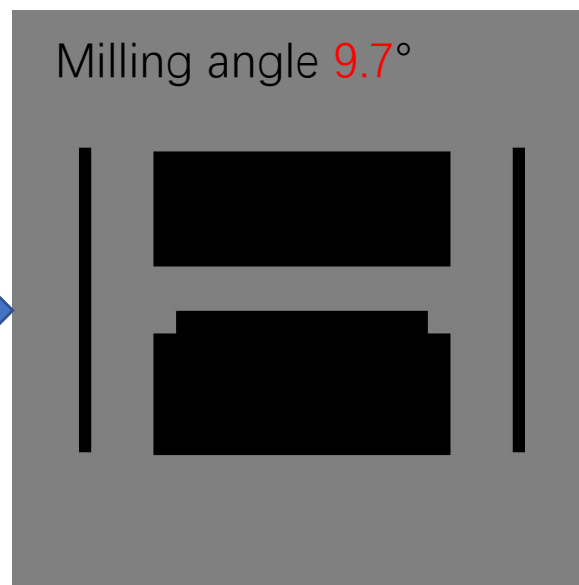
自动减薄的效果

- Milling angle 10°
- Ion beam 30kV 0.3nA $\rightarrow$ 3 μm (thickness)
- Ion beam 30kV 0.1nA $\rightarrow$ 0.8 μm (thickness)
- Actual thickness: $439 \pm 145 \text{ nm}$

自动减薄策略-变化倾转角



30kV 0.3nA -> 3 um

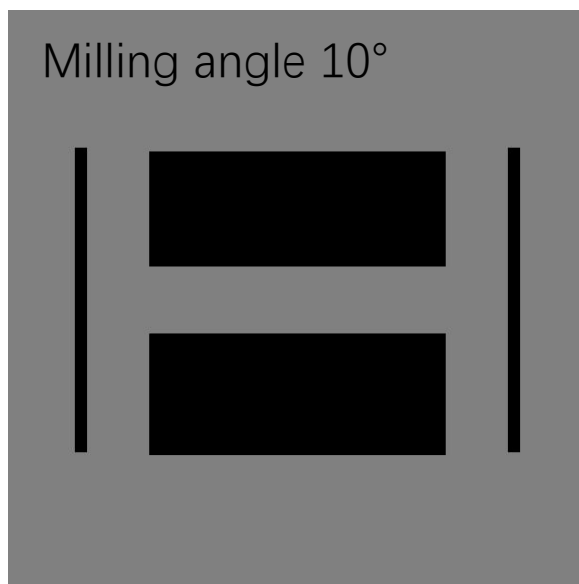
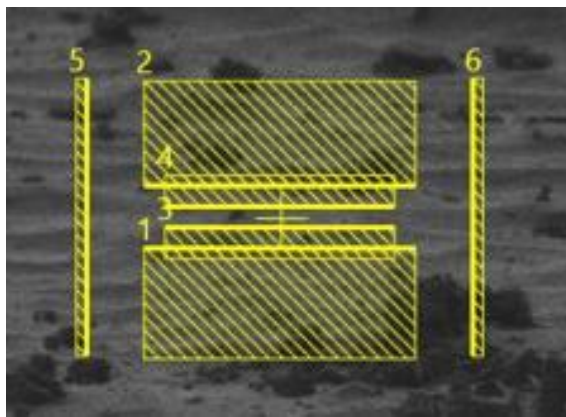


30kV 0.1nA -> 1 um

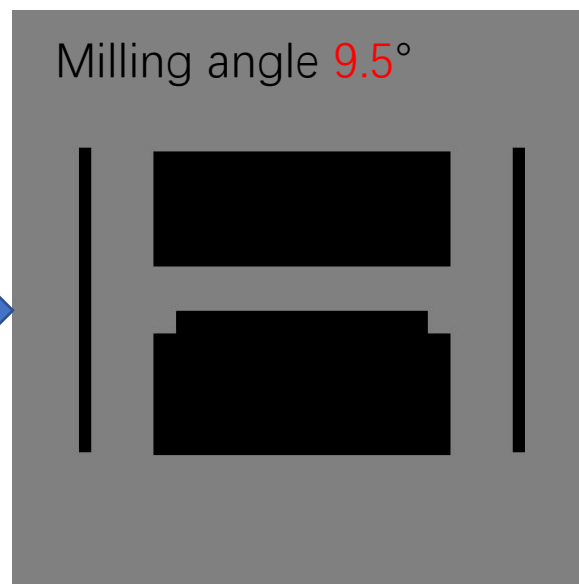
自动减薄的效果

- Milling angle 10°
- Ion beam 30kV 0.3nA $\rightarrow$ 3 μm (thickness)
- Ion beam 30kV 0.1nA $\rightarrow$ 1 μm (thickness)
- measured thickness: $675 \pm 157 \text{ nm}$

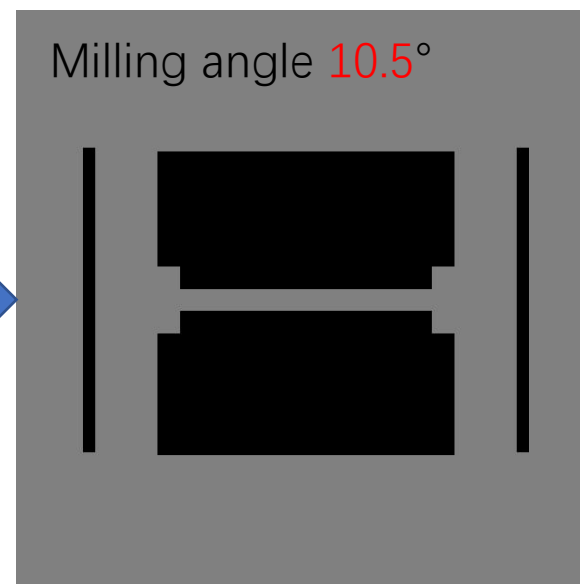
自动减薄策略-变化倾转角



30kV 0.3nA -> 3 um



30kV 0.1nA -> 1 um

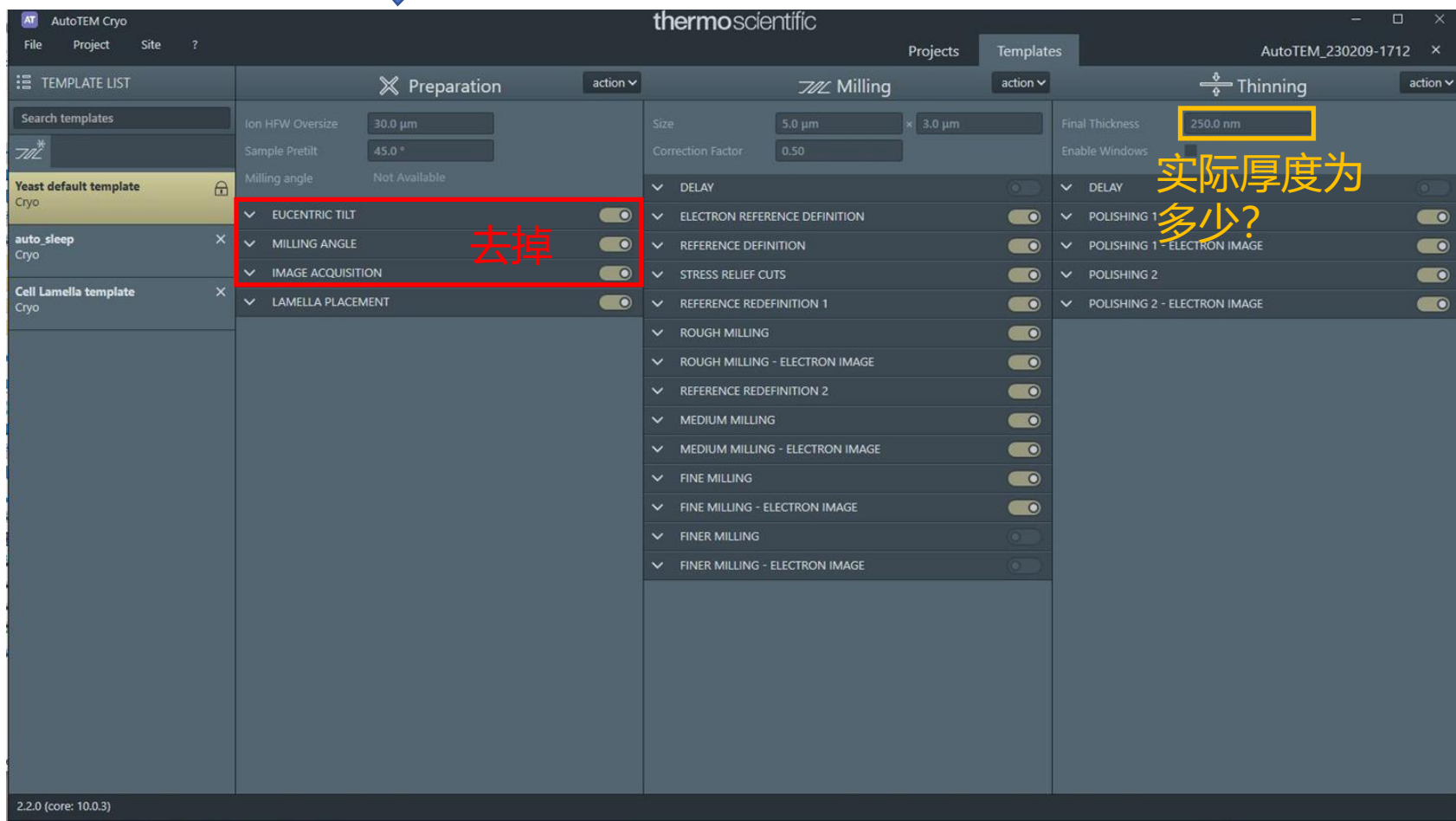
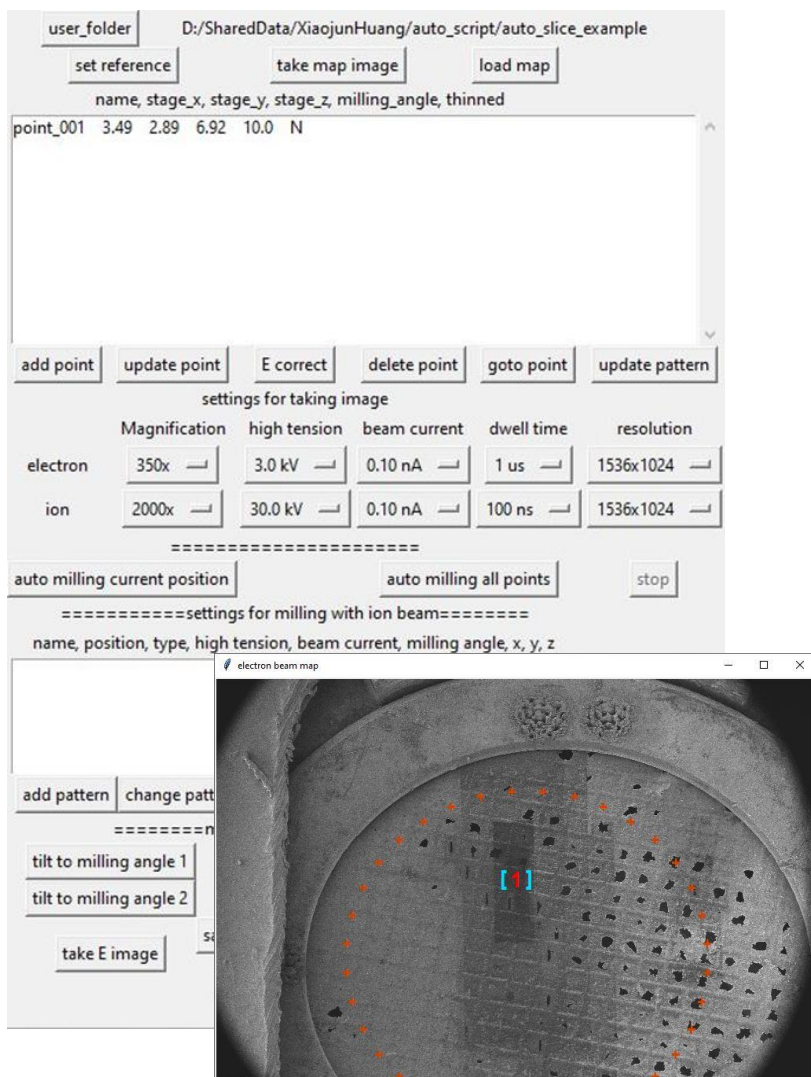


自动减薄的效果

- Milling angle 10°
- Ion beam 30kV 0.3nA $\rightarrow$ 3 μm (thickness)
- Ion beam 30kV 0.1nA $\rightarrow$ 1 μm (thickness)
- measured thickness: $845 \pm 100 \text{nm}$

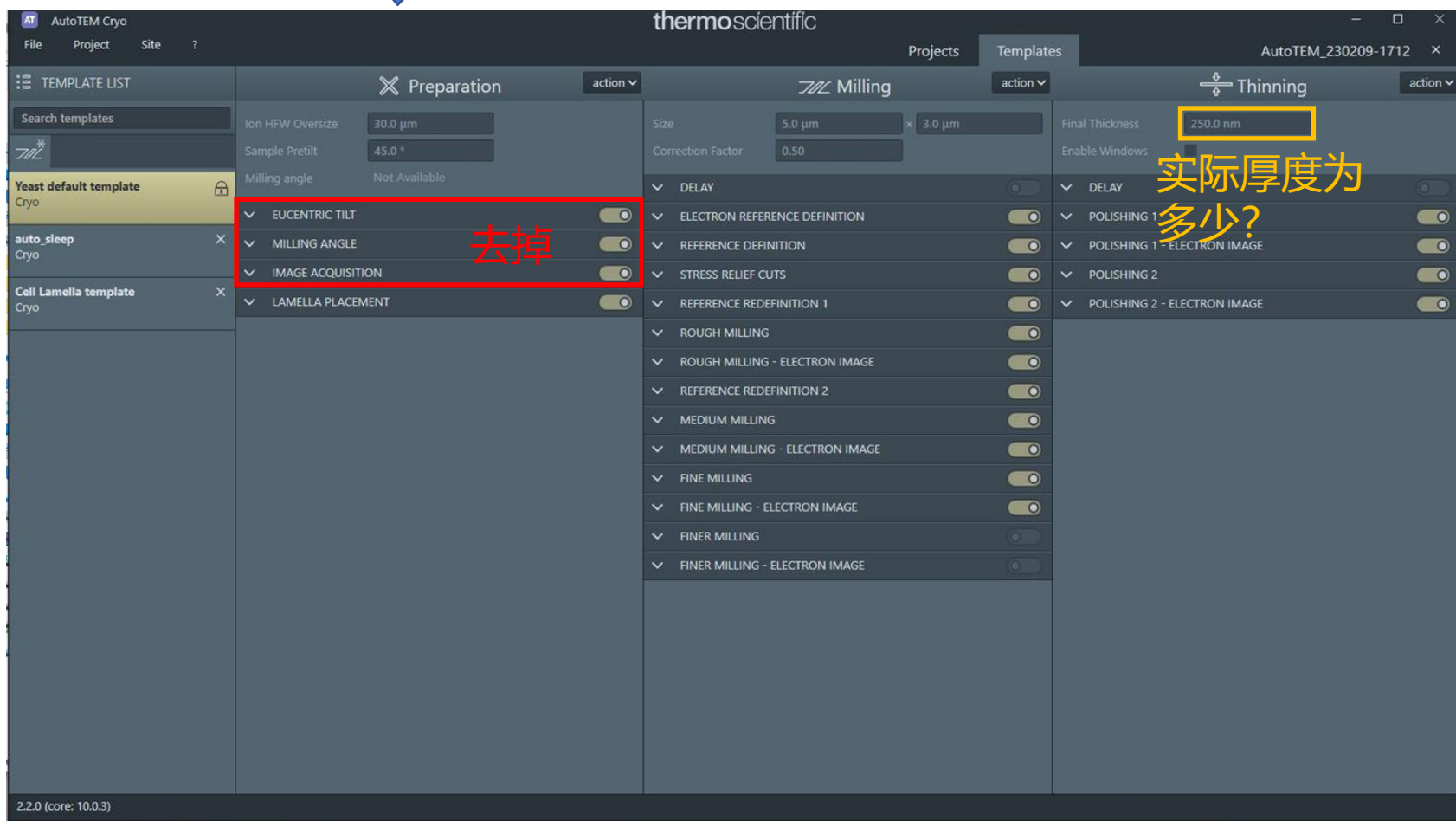
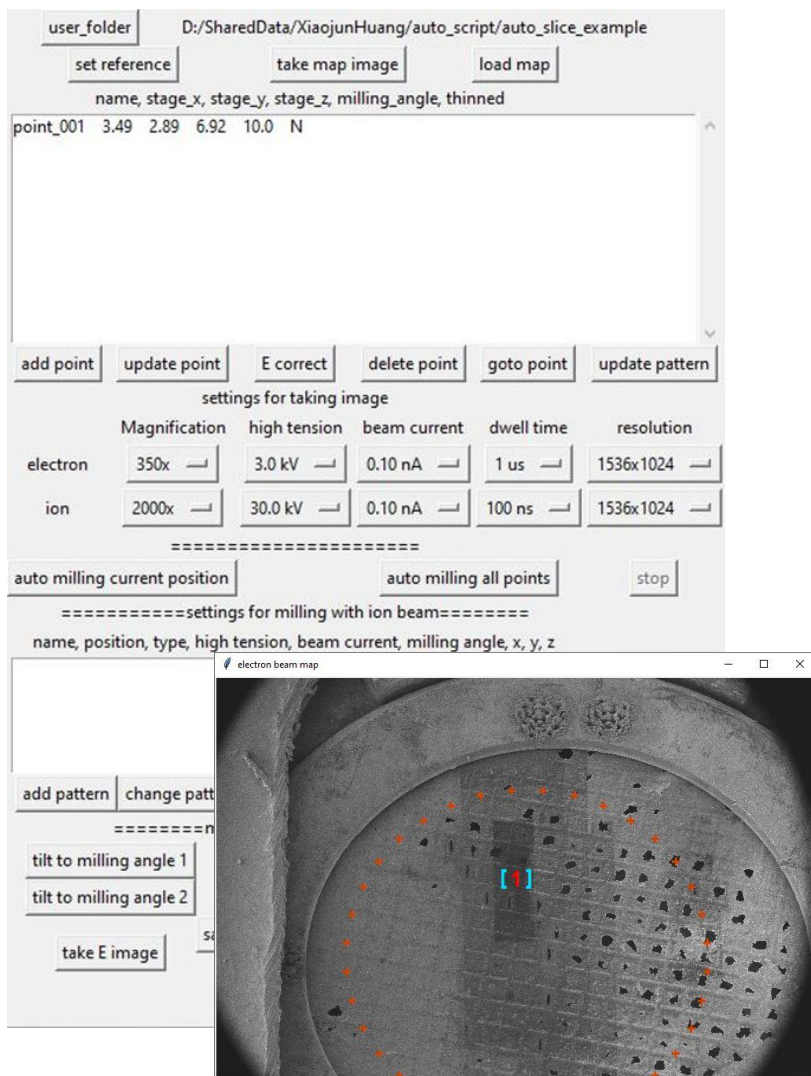
SA-FIB compatibility with AutoTEM cryo

2023年11月7日冰增长测定数据为 <10 nm/h
具备使用AutoTEM cryo进行过夜自动减薄的条件



SA-FIB compatibility with AutoTEM cryo

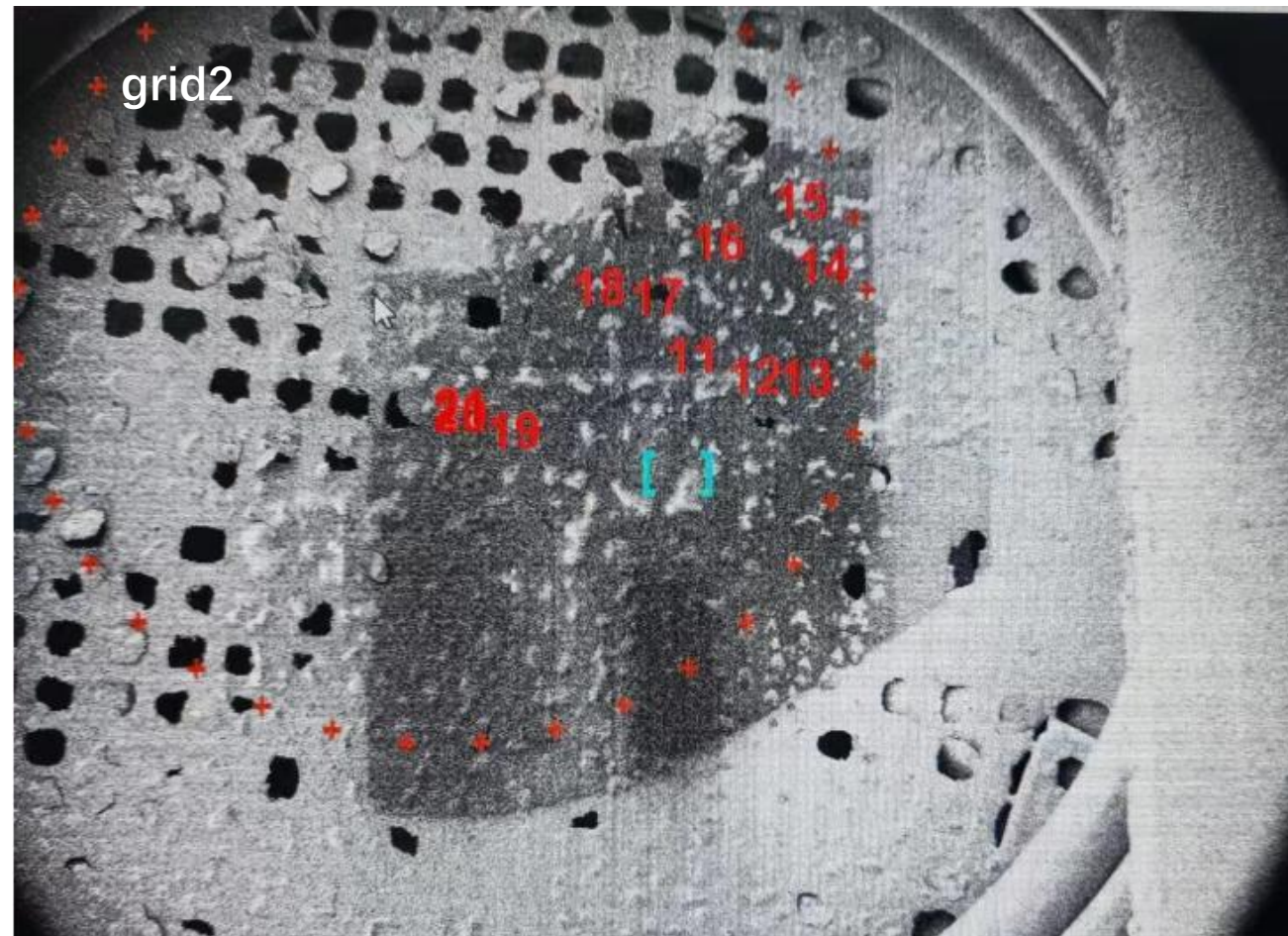
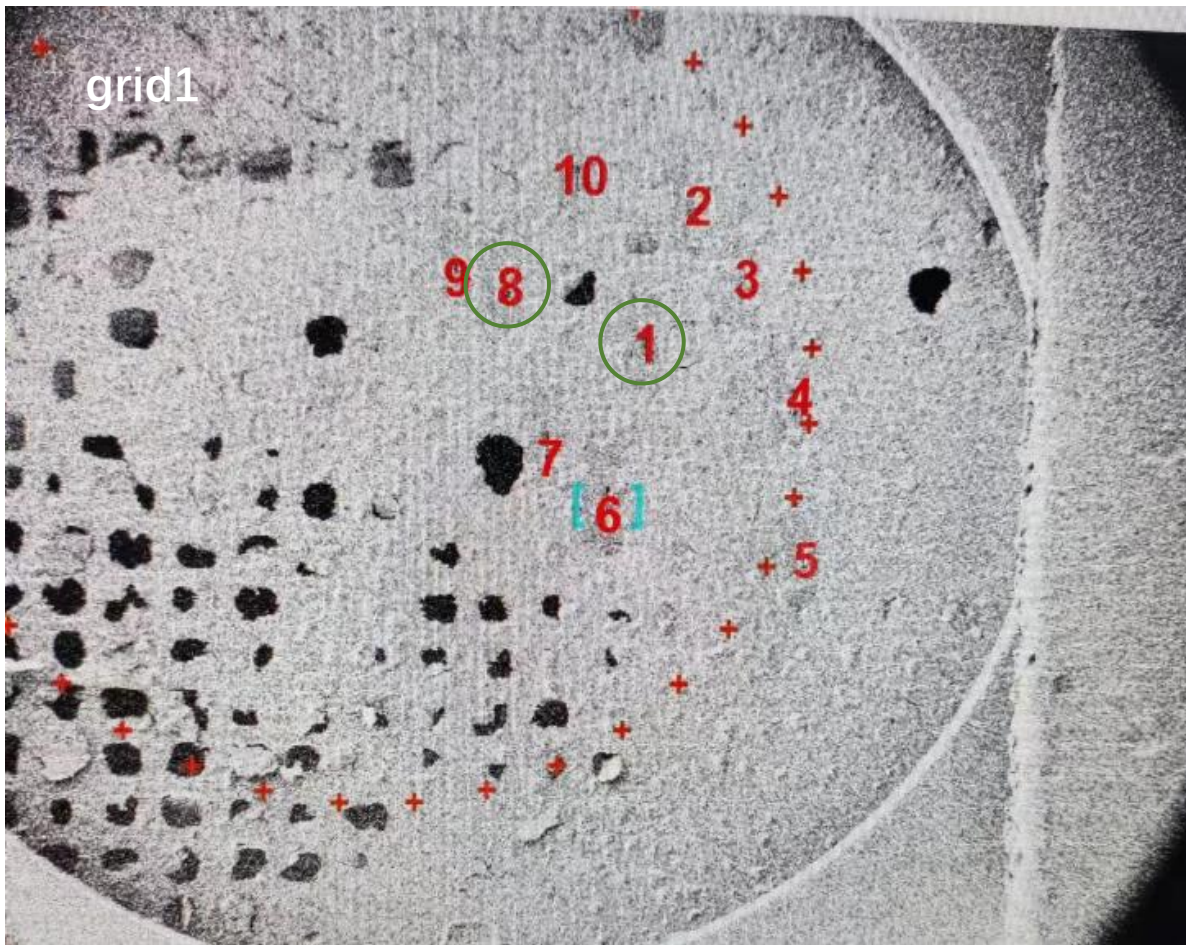
2023年11月7日冰增长测定数据为 <10 nm/h
具备使用AutoTEM cryo进行过夜自动减薄的条件



测试条件一

- Site-wise（在一个位置完成初修和精修后，再移动到下一个位置操作）
 - 减薄厚度控制最佳，但污染程度最高
- 减薄过程不进行小角度倾转
 - 切片沿减薄方向，从前往后呈现由薄到厚的渐变
- 目标厚度设置150nm
- 减薄角度15度

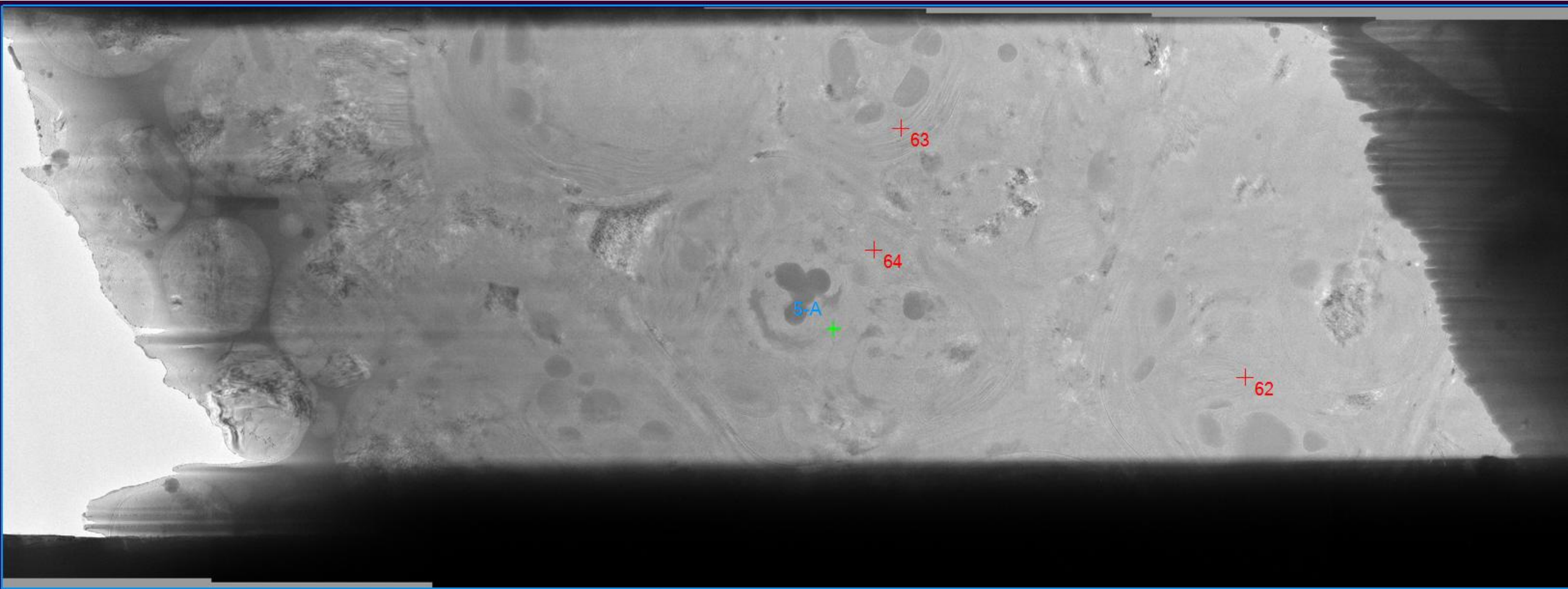
autoTEM cryo全自动减薄 (20231114)



图中显示减薄顺序，一共设置20个位置，目标厚度150nm，site-wise减薄策略
17: 00上样；19: 30开始自动减薄，显示约10小时后完成；次日8: 30检查减薄
结果；9: 00sputtering后回收

#1

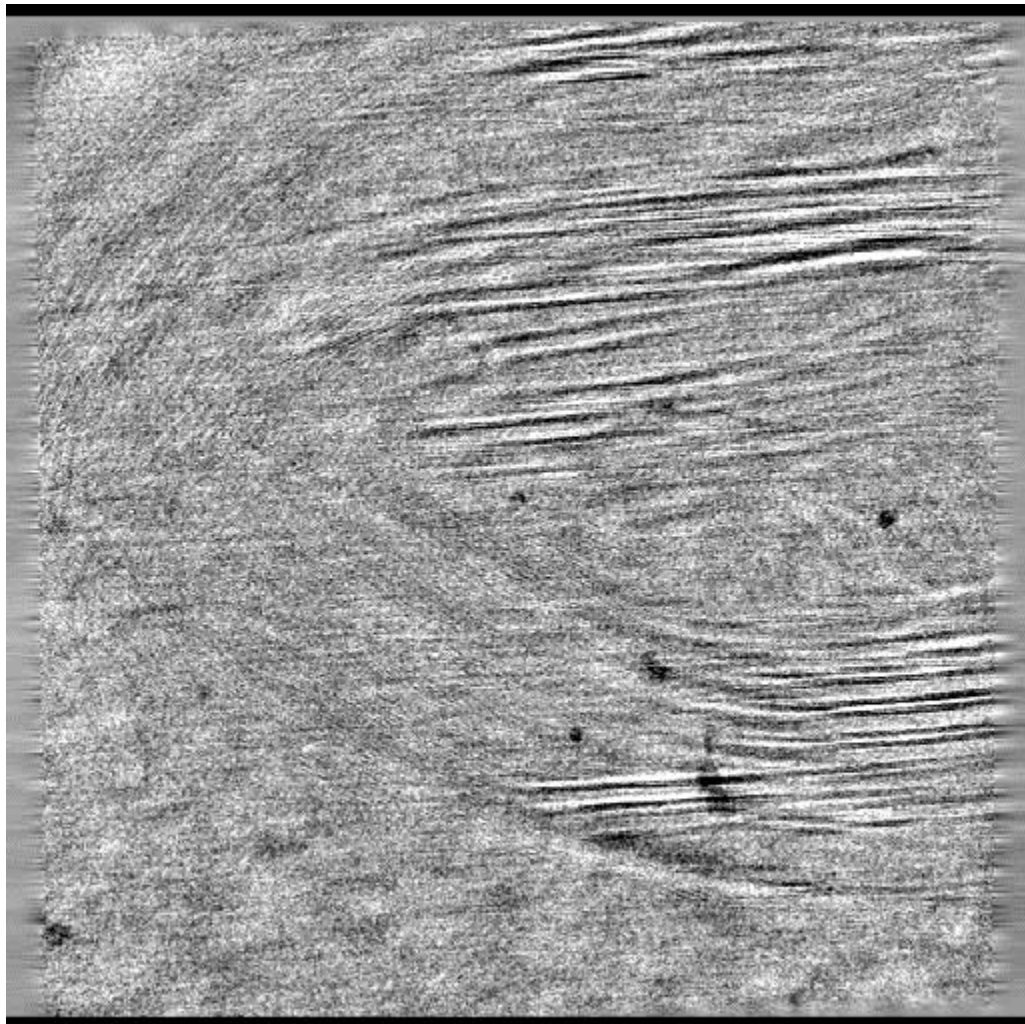
19:30-20:10进行减薄, 次日9:00取出, 在Aquilos中经历约13小时



2 um

#1-62

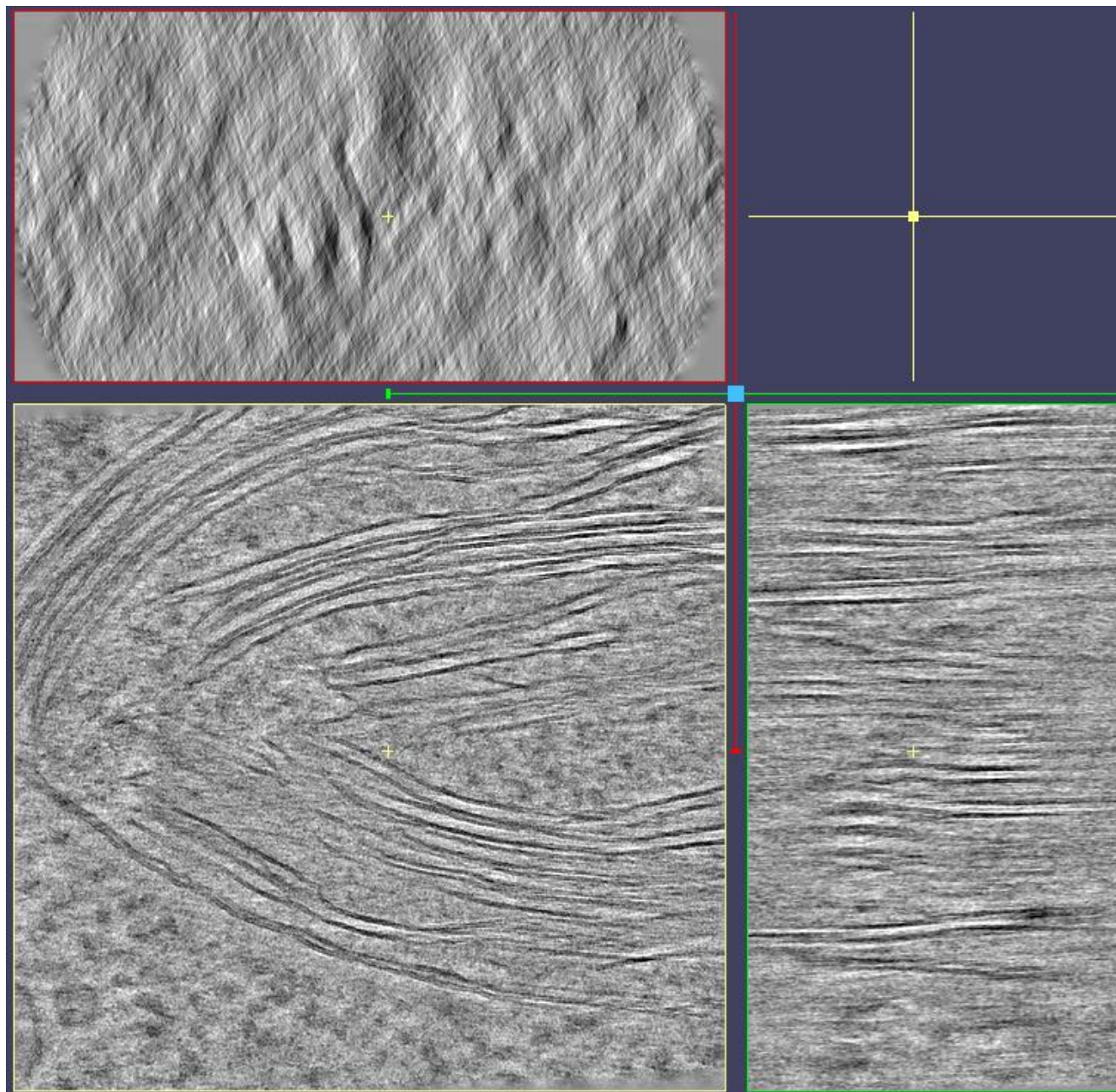
衣藻（丙烷-乙烷混合冷冻），
256nm（target 150nm）



Titan3 data collection

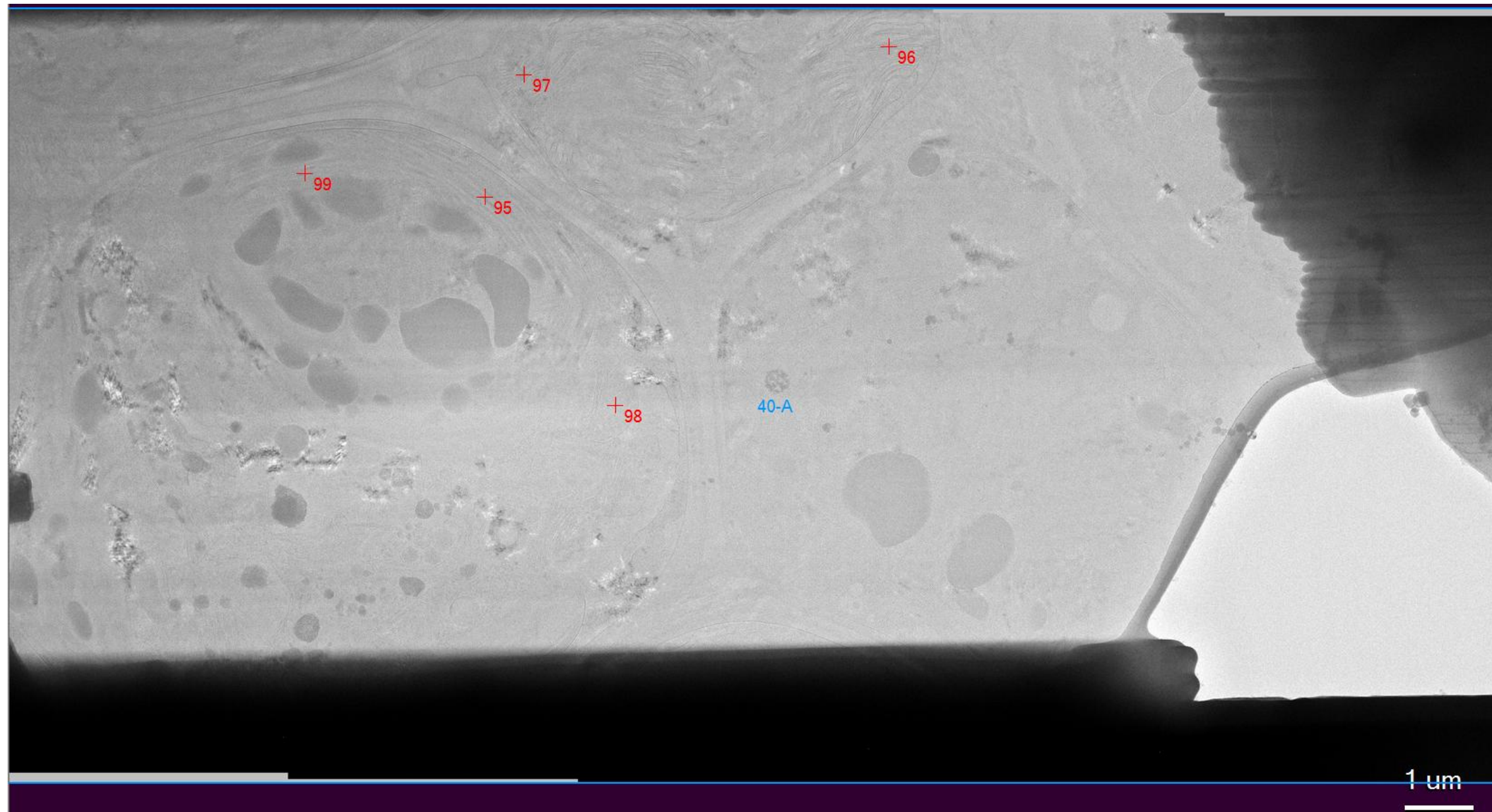
20231126

64kx, 3e/tilt, + -45度, 3度/tilt



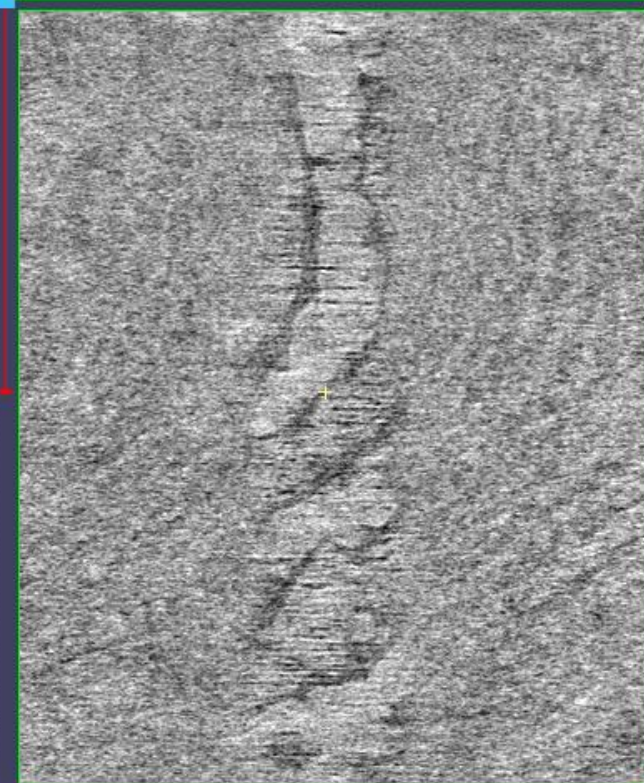
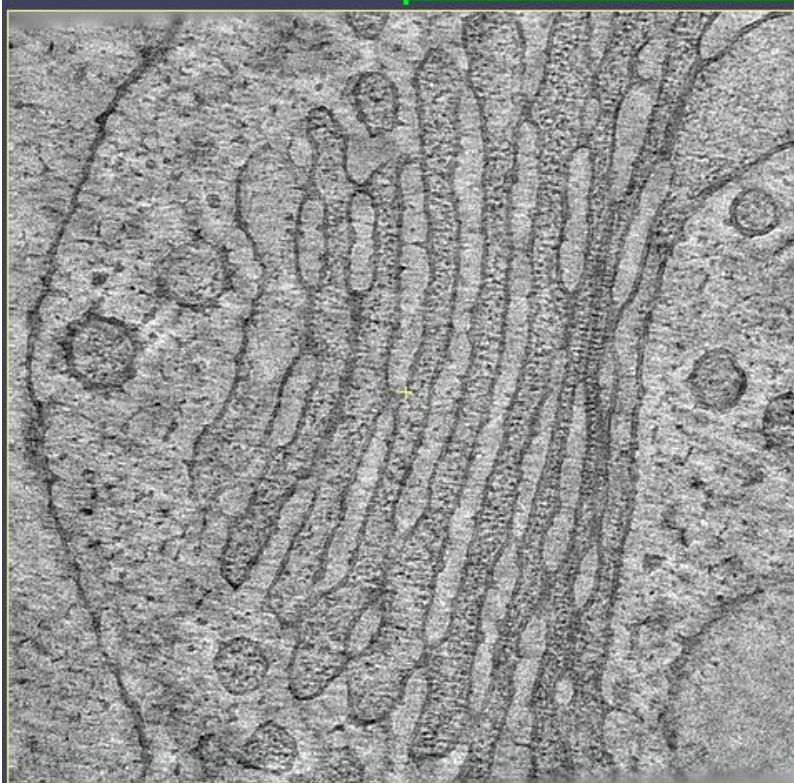
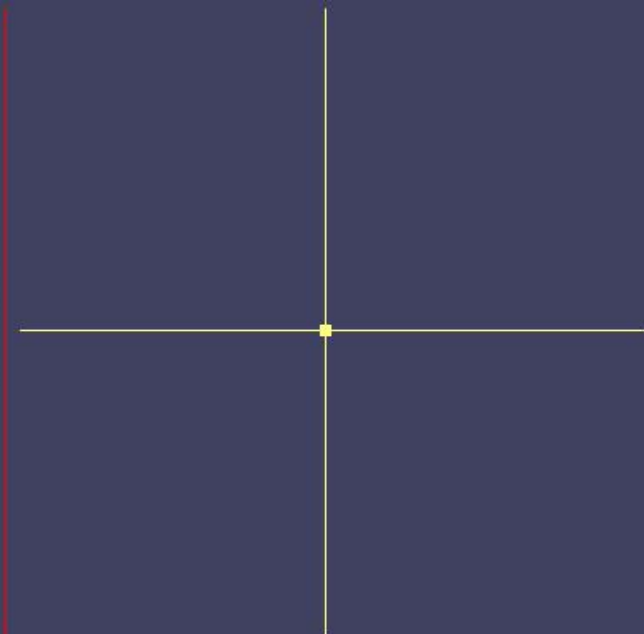
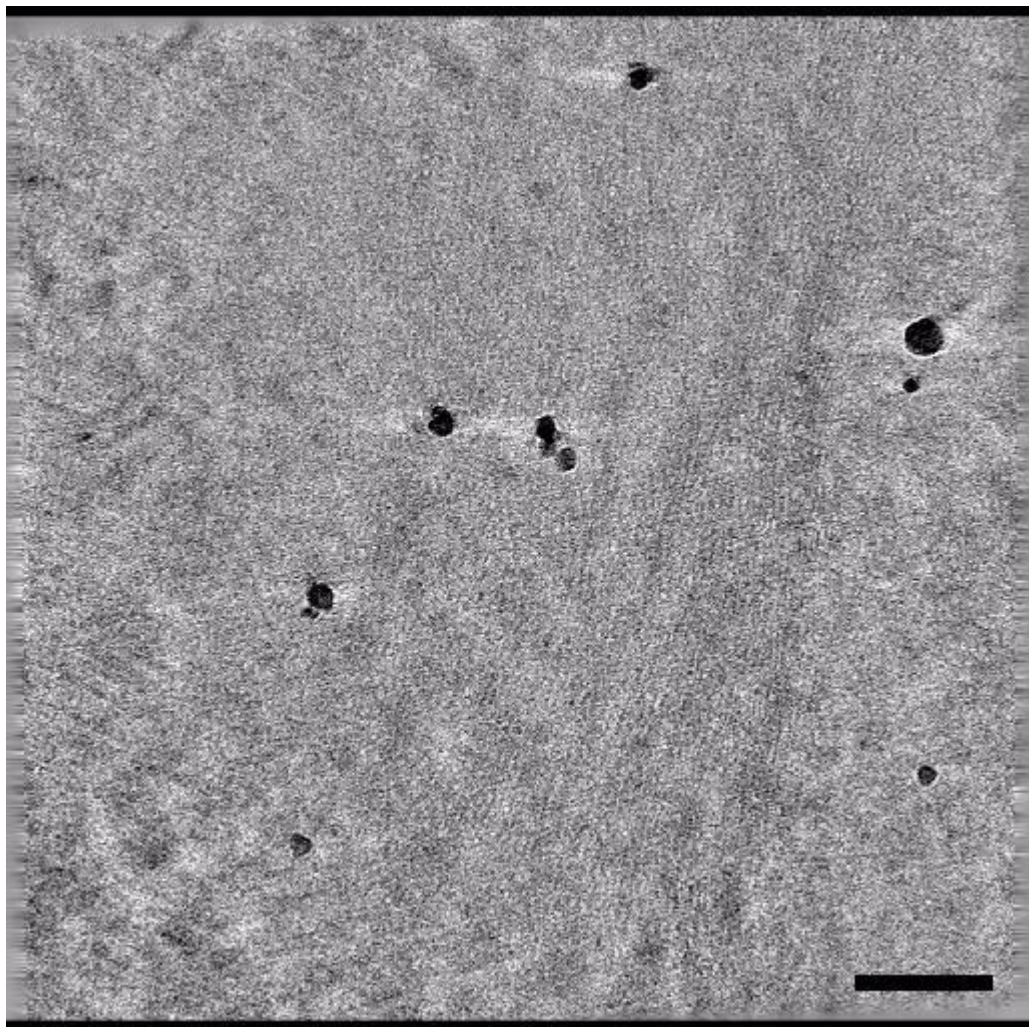
#8

0:10-0:50进行减薄, 次日9:00取出, 在Aquilos中经历约8小时



#8-98

衣藻 (丙烷-乙烷混合冷冻),
149nm (target 150nm)

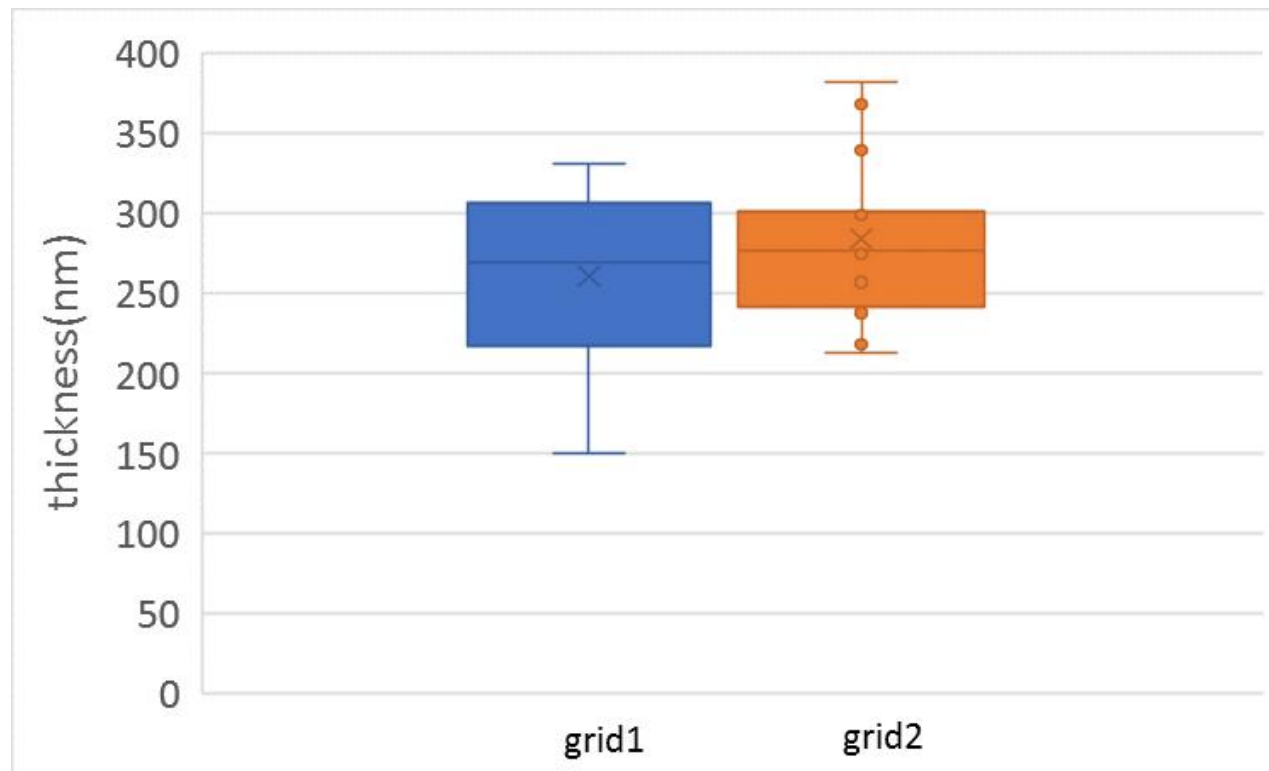


Titan3 data collection

20231126

64kx, 3e/tilt, + -45度, 3度/tilt

厚度统计



目标厚度设置150nm

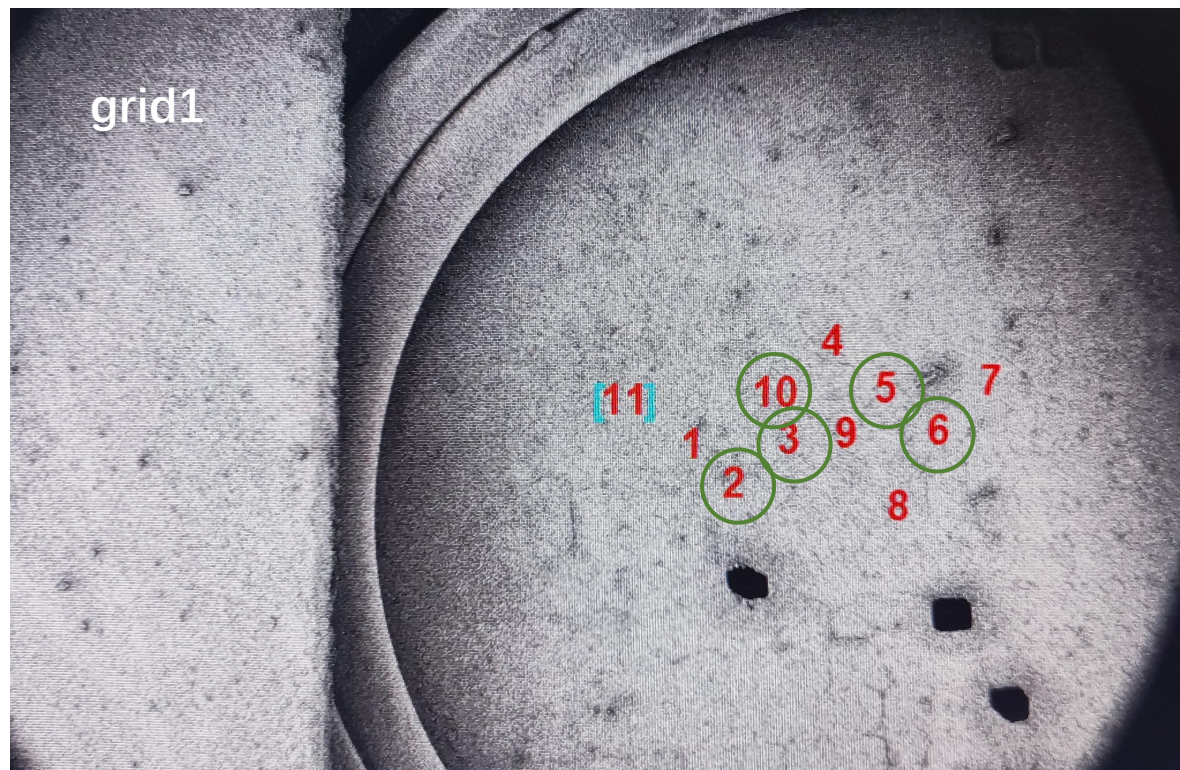
测试条件一结果

- Site-wise（在一个位置完成初修和精修后，再移动到下一个位置操作）
 - 未观察到明显污染层
- 减薄过程不进行小角度倾转
 - 切片沿减薄方向，从前往后呈现由薄到厚的渐变
- 目标厚度设置150nm
 - 厚度范围在150-370nm之间，多数在220-300nm
- 其它问题
 - 晶态冰层
 - 样品过厚，薄片在倾转过程中被遮挡，需增大减薄上下区域

测试条件二

- Site-wise (在一个位置完成初修和精修后, 再移动到下一个位置操作)
 - 减薄厚度控制最佳, 但污染程度最高
- 减薄过程不进行小角度倾转
 - 切片沿减薄方向, 从前往后呈现由薄到厚的渐变
- 目标厚度设置100nm
- 减薄角度15度

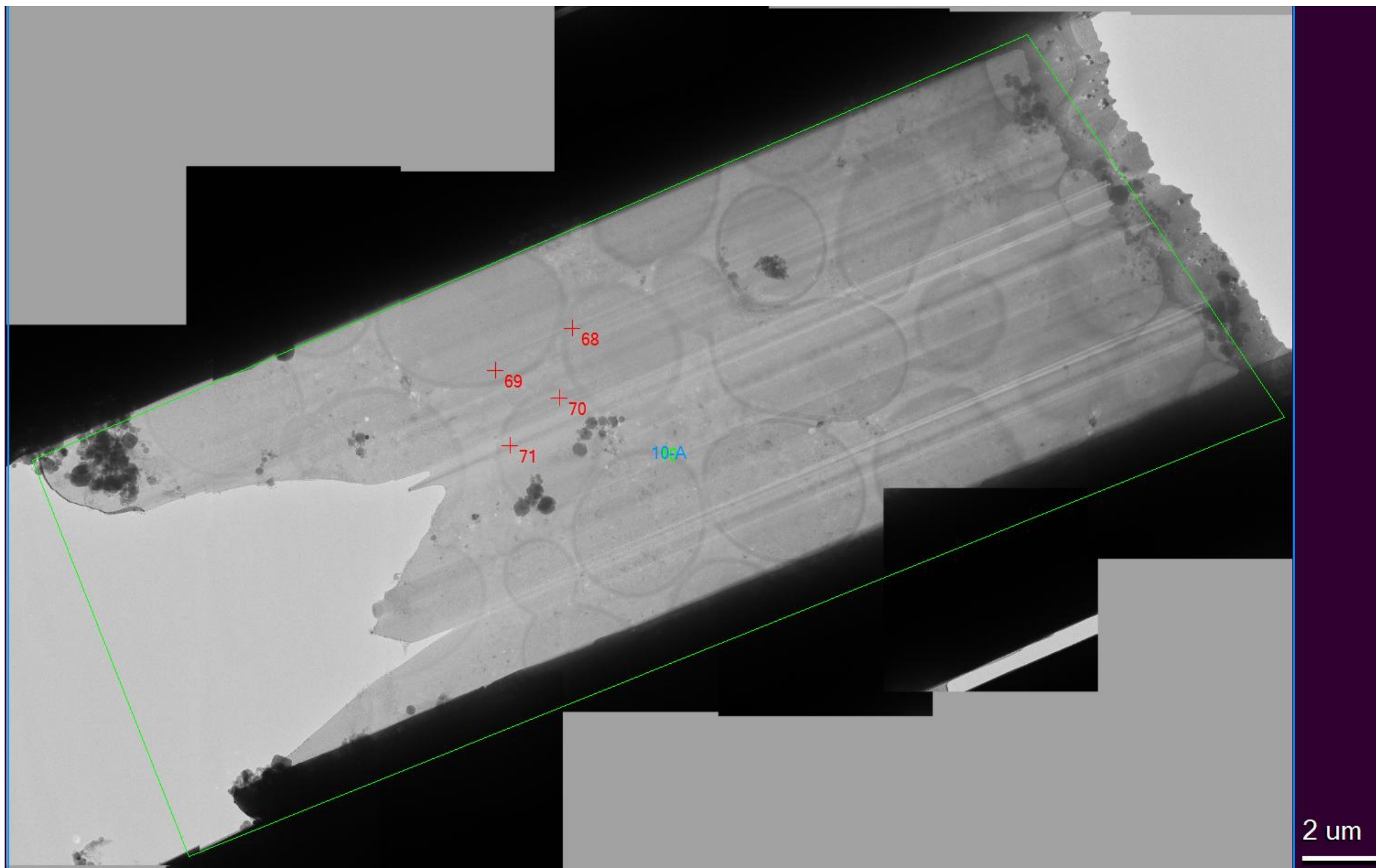
autoTEM cryo全自动减薄 (20231128)



图中显示减薄顺序，一共设置21个位置，目标厚度100nm，site-wise减薄策略
19: 30上样；21: 30开始自动减薄，显示约12小时后完成；次日8: 30除了12号
以及18-21号位以外，其余位置已完成减薄；9: 00未作sputtering就回收

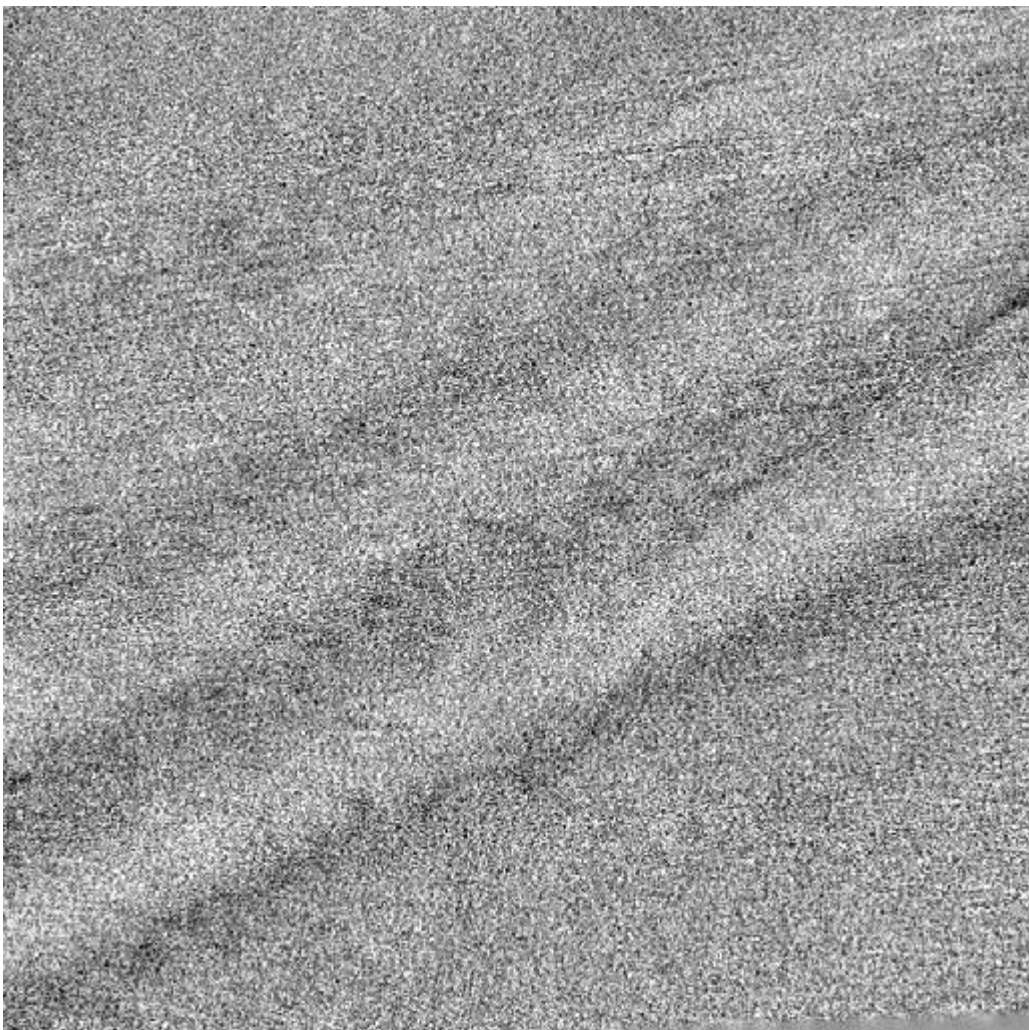
#2

22:10-22:50进行减薄，次日9:00取出，在Aquilos中经历约10小时



#2-69

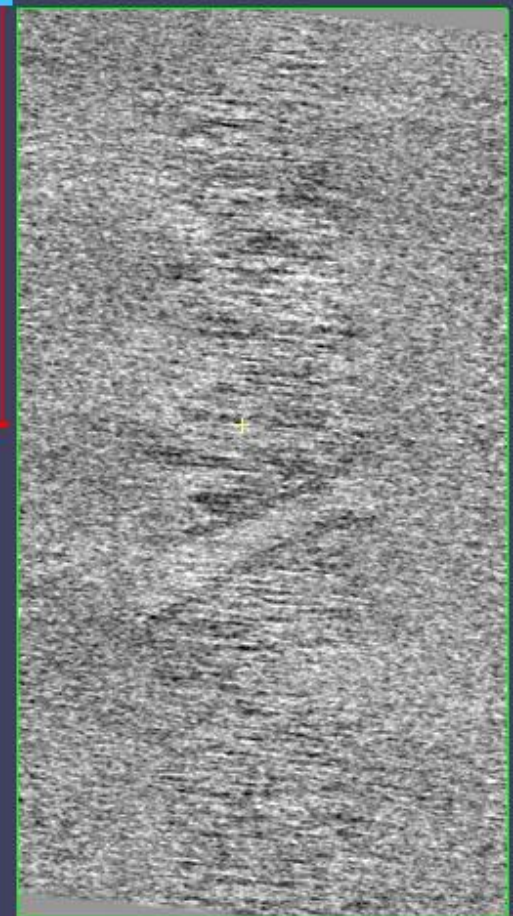
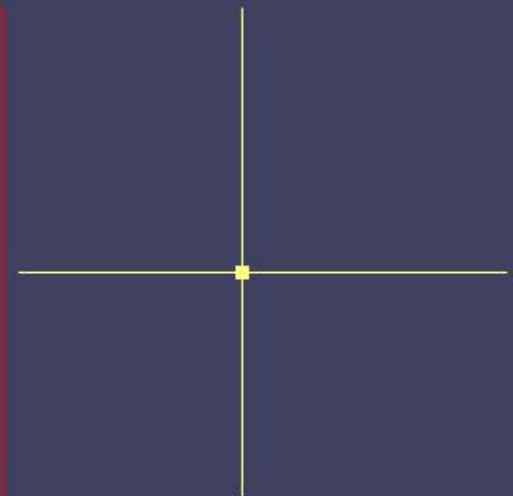
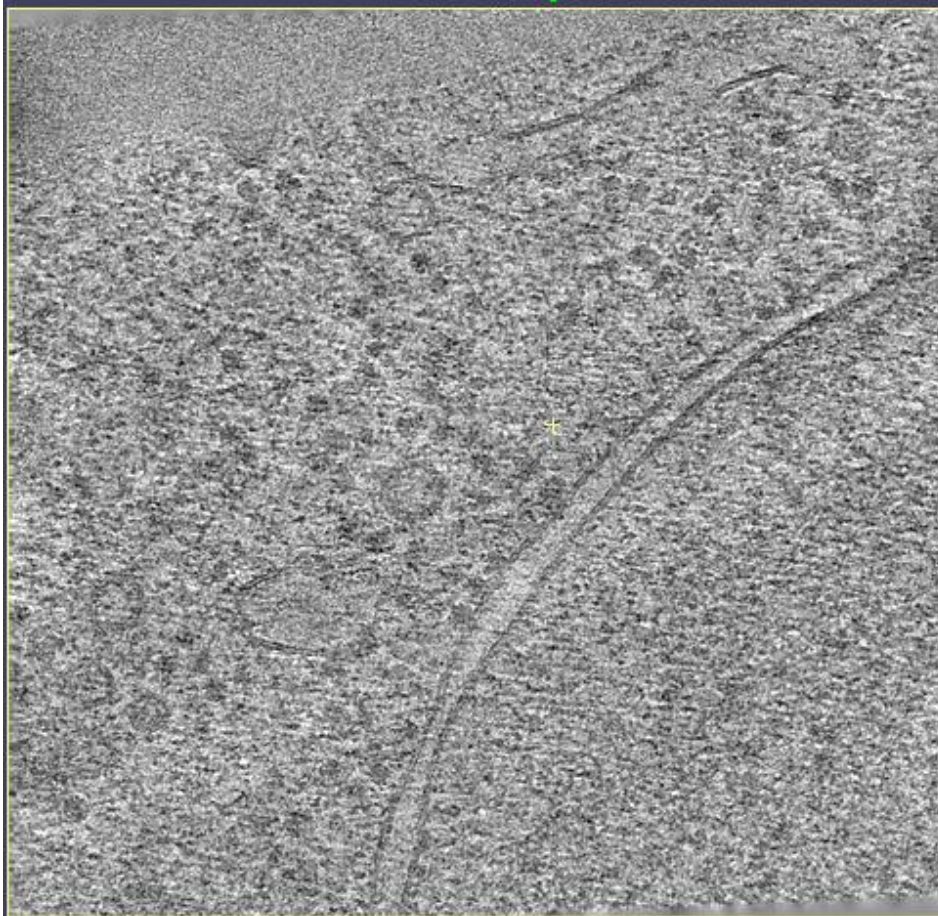
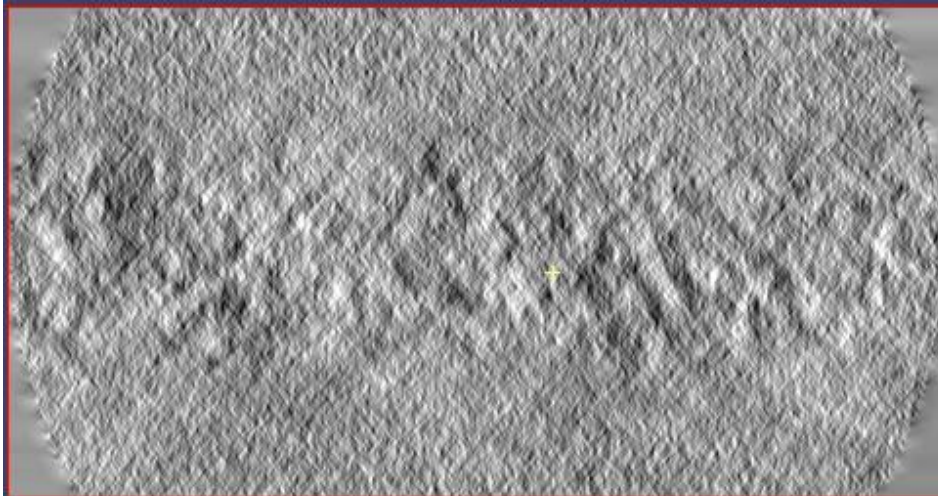
酿酒酵母（丙烷-乙烷混合冷冻），184nm（target 100nm）



Titan3 data collection

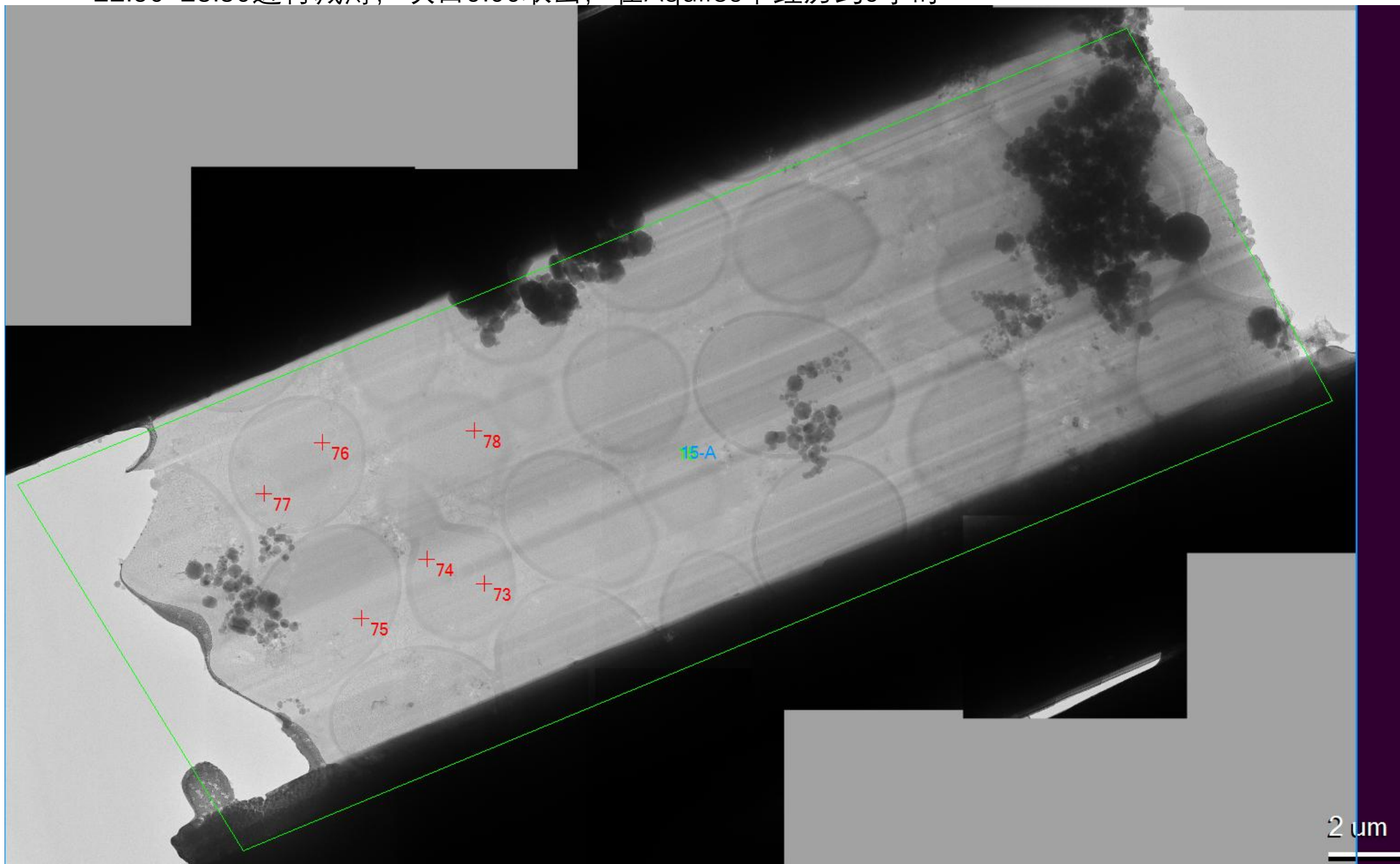
20231129

64kx, 3e/tilt, + -45度, 3度/tilt



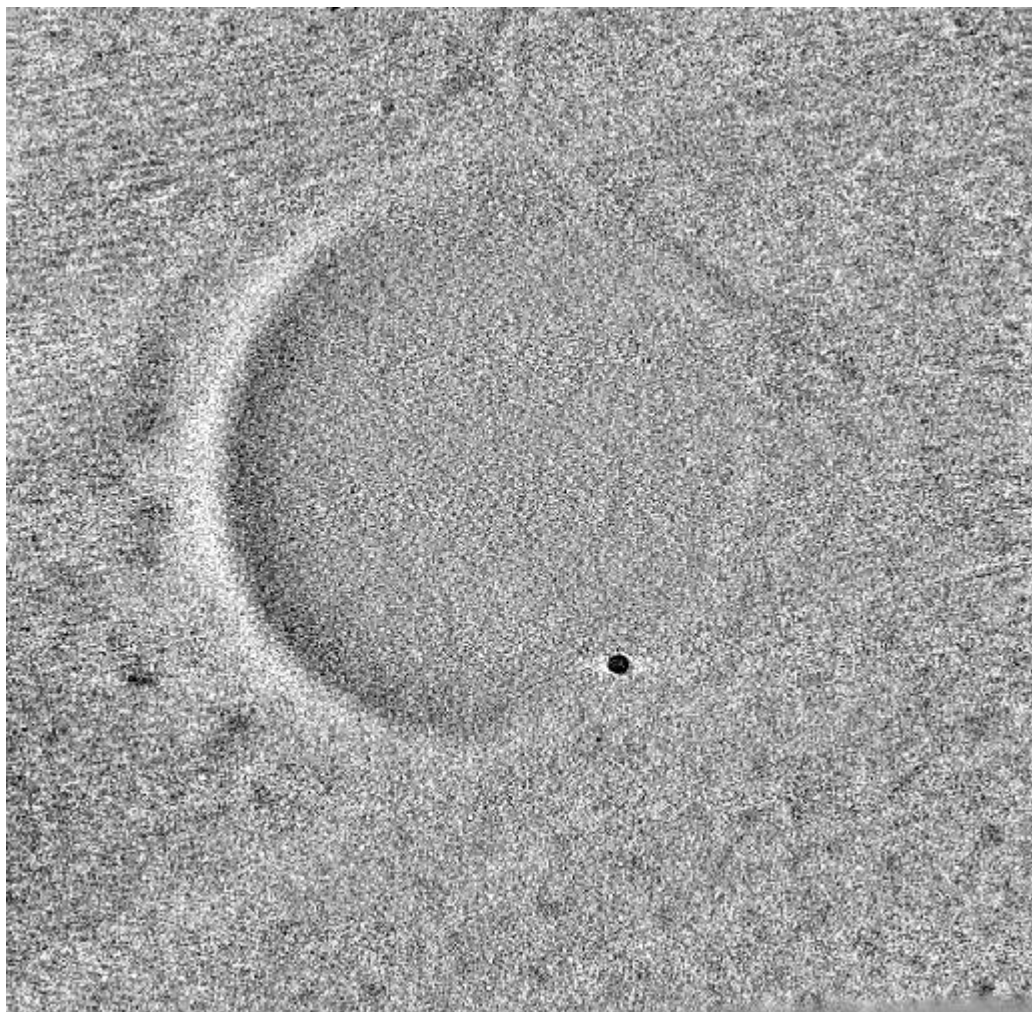
#3

22:50-23:30进行减薄，次日9:00取出，在Aquilos中经历约9小时



#3-75

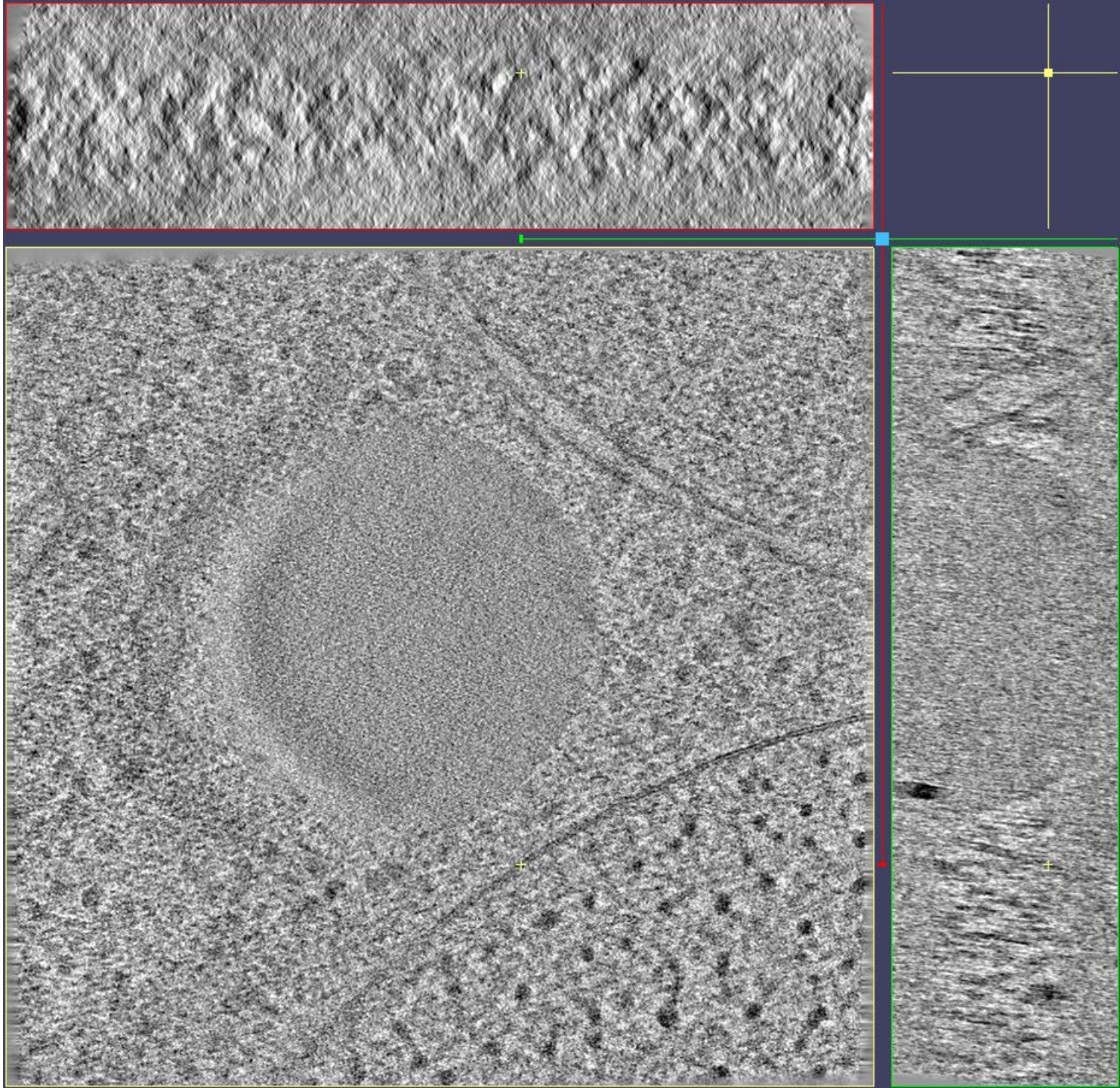
酿酒酵母（丙烷-乙烷混合冷冻），112nm（target 100nm）



Titan3 data collection

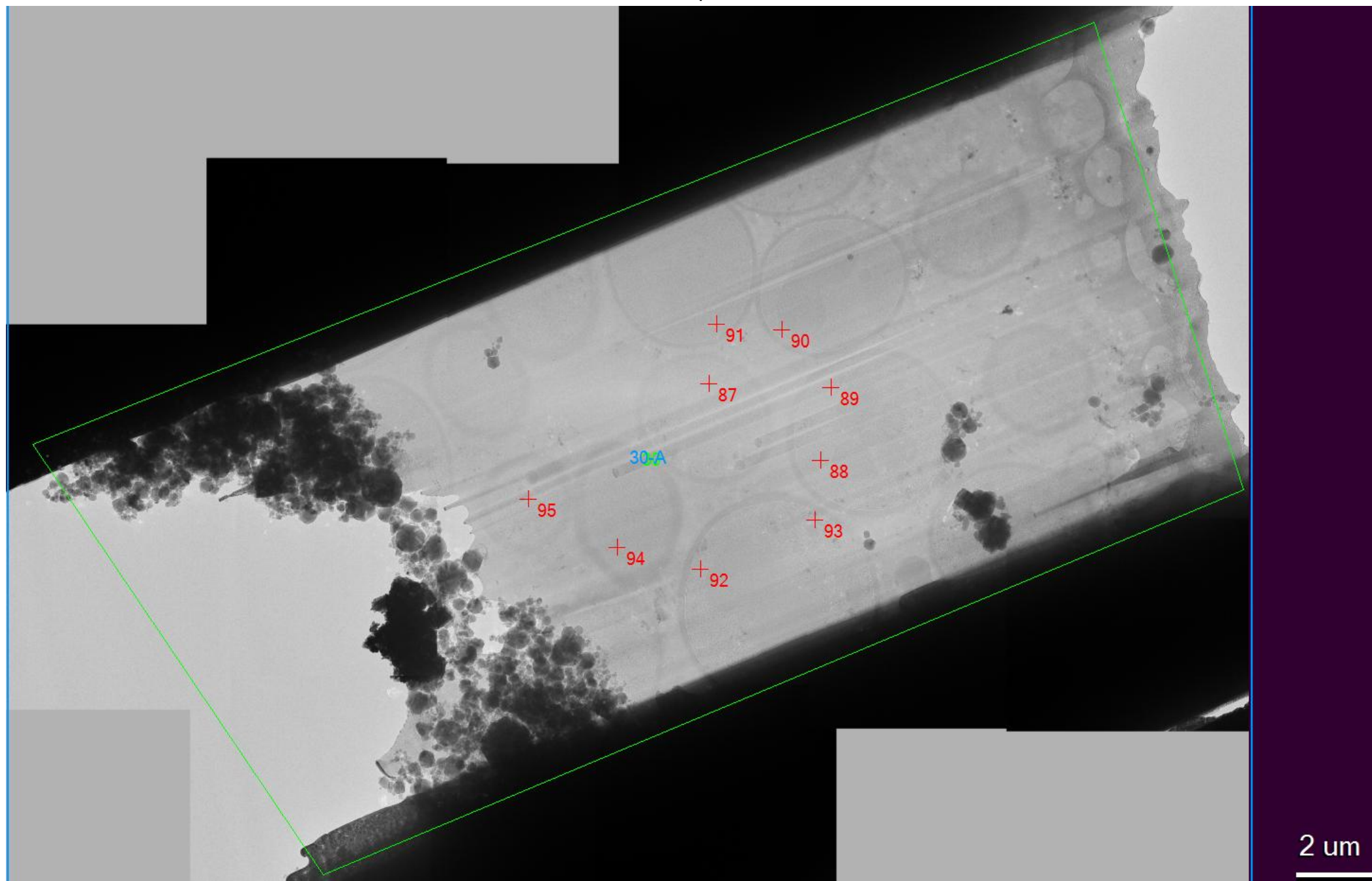
20231129

64kx, 3e/tilt, + -45度, 3度/tilt



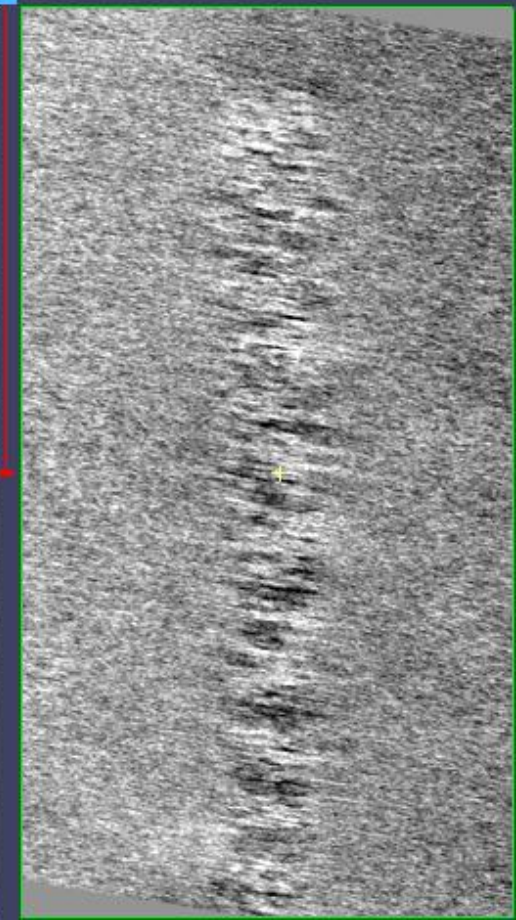
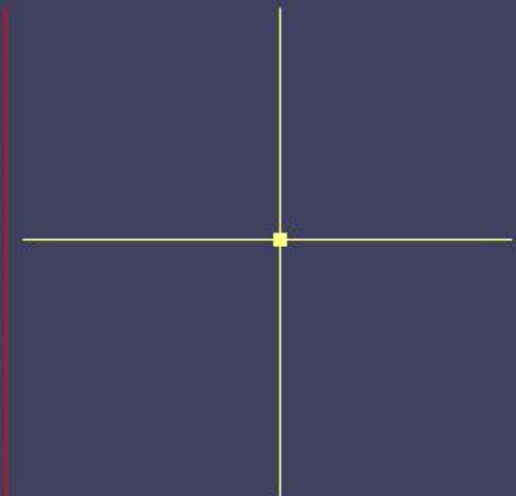
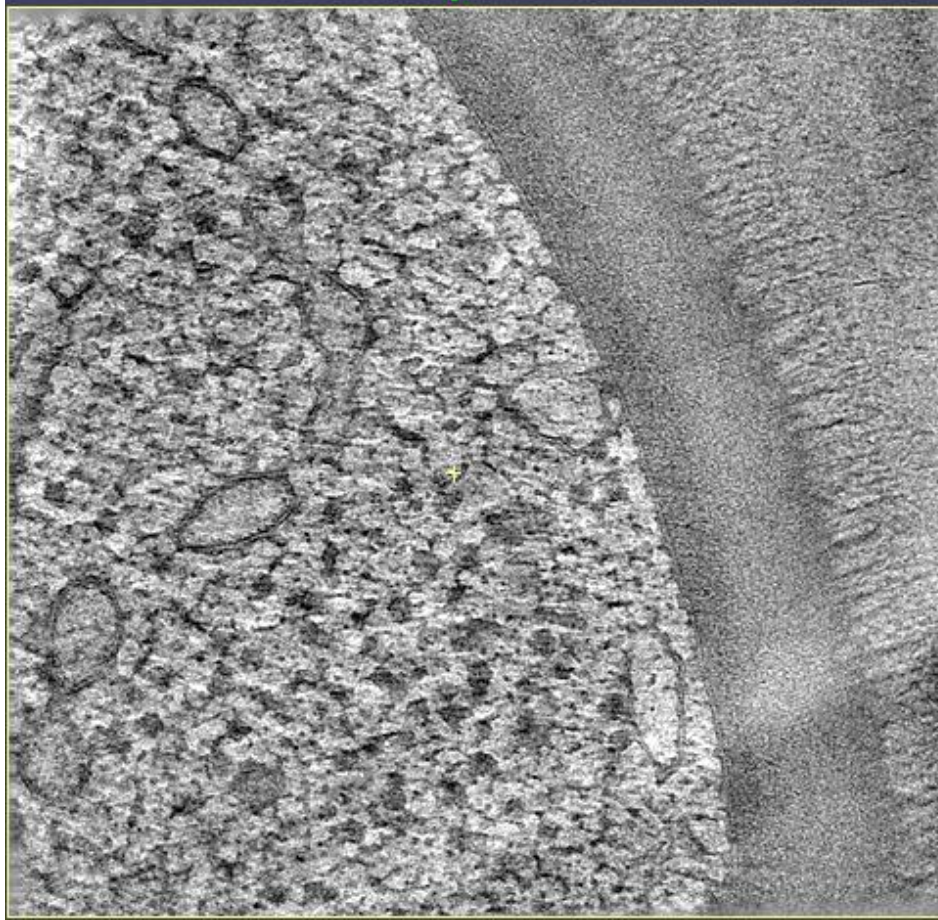
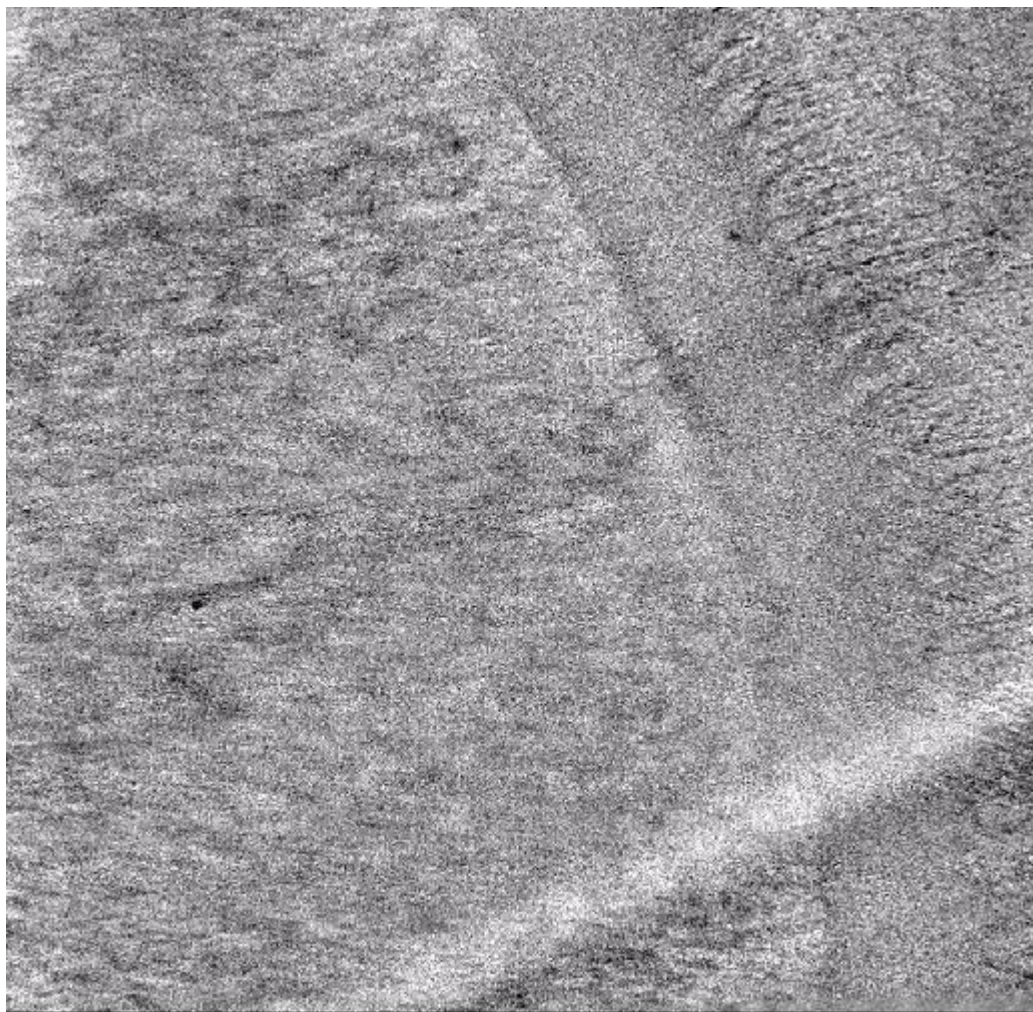
#5

0:10-0:50进行减薄, 次日9:00取出, 在Aquilos中经历约9小时



#5-88

酿酒酵母（丙烷-乙烷混合冷冻），93nm（target 100nm）



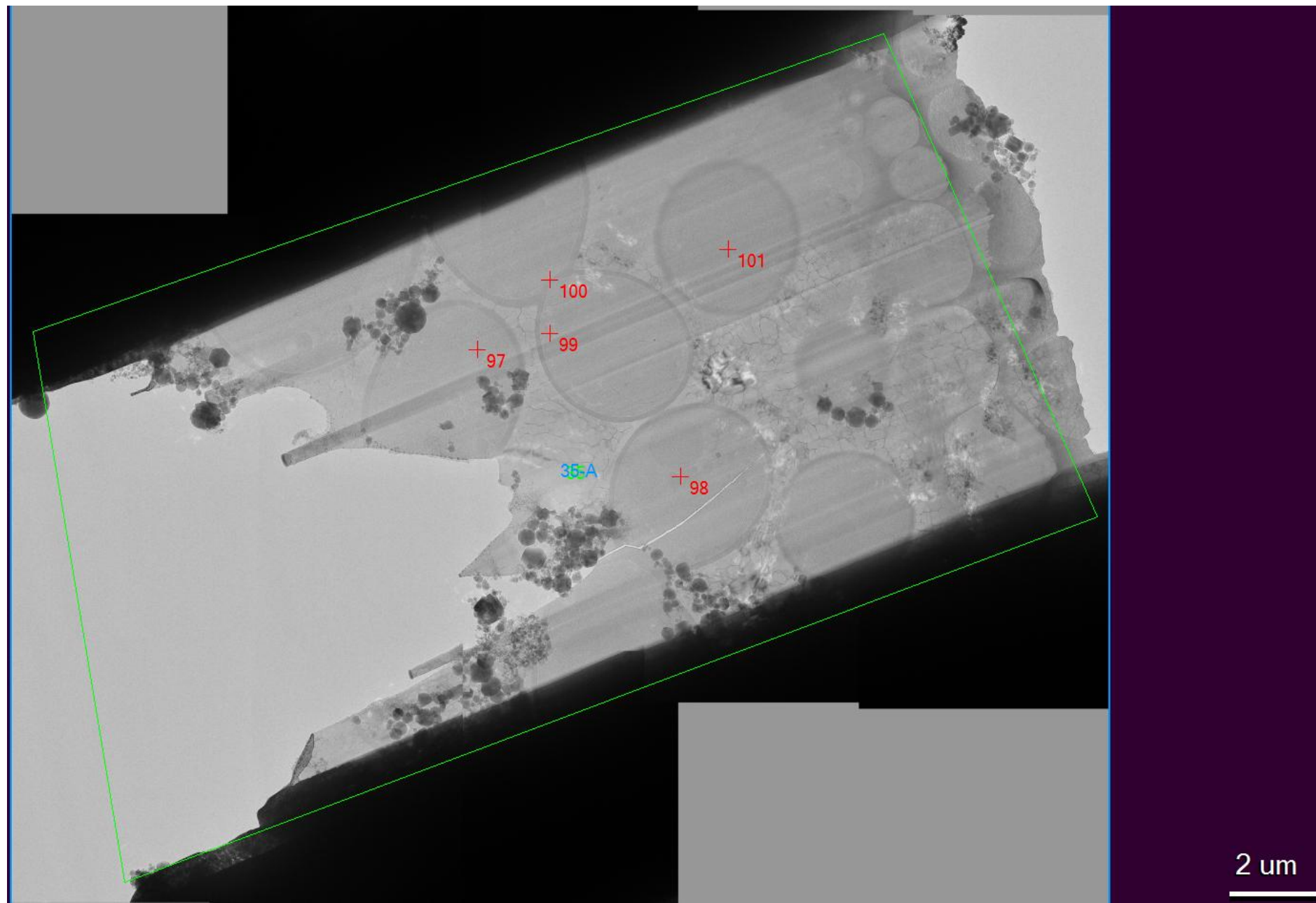
Titan3 data collection

20231129

64kx, 3e/tilt, + -45度, 3度/tilt

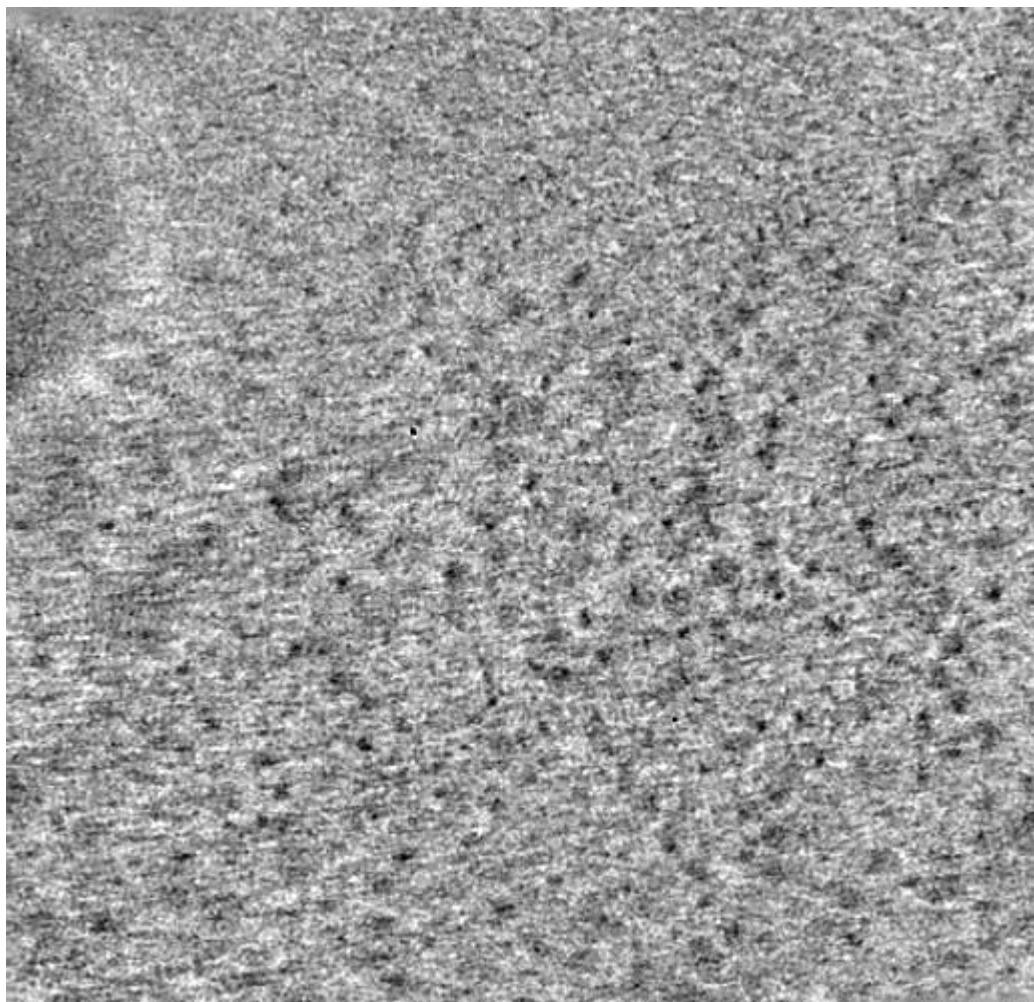
#6

0:50-1:30进行减薄, 次日9:00取出, 在Aquilos中经历约8小时



#6-97

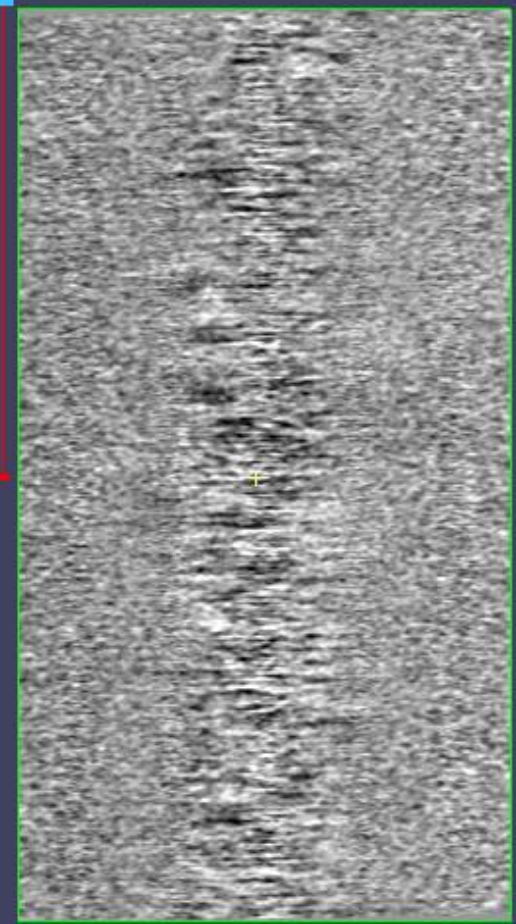
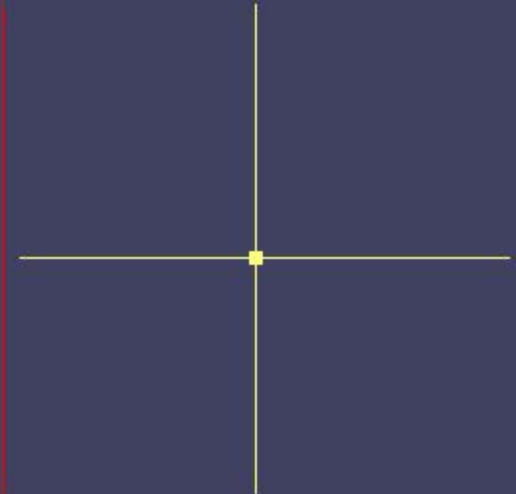
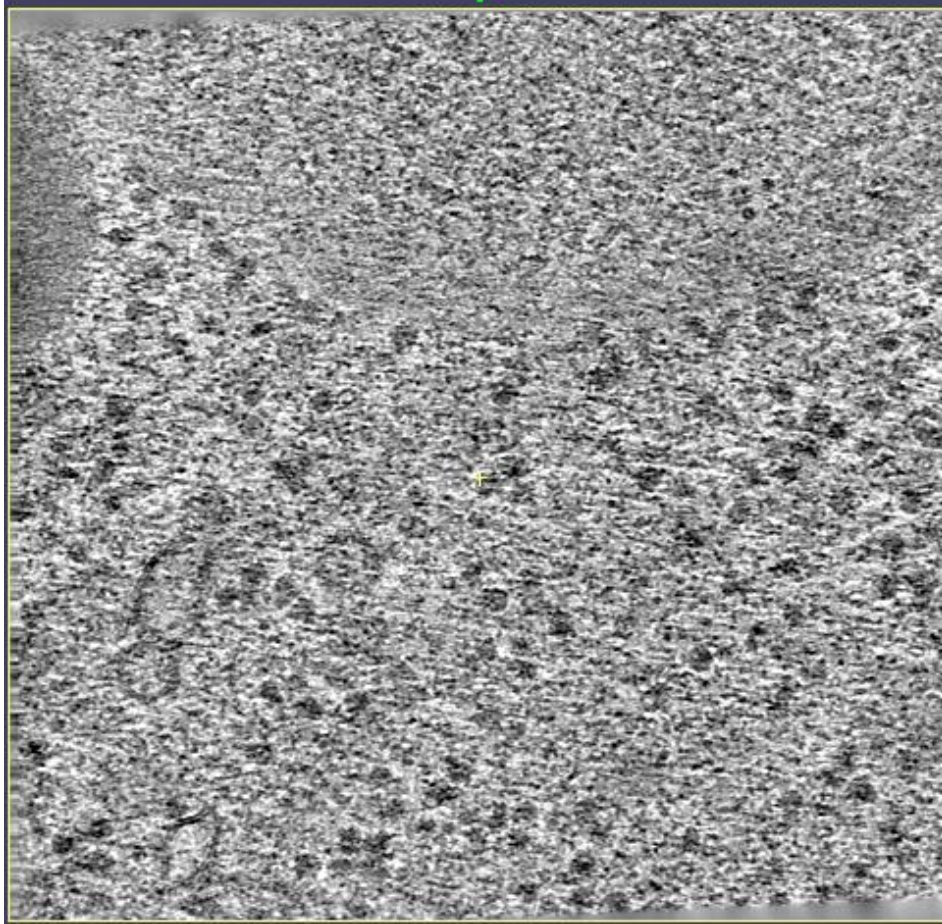
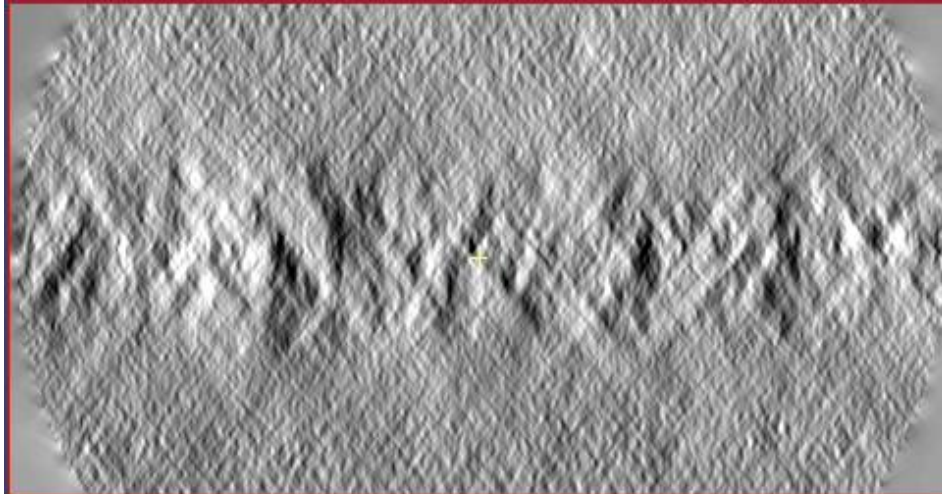
酿酒酵母（丙烷-乙烷混合冷冻），137nm（target 100nm）



Titan3 data collection

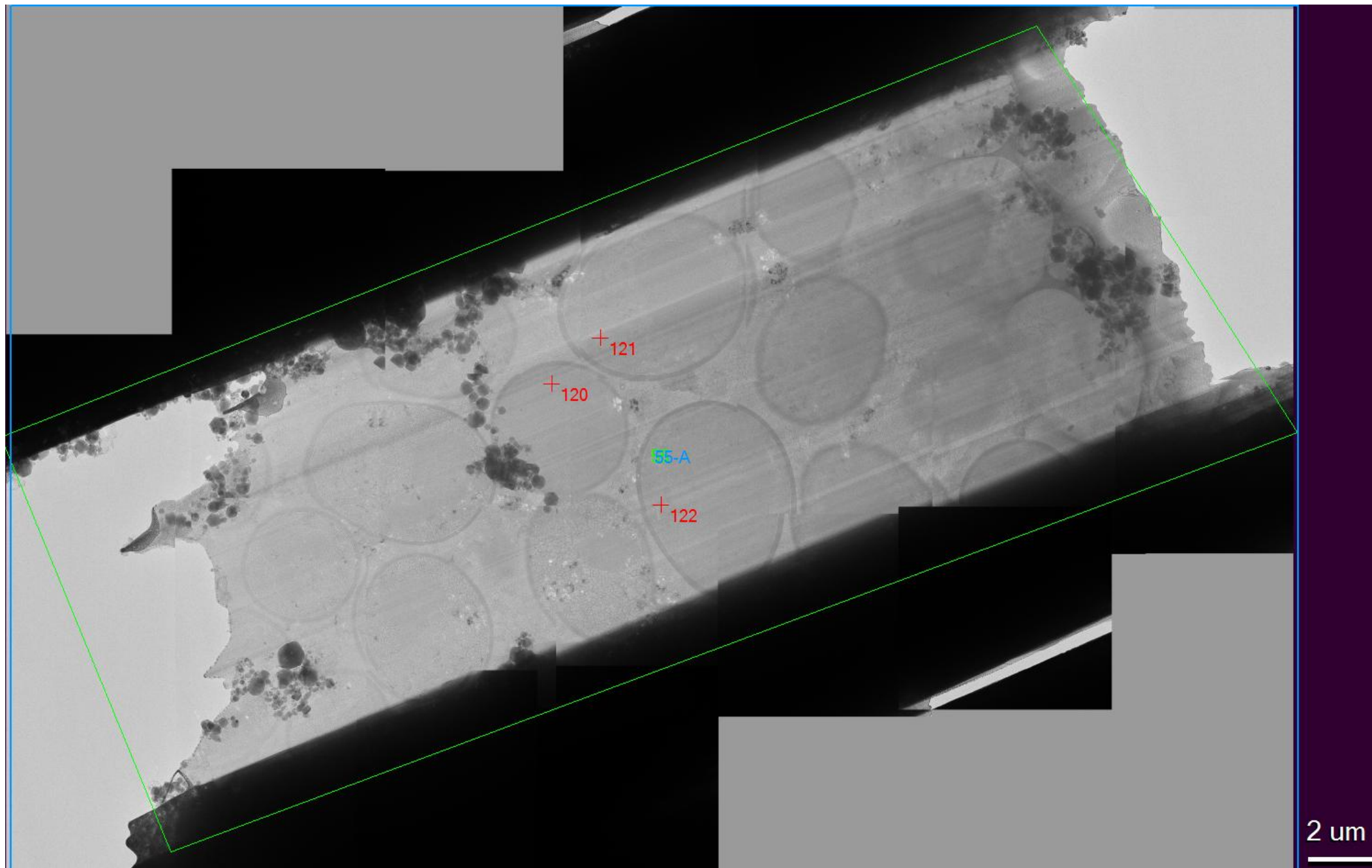
20231129

64kx, 3e/tilt, + -45度, 3度/tilt



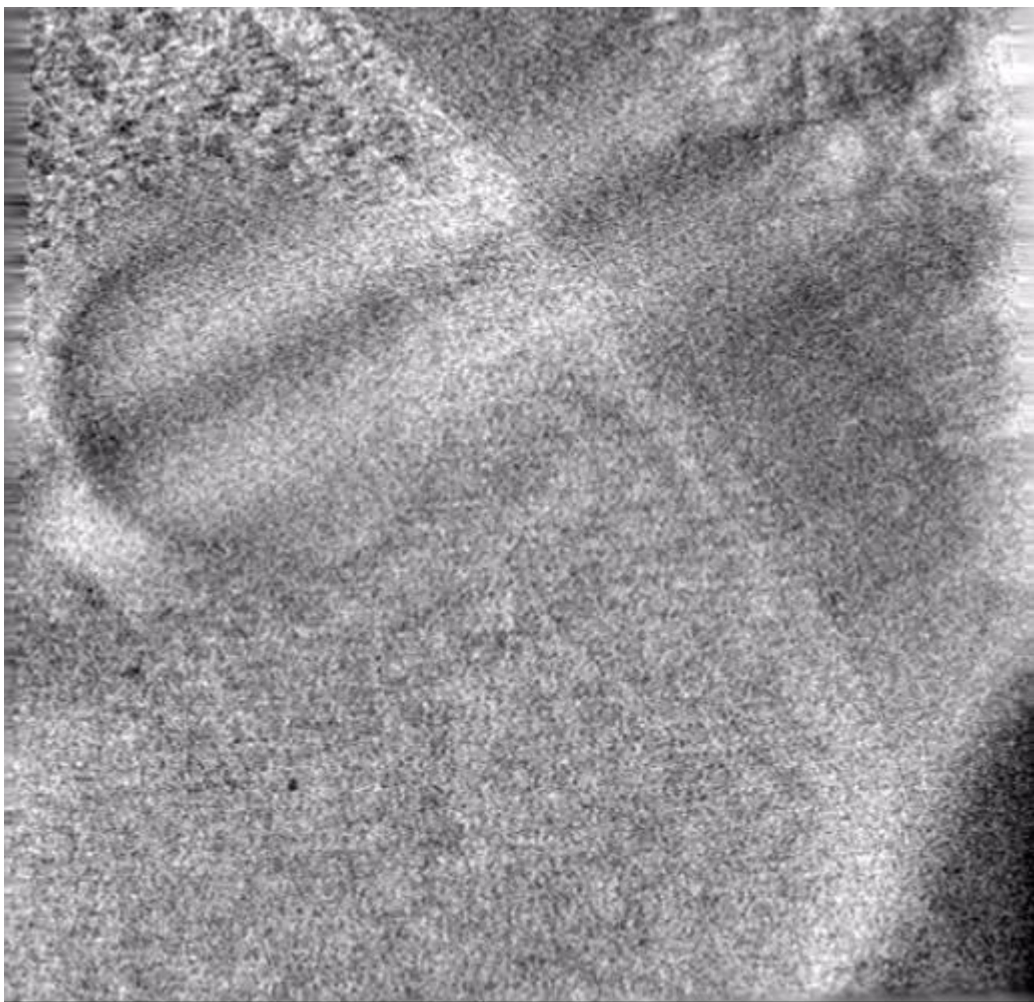
#10

3:30-4:10进行减薄, 次日9:00取出, 在Aquilos中经历约5小时



#10-121

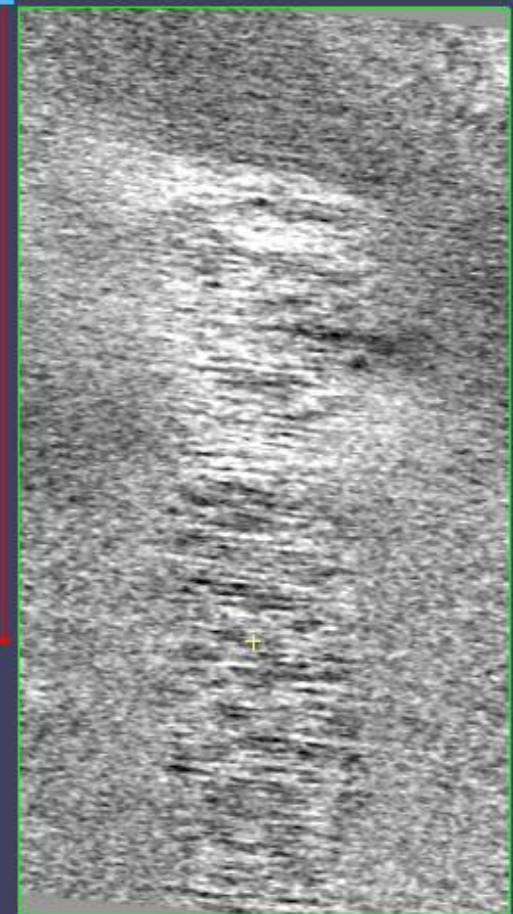
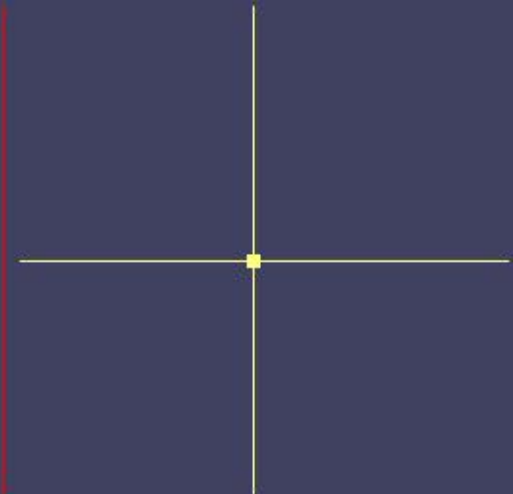
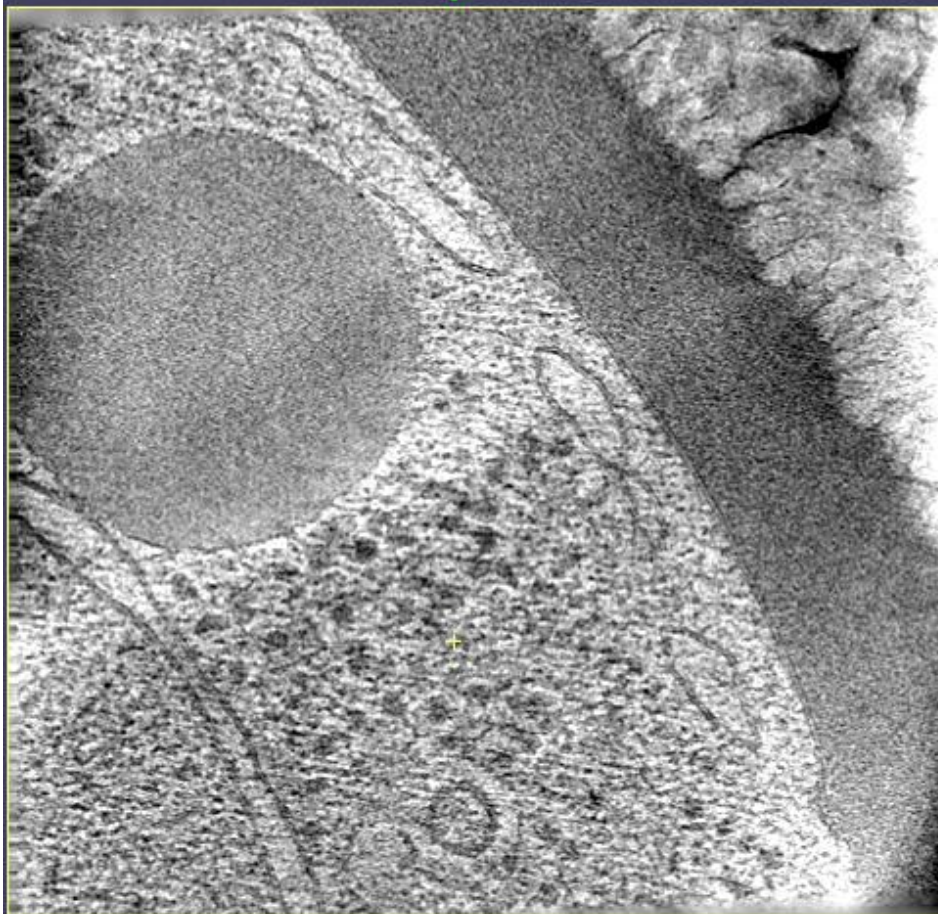
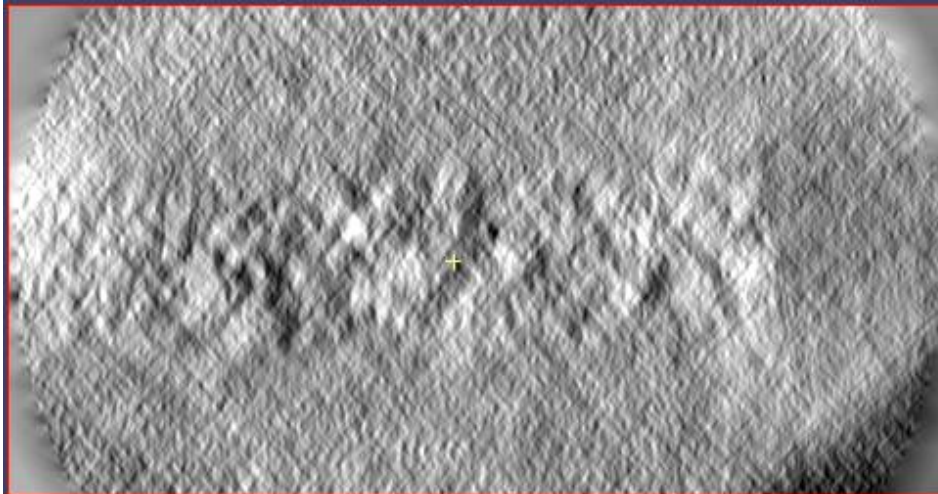
酿酒酵母（丙烷-乙烷混合冷
冻），170nm（target 100nm）



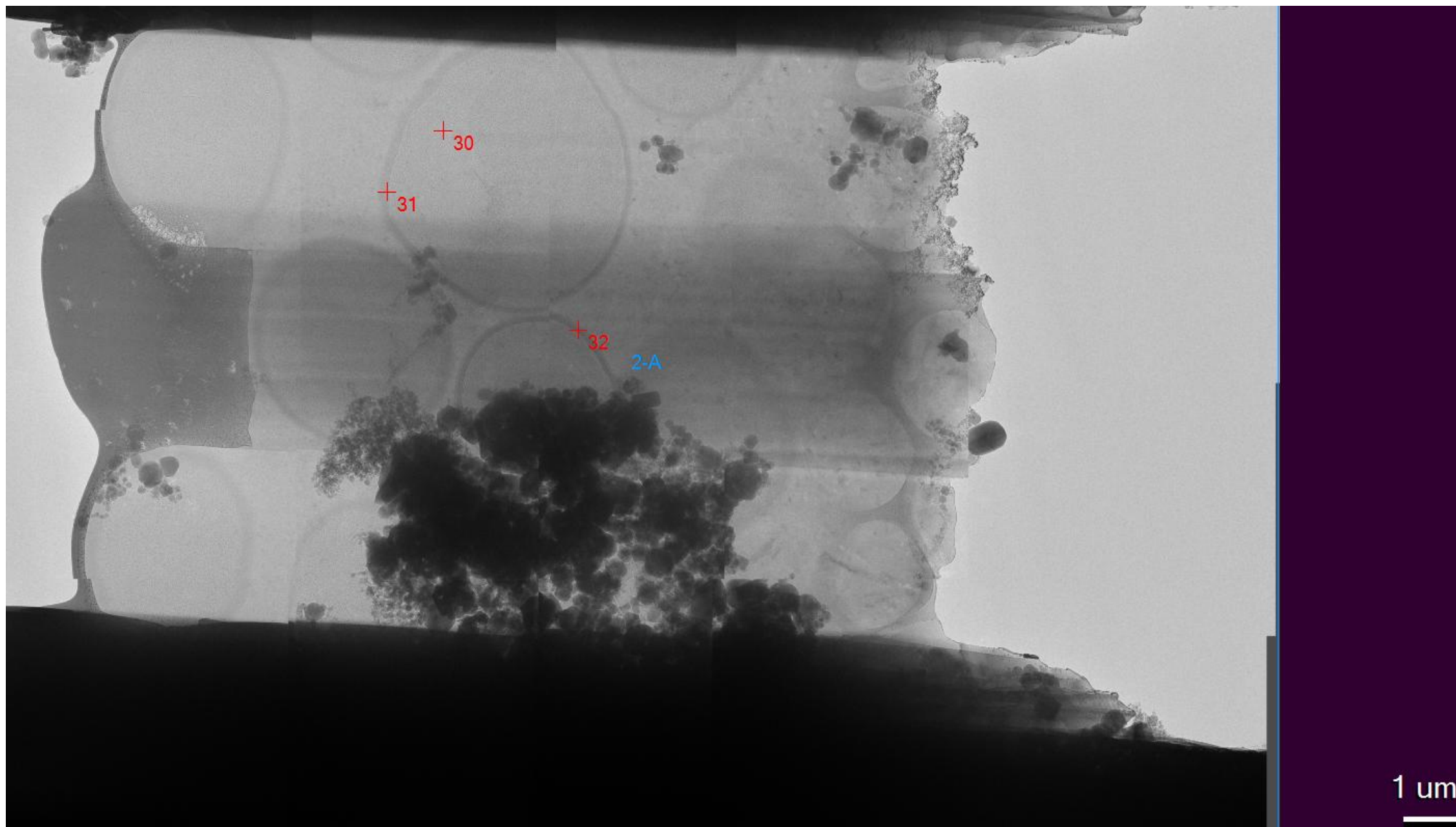
Titan3 data collection

20231129

64kx, 3e/tilt, + -45度, 3度/tilt

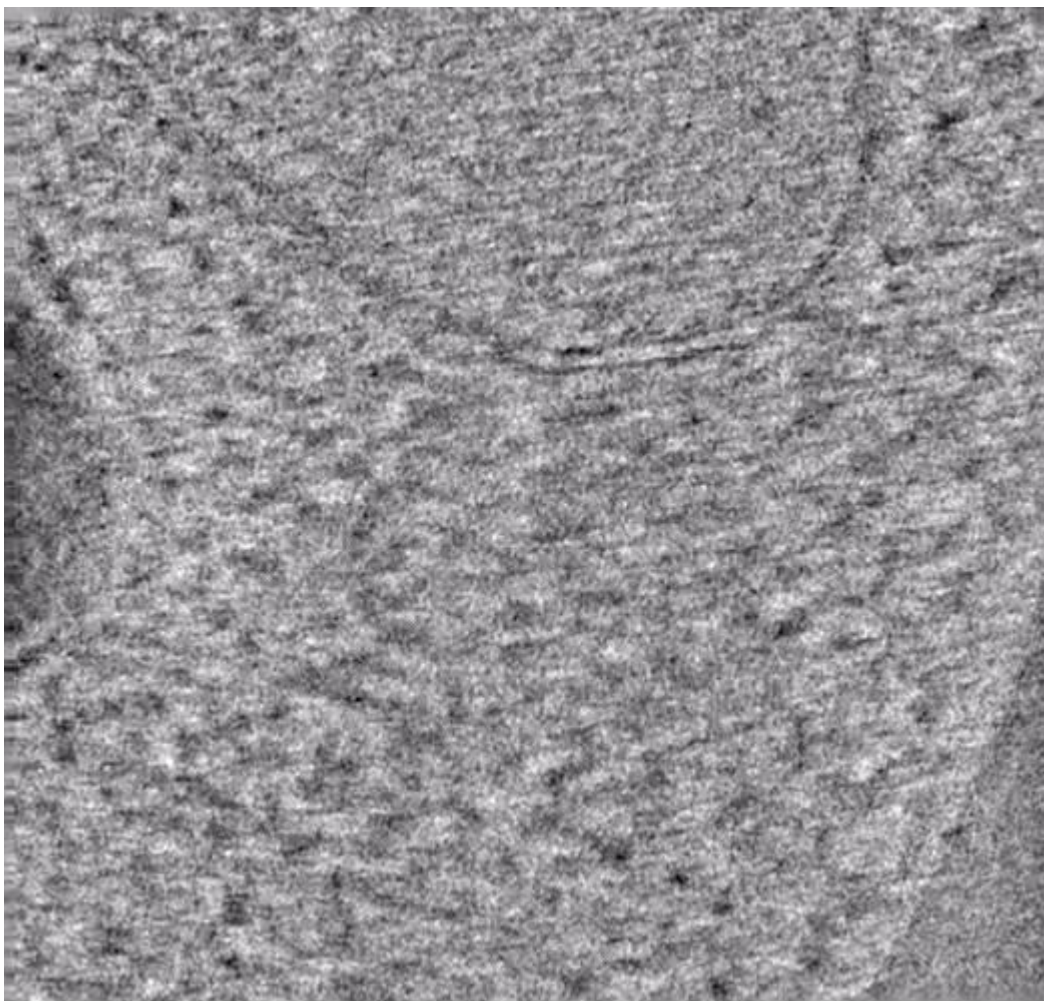


#13 3:30-4:10进行减薄，次日9:00取出，在Aquilos中经历约5小时



#13-30

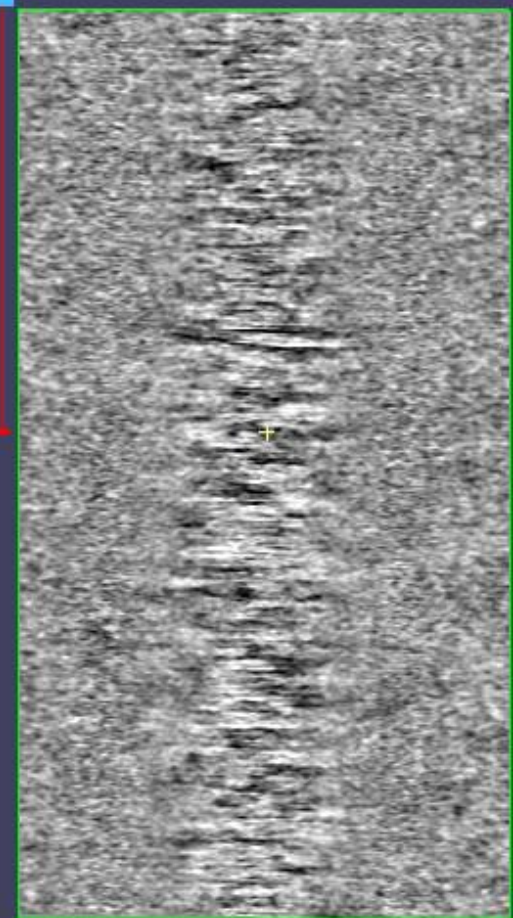
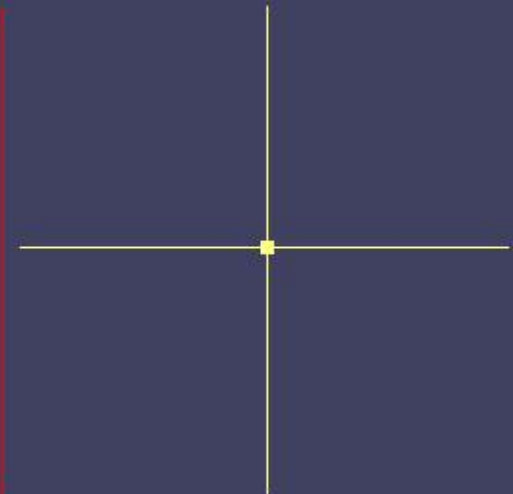
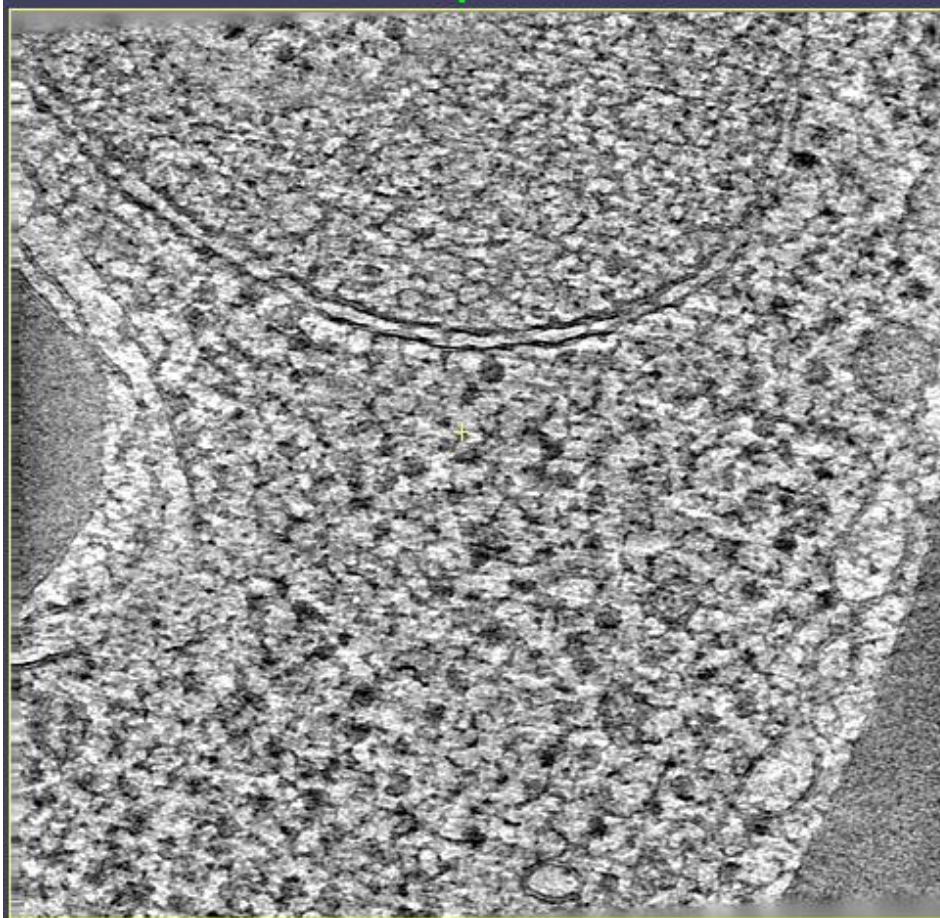
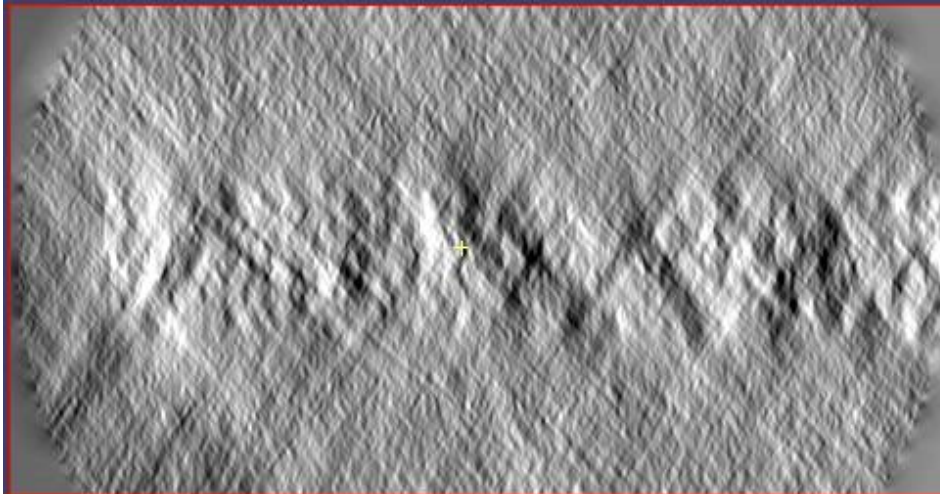
酿酒酵母（丙烷-乙烷混合冷冻），128nm（target 100nm）



Titan3 data collection

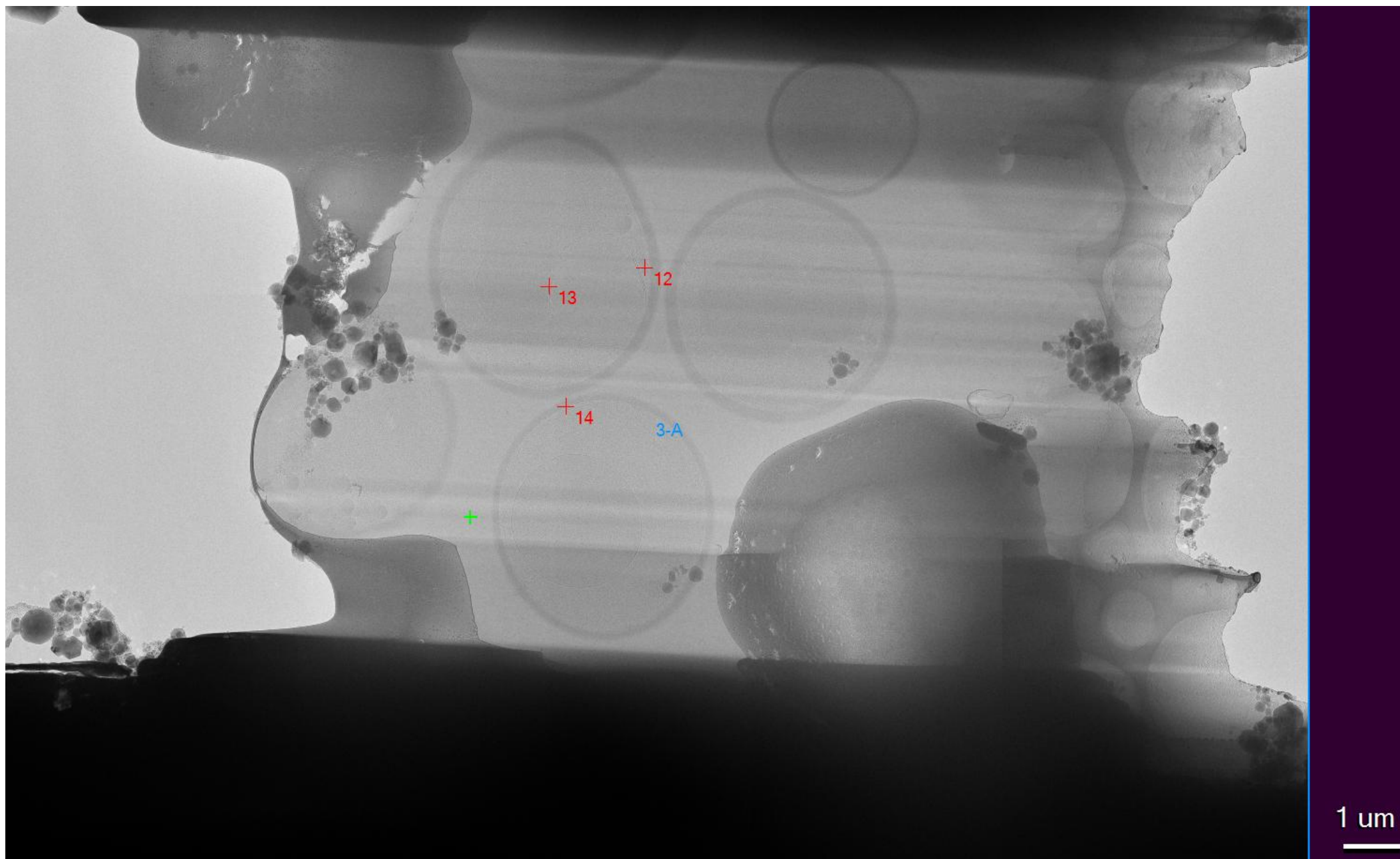
20231129

64kx, 3e/tilt, + -45度, 3度/tilt



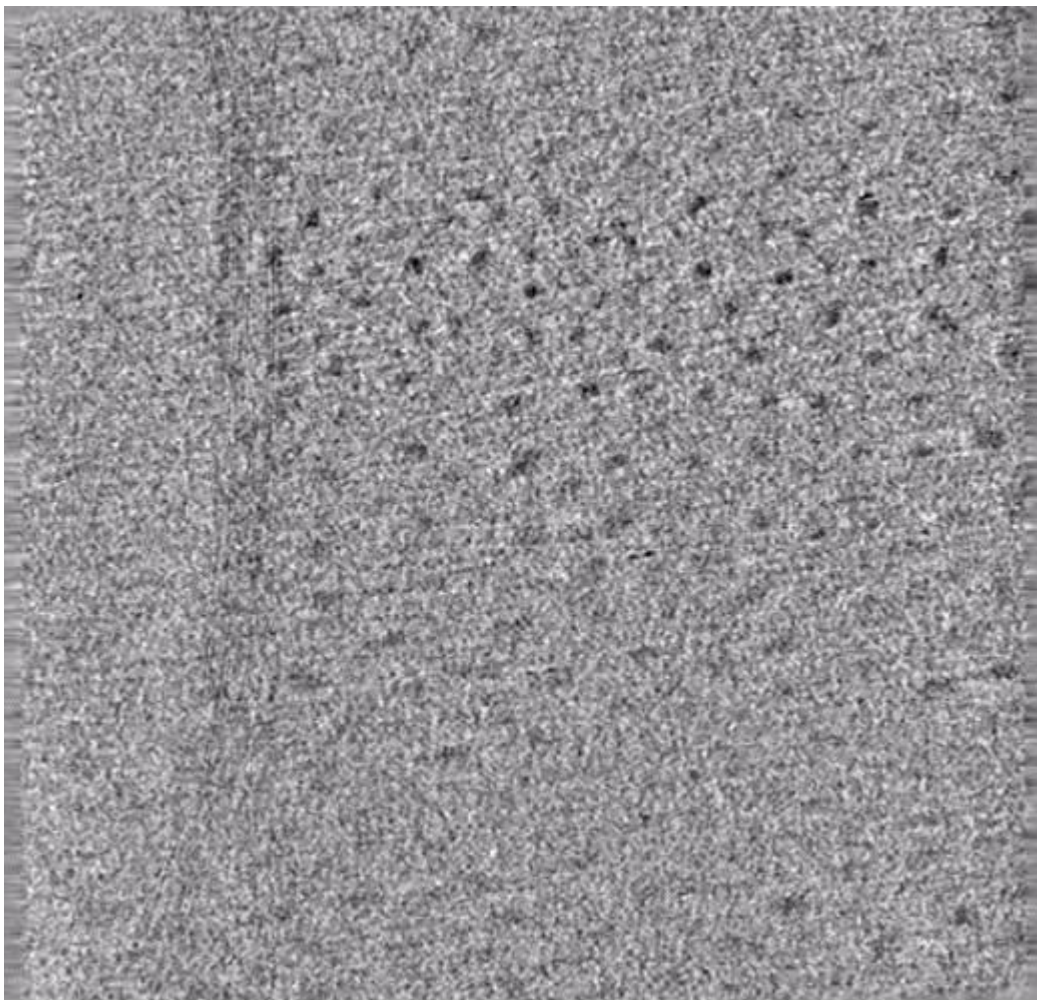
#14

5:30-6:10进行减薄，次日9:00取出，在Aquilos中经历约3小时



#14-13

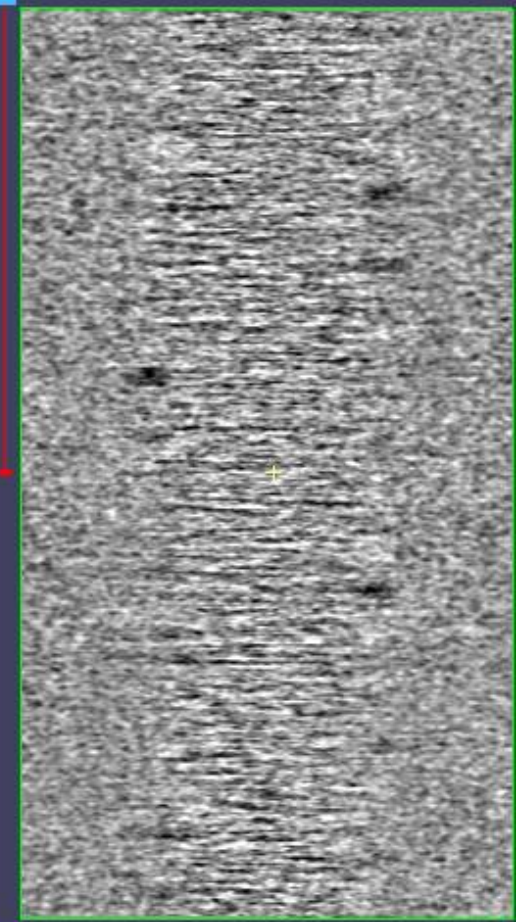
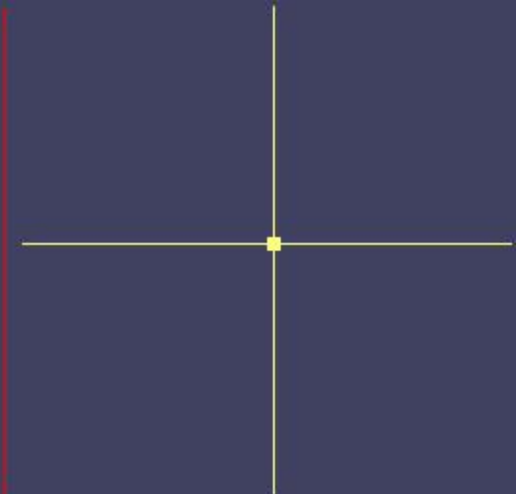
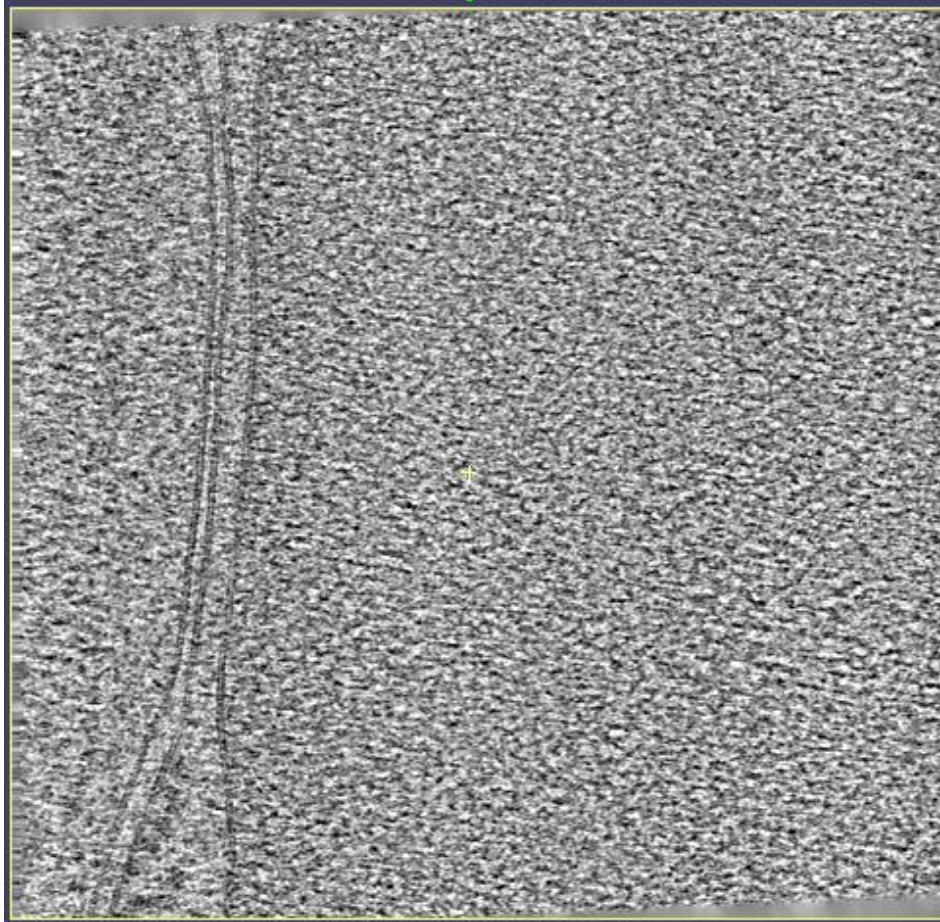
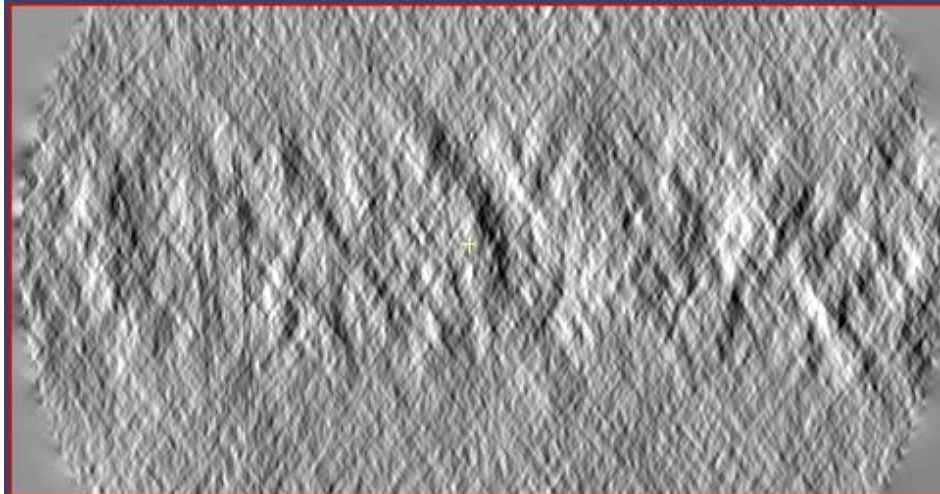
酿酒酵母（丙烷-乙烷混合冷冻），204nm（target 100nm）



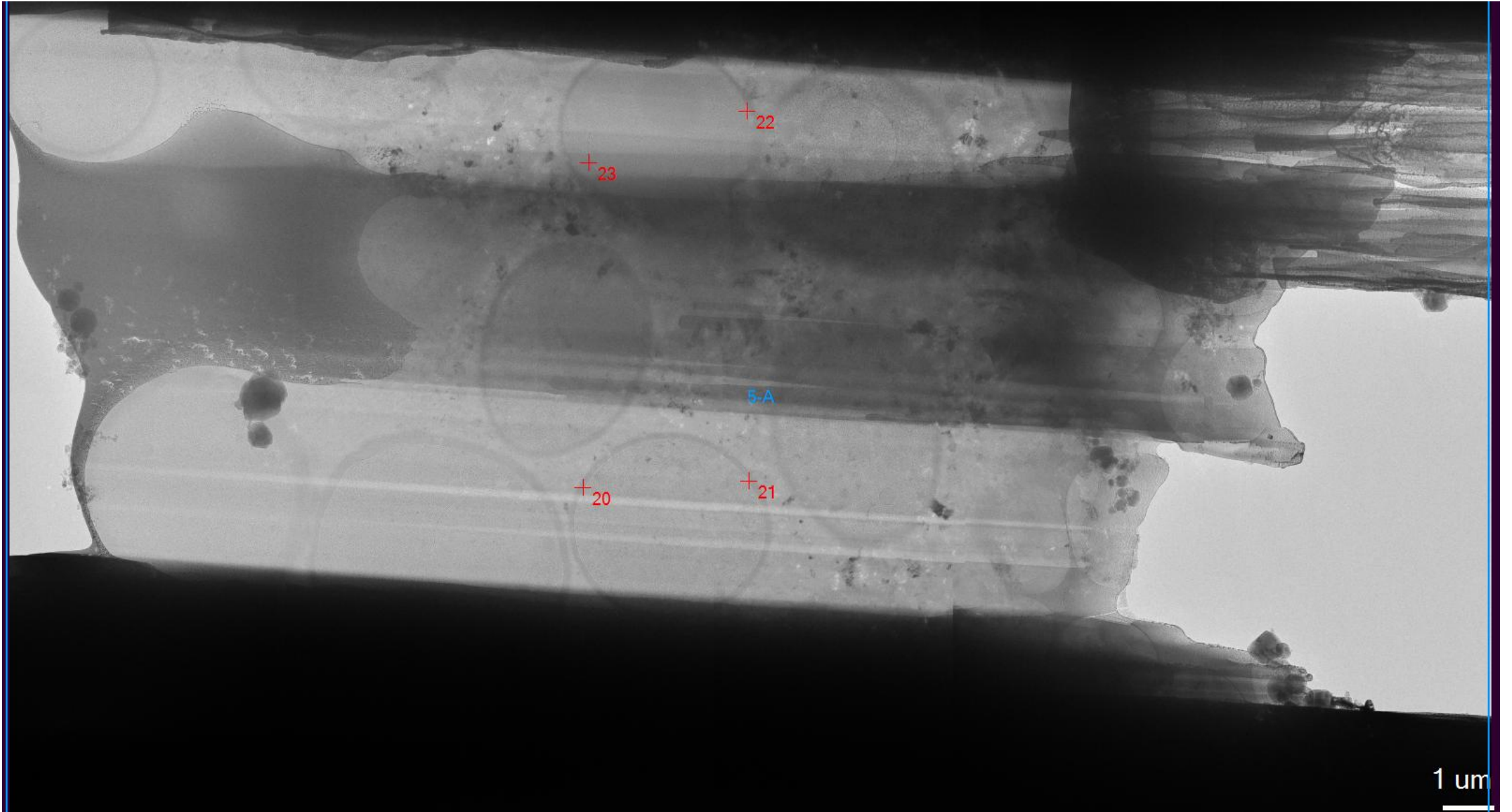
Titan3 data collection

20231129

64kx, 3e/tilt, + -45度, 3度/tilt

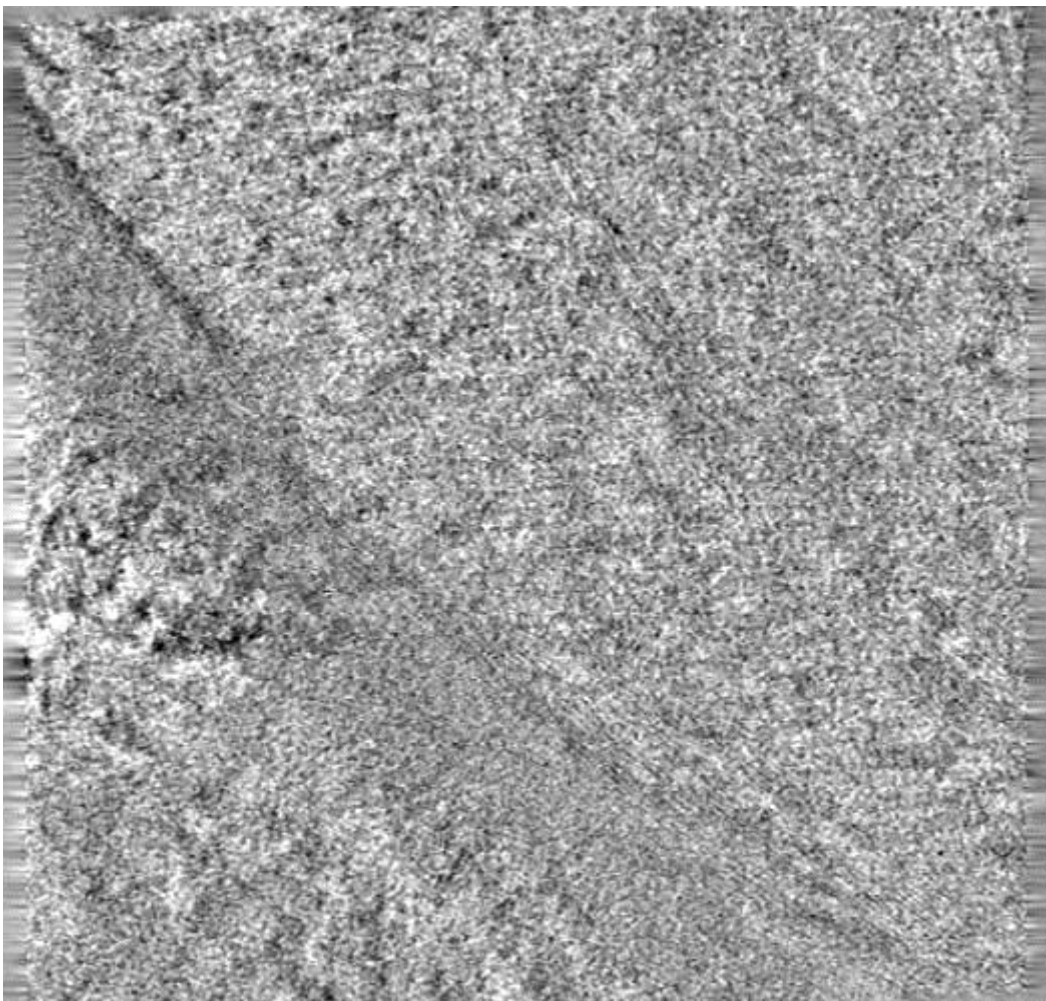


#16 6:50-7:30进行减薄，次日9:00取出，在Aquilos中经历约1小时



#16-21

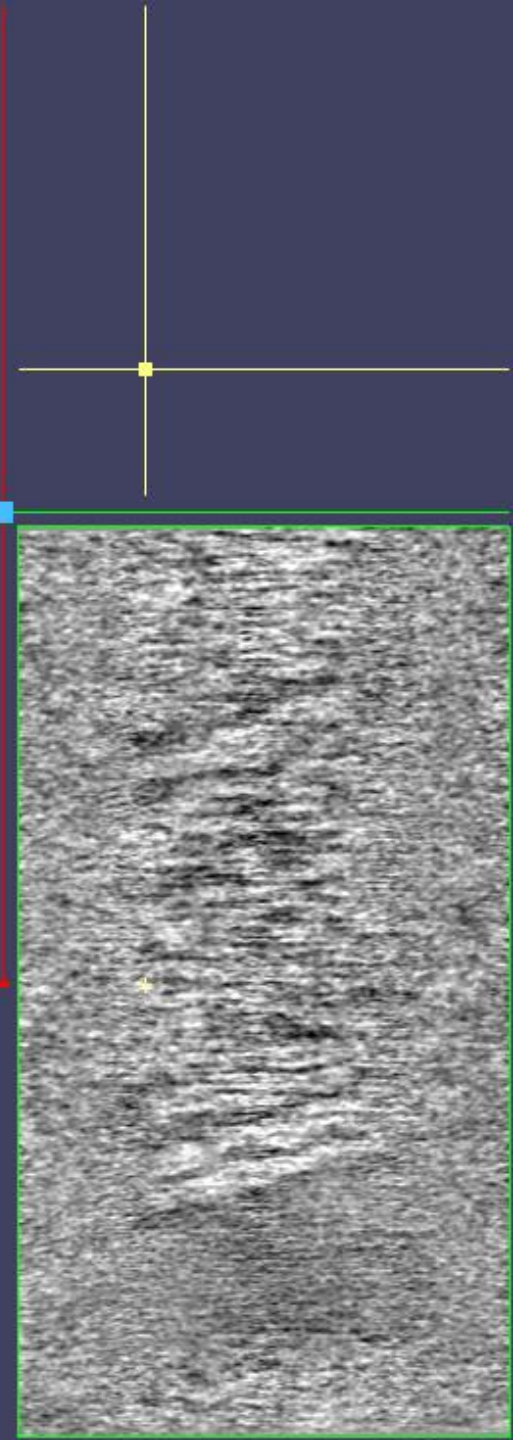
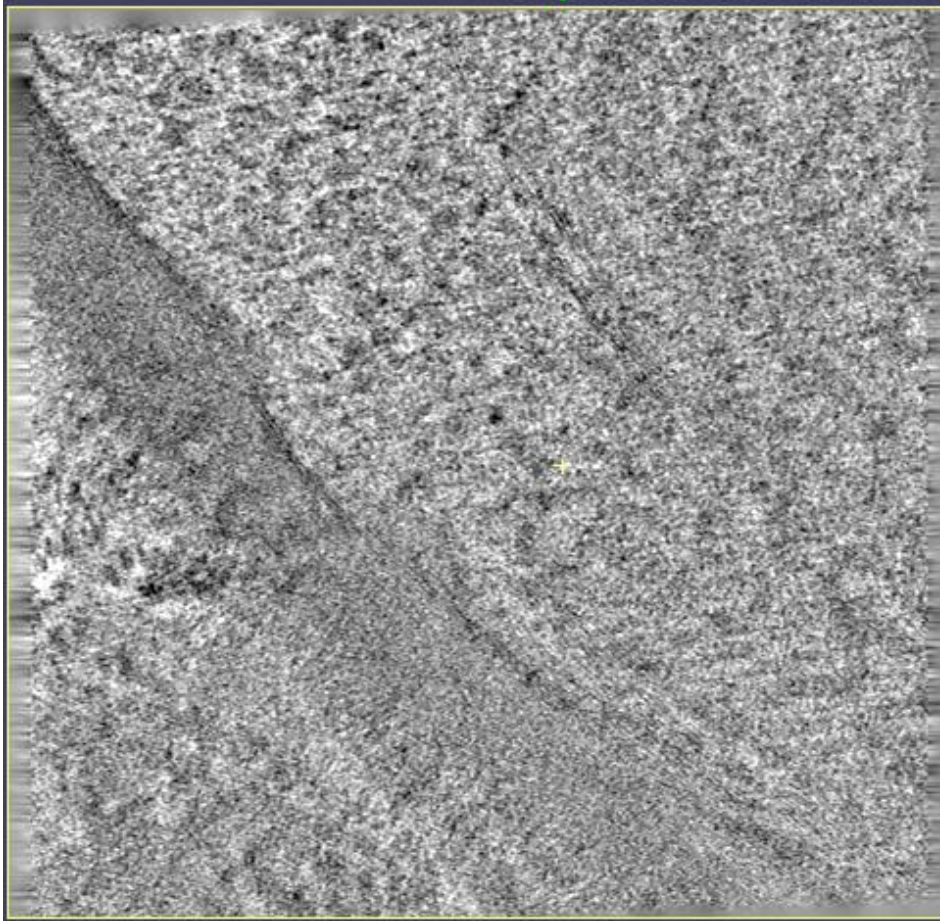
酿酒酵母（丙烷-乙烷混合冷冻），198nm（target 100nm）



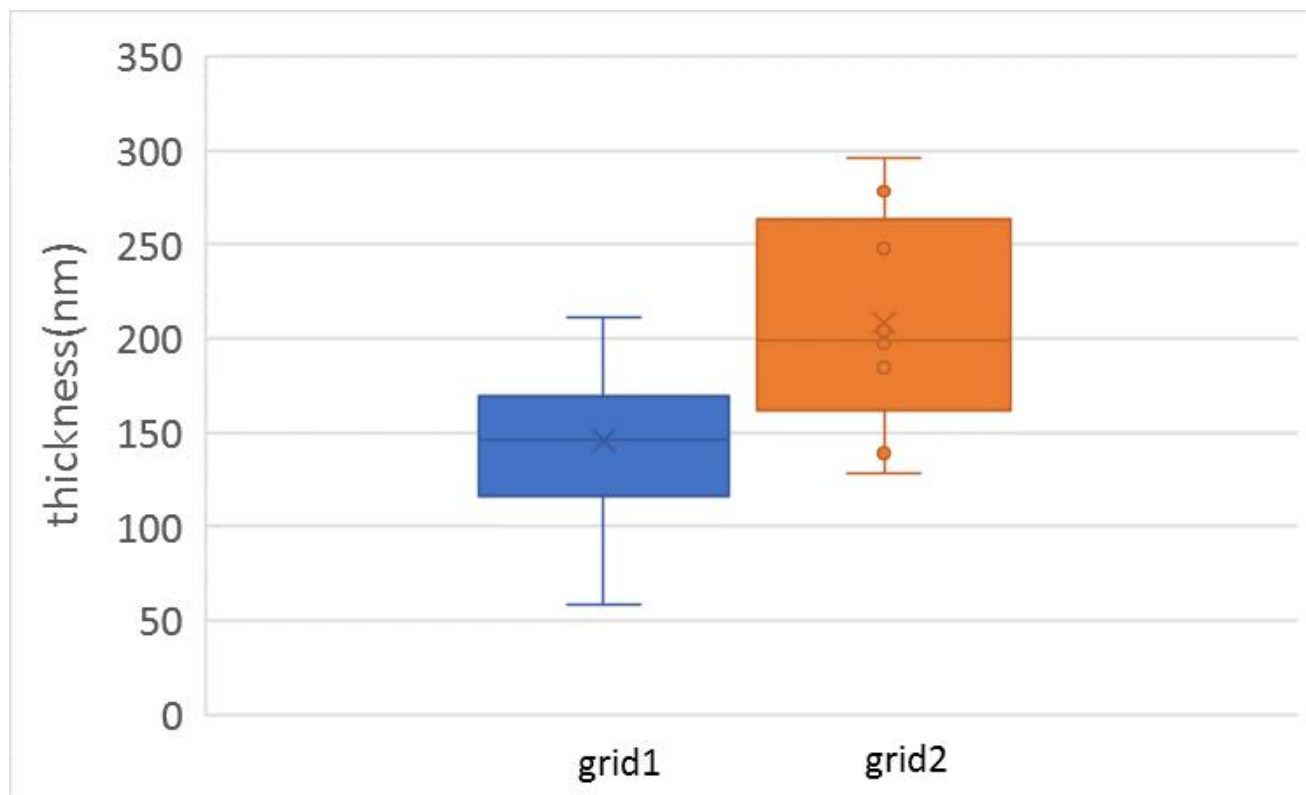
Titan3 data collection

20231129

64kx, 3e/tilt, + -45度, 3度/tilt



厚度统计



目标厚度设置100nm

测试条件二结果

- Site-wise (在一个位置完成初修和精修后, 再移动到下一个位置操作)
 - 未观察到明显污染层
- 减薄过程不进行小角度倾转
 - 切片沿减薄方向, 从前往后呈现由薄到厚的渐变
- 目标厚度设置100nm
 - 厚度范围在60-300nm之间, 多数在120-260nm
- 其它问题
 - 晶态冰层
 - 样品厚度受GIS沉积层厚度影响很大